

On Rectangularity in Distributionally Robust Asset Pricing

Master in Quantitative Economics and Finance

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Abstract

We study how imposing rectangularity of the ambiguity set affects robust superhedging prices under distributional uncertainty. Rectangularity is often used because it makes dynamic programming possible, but it also changes the underlying model of uncertainty: it allows worst-case conditional distributions to be chosen independently after each history, even when the original uncertainty links distributions across time. As a result, the computed robust price may reflect not only protection against model uncertainty, but also the additional conservativeness introduced by rectangularization.

We make two main contributions. First, we introduce and analyze the rectangularity gap, defined as the difference between the robust superhedging price under the rectangularized ambiguity set and the price under the original non-rectangular ambiguity set. We show that this gap is nonnegative and relate it to the loss of global consistency constraints across time. We also derive theoretical examples and structural results that identify when the gap can vanish and when it can become economically significant.

Second, we reformulate robust superhedging as a robust Markov decision process (MDP) and compare rectangular and non-rectangular prices for a parametric volatility uncertainty model in order to quantify the gap. The numerical results show that the rectangularity gap can be large when uncertainty is coupled across time, meaning that rectangular approximations may substantially overstate the capital required for robust hedging. Overall, we show that rectangularity is not only a tractability assumption, but a modeling choice with direct pricing consequences.

Keywords: *Distributionally Robust Optimization, Robust Asset Pricing, Rectangularity, Markov Decision Processes, Model Uncertainty*

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1 Introduction

In recent years, asset pricing theory has increasingly incorporated forms of uncertainty that go beyond classical probability models. Traditional no-arbitrage models typically assume a known probability distribution for asset returns. In practice, however, financial markets are subject to model uncertainty and ambiguity, since the underlying distribution is often unknown and may change over time. This mismatch between theoretical assumptions and empirical realities motivates more robust approaches to financial modeling.

Distributional robustness provides one way to address this problem. Instead of relying on a single fixed probability measure, robust methods consider a family of possible distributions that may represent the true data-generating process. This framework has developed along several complementary lines. The dominated multiple-priors approach of Epstein and Schneider (2003) represents ambiguity through a set of priors dominated by a common reference measure. The sublinear expectations approach of Peng (2007) specifies ambiguity locally through a convex set of admissible volatility matrices in continuous time. The non-dominated framework of Bouchard and Nutz (2015), which we adopt here, allows for ambiguity sets consisting of mutually singular probability measures and is therefore well suited to ambiguity sets defined through statistical distances such as Wasserstein balls.

Although these approaches differ in their structural assumptions, they share a common feature. Computational tractability is typically recovered by imposing rectangularity on the ambiguity set, either explicitly as in Epstein and Schneider (2003) and the robust Markov decision processes (MDP) literature of Iyengar (2005) and Nilim and El Ghaoui (2005), or implicitly through the local specification of ambiguity as in Peng (2007). Rectangularity allows robust MDP problems to be solved through dynamic programming, and Wiesemann et al. (2013) showed that without it the problem is strongly NP-hard. This helps explain why rectangularity has become standard in much of the literature. At the same time, it allows the adversary to reselect the worst-case model after every history, thereby enlarging the set of feasible adversarial strategies beyond what the modeler may have intended. The resulting prices can therefore be more conservative than those justified by the original non-rectangular ambiguity set. However, relatively little is known about the magnitude of this conservativeness in concrete pricing problems or about the conditions under which it becomes economically significant.

We study this issue in the context of robust superhedging under distributional uncertainty. Starting from classical no-arbitrage theory, we review asset pricing under model uncertainty and distributionally robust optimization before focusing on the distinction between rectangular and non-rectangular ambiguity sets. Rectangularity is the property that allows admissible one-step conditional distributions to be selected independently at each time and history, and it is precisely this property that makes robust dynamic programming possible. However, there is often no economic reason to expect the underlying ambiguity to satisfy such a structure. The central question is therefore whether imposing rectangularity for computational tractability leads to overly conservative robust superhedging prices.

The main contribution of this thesis is to identify and formalize this pricing distortion through the notion of the rectangularity gap. We define the rectangularity gap as the difference between the robust superhedging price under the original ambiguity set and the price obtained after rectangularization. We show that this gap is nonnegative, relate it to the enlargement of the ambiguity set caused by rectangularization, and provide theoretical and numerical evidence that it can become economically significant when uncertainty is coupled across time. More fundamentally, we argue that rectangularity is not merely a computational simplification enabling dynamic programming, but a modeling choice that alters the structure of uncertainty itself. In

non-rectangular models, admissible conditional distributions are linked through global consistency constraints across time. Rectangularization removes these constraints and permits conditional laws to be recombined independently after each history, thereby enlarging the set of feasible adversarial strategies. As a consequence, the resulting worst-case model under rectangularization need not correspond to any globally admissible probability measure in the original ambiguity set. To quantify the effect of this enlargement, we introduce the rectangularity gap, defined as the difference between the robust superhedging price under the rectangularized ambiguity set and the price under the original ambiguity set. We show that this gap is nonnegative, relate it to the failure of global consistency constraints under dynamic recombination, and provide theoretical and numerical examples demonstrating that the gap can become economically significant when uncertainty is temporally coupled. Furthermore, to study this effect computationally, we reformulate robust superhedging as a robust Markov decision process on a history-augmented state space. This allows us to compare rectangular and non-rectangular prices in both parametric and nonparametric ambiguity models and to use actor-critic methods to approximate non-rectangular prices for larger horizons.

The thesis is structured as follows. Section 2 reviews classical no-arbitrage pricing and superhedging duality. Section 3 introduces the non-dominated framework of Bouchard and Nutz (2015), defines rectangular and non-rectangular ambiguity sets, and formalizes the rectangularity gap together with its basic structural properties. Section 4 reformulates robust superhedging as a robust Markov decision process on a history-augmented state space and studies how rectangularity is preserved under this reformulation. It also introduces the actor-critic algorithm used to approximate non-rectangular prices at larger horizons. Section 5 presents numerical experiments comparing rectangular and non-rectangular prices in both parametric and nonparametric ambiguity models. Section 6 derives analytical limits and explicit examples illustrating the behavior of the rectangularity gap. Related work is discussed throughout the thesis where it is most relevant.

2 Classical Asset Pricing

While many researchers have contributed to the formulation of the Fundamental Theorem of Asset Pricing (FTAP), Delbaen and Schachermayer (1994) extend the Fundamental Theorem of Asset Pricing beyond complete markets—markets where every contingent claim can be replicated by a portfolio of assets. Many real-world financial markets are incomplete (e.g., not all risks can be hedged), and their framework allows for no-arbitrage to be a sufficient condition guaranteeing the existence of a risk-neutral measure. Therefore, we use their model to present the classical (non-robust) theory of no-arbitrage asset pricing.

The original framework of Delbaen and Schachermayer (1994) is formulated in continuous time and models asset prices as continuous semimartingales. While continuous time is a more realistic representation of financial markets because it allows for modeling continuous trading and precise price dynamics, once we introduce distributionally robust models and non-rectangular ambiguity sets we encounter significant tractability issues in continuous time.

Therefore, for the purpose of this thesis, we adopt a complementary discrete-time, finite-horizon setting, which allows us to explore different models of uncertainty through both numerical simulation and analytical derivation while maintaining the essential economic interpretation of our model. However, an extension of this work to continuous time could be an interesting future research direction.

2.1 Market model

We consider a discrete-time financial market model over a finite horizon T with d risky assets on the filtered probability space $(\Omega, \mathcal{F}, P, (\mathcal{F}_t)_{t=0}^T)$. We model the underlying asset prices as a stochastic process $S = (S_t)_{t=0}^T$, where $S_t = (S_t^1, \dots, S_t^d) \in \mathbb{R}^d$, which is adapted to the filtration $(\mathcal{F})_{t=0}^T$.

Definition 2.1 (Trading strategy). *A trading strategy is a predictable process*

$$H = (H_t)_{t=0}^{T-1}, \quad H_t \in \mathbb{R}^d,$$

where H_t is $\mathcal{F}_t$ -measurable.

The trading strategy is interpreted as the number of units held on $[t, t+1)$, i.e., from time t to just before time $t+1$.

Definition 2.2 (Wealth process). *Given initial capital $x \in \mathbb{R}$ and a trading strategy H , the corresponding wealth process is defined by*

$$V_0^{x,H} = x,$$

and

$$V_{t+1}^{x,H} = V_t^{x,H} + H_t \cdot (S_{t+1} - S_t), \quad \forall t = 0, \dots, T-1$$

Equivalently,

$$V_t^{x,H} = x + (H \cdot S)_t, \quad \text{where} \quad (H \cdot S)_t = \sum_{k=0}^{t-1} H_k \cdot (S_{k+1} - S_k).$$

The term $(H \cdot S)_t$ represents the cumulative return of the strategy up to time t depending on the changes in the price of the underlying between each period. This formulation corresponds to the self-financing condition which ensures that the strategy does not involve any external cash flows, and all trading is funded purely by the assets already held in the portfolio.

Definition 2.3 (Admissible strategy). *A predictable strategy H is called admissible if there exists $C > 0$ such that*

$$V_t^{0,H} \geq -C \quad \forall t = 0, \dots, T \quad P\text{-a.s.}$$

We denote the set of admissible strategies by $\mathcal{H}$.

This can be interpreted as a strategy being admissible if its wealth process is uniformly bounded from below for all periods t , preventing the use of strategies that allow for unbounded losses.

No-arbitrage is the condition that allows for the asset to be fairly priced according to its level of risk and the other assets available on the market. The fundamental idea is that the prices in the market should be set so that one cannot generate a strictly positive payoff at zero cost.

Definition 2.4 (No-Arbitrage (NA)). *The market satisfies NA if for every admissible strategy $H \in \mathcal{H}$,*

$$(H \cdot S)_T \geq 0 \quad P\text{-a.s.} \quad \Rightarrow \quad (H \cdot S)_T = 0 \quad P\text{-a.s.}$$

An equivalent formulation due to Delbaen and Schachermayer (1994) allows one to reinterpret the asset pricing problem as an optimization problem over a convex cone and to view the

Fundamental Theorem of Asset Pricing as a separation result between this cone and the positive orthant.

Let $(\Omega, \mathcal{F}, P)$ be a probability space. For $p \in [1, \infty)$, define

$$L^p(P, \Omega) = \left\{ X : \Omega \rightarrow \mathbb{R} \text{ } \mathcal{F}\text{-measurable} : \int_{\Omega} |X|^p dP < \infty \right\},$$

where two functions $X, Y \in L^p(P, \Omega)$ are considered the same element if

$$X = Y \quad P - a.s.$$

For $p = \infty$,

$$L^\infty = \{X : \Omega \rightarrow \mathbb{R} \text{ } \mathcal{F}\text{-measurable} : \|X\|_\infty < \infty\},$$

where

$$\|X\|_\infty = \operatorname{ess\,sup}_{\omega \in \Omega} |X(\omega)|.$$

Again, two functions $X, Y \in L^\infty$ are considered the same element if

$$X = Y \quad P - a.s.$$

We also introduce the space of all real-valued $\mathcal{F}$ -measurable functions (random variables),

$$L^0(P, \Omega) = \{X : \Omega \rightarrow \mathbb{R} \mid X \text{ is } \mathcal{F}\text{-measurable}\},$$

where two functions $X, Y \in L^0(P, \Omega)$ are considered the same element if

$$X = Y \quad P - a.s.$$

(Brézis, 2011, Chapter 4)

Let $\mathcal{C} = \{(H \cdot S)_T : H \in \mathcal{H}\}$ denote the set of attainable claims. Then NA is equivalent to

$$\mathcal{C} \cap L_+^0 = \{0\},$$

that is, no nontrivial nonnegative payoff can be generated at zero cost.

2.2 Asset pricing: the primal problem

The financial instrument (derivative) that we want to price is called a contingent claim, i.e., a financial contract whose payoff at time T depends on the realized path or terminal value of the underlying assets. Formally, a contingent claim is an $\mathcal{F}_T$ -measurable random variable $\xi : \Omega \rightarrow \mathbb{R}$ such that $\xi \in L^\infty(P, \Omega)$.

Then the pricing question boils down to: What is a fair price today for a random payoff ξ delivered at time T ? The no-arbitrage condition constrains which payoffs can be generated at zero cost, but it does not directly tell us how to price a contingent claim that cannot be perfectly replicated.

The appropriate fair upper bound on the price of a claim ξ is given by the superhedging price: the minimal initial capital that allows the seller to offset the risk of ξ by trading appropriately in the given stocks.

Definition 2.5 (Superhedging price). *For a contingent claim $\xi \in L^\infty(P, \Omega)$, the superhedging*

price is defined as

$$\pi_s(\xi) = \inf \{x \in \mathbb{R} : \exists H \in \mathcal{H} \text{ such that } x + (H \cdot S)_T \geq \xi \quad P\text{-a.s.}\}.$$

The superhedging price is the smallest amount of initial capital with which one can construct a portfolio that, regardless of how prices evolve, ends up being worth at least ξ at time T . Any seller charging less than $\pi_s(\xi)$ cannot guarantee delivery of ξ ; any seller charging more may be undercut by a competitor who is able to offer a lower price.

When ξ is attainable, i.e., $\xi = x + (H \cdot S)_T$ for some $H \in \mathcal{H}$, the superhedging price reduces to x , the cost of replicating the terminal payoff of the portfolio (i.e., the initial capital).

The primal superhedging problem can thus be written equivalently as

$$\pi_s(\xi) = \inf_{x, H \in \mathcal{H}} \{x : x + (H \cdot S)_T \geq \xi \quad P\text{-a.s.}\}.$$

This is a linear program in the pair (x, H) , where the objective is affine in x and the constraint is linear in both x and the trading strategy H . In a finite-state setting, this reduces to a finite-dimensional linear program. We exploit this structure in the next subsection to derive a dual representation via linear programming duality.

2.3 Asset pricing: the dual problem

The no-arbitrage condition can be shown to be equivalent to the existence of an equivalent martingale measure.

Let $(\Omega, \mathcal{F})$ be a measurable space and let P and Q be probability measures on $(\Omega, \mathcal{F})$. We say that Q is *absolutely continuous* with respect to P , written $Q \ll P$, if

$$P(A) = 0 \Rightarrow Q(A) = 0 \quad \text{for all } A \in \mathcal{F}.$$

Two probability measures P and Q are called *equivalent*, written $P \sim Q$, if they are mutually absolutely continuous, i.e.

$$P \ll Q \quad \text{and} \quad Q \ll P.$$

Equivalently,

$$P(A) = 0 \iff Q(A) = 0 \quad \text{for all } A \in \mathcal{F}.$$

A probability measure Q on Ω is a martingale measure if S is a Q -martingale in the filtration $(\mathcal{F}_t)$. Two measures P and Q on the same measurable space are said to be equivalent if each one is absolutely continuous with respect to the other $P \sim Q$.

To make the dual structure transparent, we first consider a one-period finite-state model. Let

$$\Omega = \{\omega_1, \dots, \omega_K\},$$

and let $\Delta S(\omega_u) = S_1(\omega_u) - S_0(\omega_u)$ denote the price increment in state $\omega_u \in \Omega$. For a contingent claim $X : \Omega \rightarrow \mathbb{R}$, the superhedging price is defined by

$$\pi_s(X) = \inf \left\{ x \in \mathbb{R} : \exists H \in \mathcal{H} \text{ such that } x + H^\top \Delta S(\omega) \geq X(\omega) \quad \forall \omega \in \Omega \right\}.$$

Since Ω is finite, this is a linear program:

$$\min_{x, H}$$

subject to

$$x + H^\top \Delta S(\omega_u) \geq X(\omega_u), \quad u = 1, \dots, K.$$

Define the decision vector

$$z = \begin{pmatrix} x \\ H \end{pmatrix} \in \mathbb{R}^{d+1}, \quad c = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad b = \begin{pmatrix} X(\omega_1) \\ \vdots \\ X(\omega_K) \end{pmatrix},$$

and let

$$A = \begin{pmatrix} 1 & \Delta S(\omega_1)^\top \\ \vdots & \vdots \\ 1 & \Delta S(\omega_K)^\top \end{pmatrix}.$$

Then the primal problem can be written as

$$\min_{z \in \mathbb{R}^{d+1}} c^\top z \quad \text{subject to} \quad Az \geq b.$$

Its dual linear program is

$$\max_{y \in \mathbb{R}^K} b^\top y$$

subject to

$$\begin{aligned} A^\top y &= c, \\ y &\geq 0. \end{aligned}$$

Writing $y = (y_1, \dots, y_K)^\top$, the dual feasibility conditions become

$$\sum_{u=1}^K y_u = 1, \quad \sum_{u=1}^K y_u \Delta S(\omega_u) = 0, \quad y_u \geq 0 \text{ for all } u.$$

Hence y is a probability vector on Ω . Defining

$$Q(\omega_u) = y_u, \quad u = 1, \dots, K,$$

we may identify the dual variables with a probability measure Q on Ω . The second dual constraint is then exactly

$$\mathbb{E}_Q[\Delta S] = 0,$$

or equivalently,

$$\mathbb{E}_Q[S_1] = S_0,$$

which means that S is a martingale under Q . Therefore, the dual problem can be written as

$$\sup_{Q \in \mathcal{M}} \mathbb{E}_Q[X],$$

where

$$\mathcal{M} = \{Q : \mathbb{E}_Q[\Delta S] = 0\}$$

is the set of one-period martingale measures. Under the no-arbitrage condition, the superhedging problem is feasible and bounded from below. Therefore, by strong duality for linear programs, the optimal values of the primal and dual problems coincide, and the superhedging price admits

the dual representation

$$\pi_s(X) = \sup_{Q \in \mathcal{M}} \mathbb{E}_Q[X].$$

The optimal Q is called an equivalent martingale measure, and its relationship to P is given by the Radon-Nikodym theorem

$$Q(A) = \int_A Z dP \quad \forall A \in \mathcal{F},$$

where $Z : \Omega \rightarrow \mathbb{R}_+$ is the Radon-Nikodym derivative of Q with respect to P

$$Z = \frac{dQ}{dP}.$$

Formally, we define an equivalent martingale measure as follows.

Definition 2.6 (Equivalent martingale measure). *A probability measure Q is called an equivalent martingale measure (EMM) if*

- $Q \sim P$, and
- S is a Q -martingale

$$\mathbb{E}_Q[S_{t+1} | \mathcal{F}_t] = S_t.$$

In the multi-period setting, trading strategies are dynamic and depend on the available information at each time. As a result, the absence of arbitrage must be enforced not only at maturity but at every intermediate time step. The equivalent martingale measure then belongs to the set of martingale measures which are equivalent to P

$$\mathcal{M} = \{Q \sim P : \mathbb{E}_Q[S_{t+1} | \mathcal{F}_t] = S_t \text{ for all } t = 0, \dots, T-1\}$$

Under no-arbitrage, the superhedging price admits the dual representation

$$\pi_s(X) = \sup_{Q \in \mathcal{M}} \mathbb{E}_Q[X], \quad \text{for all } X \in L^\infty(P, \Omega).$$

Remark 1 (From finite-state LP to general duality). *The finite-state derivation above identifies the dual variables of the superhedging LP with probability vectors satisfying the martingale condition. This is not an artifact of the finite-state assumption: the same structure arises in the general case via convex duality. Specifically, the set of attainable claims*

$$\mathcal{C} = \{(H \cdot S)_T : H \in \mathcal{H}\}$$

is a convex cone in L^∞ , and the superhedging price can be written as

$$\pi(\xi) = \inf\{x \in \mathbb{R} : \xi - x \in \mathcal{C} - L_+^\infty\}.$$

By the Hahn–Banach separation theorem applied in the L^∞ – L^1 duality, continuous linear functionals supporting $\mathcal{C}$ correspond precisely to probability measures under which S is a martingale (Delbaen and Schachermayer, 1994). The detailed proof in the general case can be found in (Föllmer and Schied, 2004, Corollary 7.18).

This can be summed up in the following result due to El Karoui and Quenez (1995)

Theorem 2.1 (Superhedging duality). *Assume NA holds. Then for every bounded claim $\xi \in L^\infty$,*

$$\pi(\xi) = \sup_{Q \in \mathcal{M}} \mathbb{E}_Q[\xi].$$

This relationship between the no-arbitrage condition and the existence of an equivalent martingale measure is formalized by the First Fundamental Theorem of Asset Pricing

Theorem 2.2 (First Fundamental Theorem of Asset Pricing (Discrete Time)). *The following are equivalent:*

- (i) *NA holds;*
- (ii) $\exists Q \in \{Q \sim P : \mathbb{E}_Q[S_{t+1} | \mathcal{F}_t] = S_t \text{ for all } t = 0, \dots, T-1\}$

Definition 2.7 (Market completeness). *The market is called complete if for every contingent claim $X \in L^0(P, \Omega)$ there exists an admissible trading strategy H such that the terminal wealth satisfies*

$$(H \cdot S)_T = X \quad P\text{-a.s.}$$

In other words, every contingent claim is exactly replicable.

The second fundamental theorem shows that uniqueness of the equivalent martingale measure is equivalent to market completeness, in which case superhedging prices coincide with expectations under a single pricing measure; in contrast, in incomplete markets, prices are characterized by a supremum over a set of martingale measures.

Theorem 2.3 (Second Fundamental Theorem of Asset Pricing). *Under no-arbitrage, the following are equivalent:*

- (i) *The market is complete*
- (ii) $|\mathcal{M}| = 1$, *i.e., the equivalent martingale measure is unique*

In the presence of market completeness, i.e., when $|\mathcal{M}| = 1$, the superhedging duality reduces to a single expectation $\pi(\xi) = \mathbb{E}_Q[\xi]$, yielding a unique fair price that coincides with the cost of replication. In contrast, in an incomplete market, $\mathcal{M}$ contains infinitely many measures and the superhedging price $\pi(\xi) = \sup_{Q \in \mathcal{M}} \mathbb{E}_Q[\xi]$ is generally strictly greater than the lower hedging price $\inf_{Q \in \mathcal{M}} \mathbb{E}_Q[\xi]$. The interval $[\inf_Q \mathbb{E}_Q[\xi], \sup_Q \mathbb{E}_Q[\xi]]$ then characterizes the range of arbitrage-free prices, and the choice of a specific pricing measure within this interval reflects the agent's preferences or a calibration to market data rather than a no-arbitrage requirement alone.

3 Robust Asset Pricing

There is a fundamental difference between the concepts of risk and uncertainty in finance. Risk refers to a measurable uncertainty which one can assign probabilities to. Uncertainty on the other hand is unquantifiable, it is an uncertainty about the level of uncertainty. In the language of Bouchard and Nutz (2015), whose robust asset pricing framework we adopt in this thesis, risk is generally referred to as uncertainty which can be fully characterized by one dominating probability measure. Uncertainty, or Knightian uncertainty is instead characterized by a set of non-dominated probability measures $\mathcal{P}$.

Optimization under model uncertainty in economics is often studied in the context of utility maximization such as the multiple priors framework of Epstein and Schneider (2003), who consider a discrete-time utility maximization problem under ambiguity, where uncertainty is modeled by a set of probability measures $\mathcal{P}$ rather than a single reference measure. At each time t and history $\omega \in \Omega_t$, the agent faces a set of conditional probability measures $\mathcal{P}_t(\omega)$ governing the distribution of future states. Preferences are defined recursively via a worst-case expected utility criterion. A central structural assumption is *rectangularity*, which requires that the global set of priors is generated by independent selections of conditional measures at each time and state, i.e.

$$\mathcal{P} = \{P : P = P_0 \otimes P_1(\cdot|\omega_1) \otimes \cdots \otimes P_{T-1}(\cdot|\omega_{T-1}), \text{ with } P_t(\cdot|\omega) \in \mathcal{P}_t(\omega)\}.$$

However, this framework is based on a set of dominated probability measures, i.e. there exists a reference measure with respect to which all priors admit densities. A set of priors $\mathcal{P}$ is said to be dominated if there exists a probability measure P^* such that

$$P \ll P^* \quad \text{for all } P \in \mathcal{P}.$$

If no such dominating measure exists, $\mathcal{P}$ is called non-dominated. This enforces structural assumptions and excludes nonparametric ambiguity sets such as Wasserstein balls, where measures may have different supports and can be mutually singular without a common dominating measure. A commonly adopted method for worst-case decision making in continuous-time finance relies on Peng's sublinear expectations. Peng (2007) considers a framework where ambiguity is modeled via a non-dominated set of probability measures $\mathcal{P}$ on the canonical path space. Instead of working with a single linear expectation, he introduces a *sublinear expectation* defined by

$$\mathcal{E}[X] = \sup_{P \in \mathcal{P}} \mathbb{E}_P[X],$$

which captures worst-case evaluation over all plausible models. The canonical process $(B_t)_{t \geq 0}$, called *G-Brownian motion*, generalizes classical Brownian motion by allowing its quadratic variation to vary within a given set. In particular, uncertainty is specified through a bounded, convex set of volatility matrices $\Sigma \subset \mathbb{R}^{d \times d}$, and the set of priors $\mathcal{P}$ consists of all probability measures under which

$$dB_t = \sigma_t dW_t, \quad \sigma_t \in \Sigma,$$

for some progressively measurable process $(\sigma_t)_{t \geq 0}$. The resulting expectation $\mathcal{E}$ is time-consistent and admits a dynamic representation via conditional sublinear expectations

$$\mathcal{E}_t[X] = \sup_{P \in \mathcal{P}_t(\omega)} \mathbb{E}_P[X | \mathcal{F}_t],$$

where $\mathcal{P}_t(\omega)$ denotes the set of conditional laws consistent with the history up to time t .

In this setting, uncertainty is specified locally in time through the choice of the volatility process at each time step, and the resulting set of measures is stable under concatenation of conditional laws. In this sense, the ambiguity set satisfies a rectangular-type structure, which ensures time consistency and allows for a well-defined nonlinear expectation. In contrast, this thesis relaxes that assumption in order to obtain less conservative superhedging prices by considering non-rectangular ambiguity sets considered, which are defined globally on the path space and do not admit such a decomposition. Furthermore, while Peng's sublinear expectation model can be calibrated using empirical data (for instance defining the diffusion set as an interval around data-derived implied volatility), it imposes structural assumptions about the way that the price

of the underlying behaves and does not offer the flexibility of nonparametric ambiguity sets over the full path law.

An alternative approach to modeling uncertainty avoids specifying probability measures altogether and instead encodes beliefs directly at the level of price paths. In the model-free literature, it has been observed by Vovk (2012) that requiring superhedging to hold for all possible paths leads to overly conservative prices leaving arbitrage opportunities in the case of vanilla European and American calls and puts. To address this issue, several works propose restricting the set of admissible trajectories to a prediction set $\Theta \subset \Omega$, representing paths that are deemed plausible by the agent. In the superhedging framework of Bartl et al. (2019), the superhedging condition is imposed pathwise on Θ , leading to the pricing problem

$$\pi(\xi) = \inf \{x \in \mathbb{R} : \exists H \in \mathcal{H} \text{ such that } x + (H \cdot S)_T(\omega) \geq \xi(\omega) \ \forall \omega \in \Theta\}.$$

Under suitable conditions, this admits a dual representation

$$\pi(\xi) = \sup_{Q \in \mathcal{M}(\Theta)} \mathbb{E}_Q[\xi],$$

where $\mathcal{M}(\Theta)$ is the set of martingale measures supported on Θ . This approach reduces the conservativeness of model-free superhedging by incorporating structural beliefs on price paths. However, model-free approaches such as the one of Bartl et al. (2019) avoid specifying an ambiguity set altogether and instead encode beliefs via a prediction set of admissible price paths. While this yields a pricing-hedging duality over martingale measures supported on the prediction set, it does not provide a way to relate the uncertainty to empirical data and fails to provide a data-driven decision making mechanism.

We adopt the framework of Bouchard and Nutz (2015) as the foundation of our analysis because it provides a general discrete-time model of asset pricing under non-dominated uncertainty. In contrast to dominated models, this framework allows for ambiguity sets consisting of mutually singular probability measures, which is essential for accommodating distributional uncertainty defined via statistical distances such as Wasserstein balls. Moreover, unlike parametric approaches based on sublinear expectations, it does not impose a specific structural form on the dynamics and does not enforce independence of timesteps (rectangularity). This flexibility is central to our contribution: it enables us to consider ambiguity sets that are inherently non-rectangular and to quantify the resulting gap in superhedging prices relative to their rectangular counterparts.

In classical asset pricing, a single probability measure P is fixed on $(\Omega, \mathcal{F})$, assigning probabilities to all price paths, and the superhedging property is required to hold P -almost surely. In contrast, the robust asset pricing framework of Bouchard and Nutz (2015) considers a model-independent approach in which uncertainty is described by a family $\mathcal{P}$ of probability measures. We work in a discrete-time, finite-horizon setting where the market is governed by a non-dominated set $\mathcal{P}$, i.e. there does not exist a reference probability measure $\bar{P}$ such that $P \ll \bar{P}$ for all $P \in \mathcal{P}$. As a consequence, all equalities and inequalities in this section are to be understood as holding $\mathcal{P}$ -quasi-surely, that is, they hold P -almost surely for all $P \in \mathcal{P}$. Extensions of this framework to continuous time exist, but are outside the scope of this thesis (Biagini et al., 2017).

3.1 Market model under model uncertainty

We consider a discrete-time financial market with d stocks and e options over a finite horizon T with discounted asset price process $S = (S_t)_{t=0}^T$. Uncertainty is modeled by a nonempty family of probability measures $\mathcal{P} \subseteq \mathcal{B}(\Omega)$, which is not necessarily dominated by a single reference measure. We call this set of multiple priors the ambiguity set. A property is said to hold $\mathcal{P}$ -quasi-surely

(q.s.) if it holds P -almost surely for every $P \in \mathcal{P}$.

$$A \text{ holds } \mathcal{P}\text{-q.s.} \iff P(A) = 1 \text{ for all } P \in \mathcal{P}.$$

This notion replaces the classical almost-sure statements in the absence of a dominating reference measure: when $\mathcal{P} = \{P\}$, it reduces to P -a.s. A set $A \subseteq \Omega$ is called $\mathcal{P}$ -polar if

$$P(A) = 0 \quad \text{for all } P \in \mathcal{P}.$$

Equivalently, a property holds $\mathcal{P}$ -quasi-surely if it fails only on a $\mathcal{P}$ -polar set. Two families of probability measures $\mathcal{P}$ and $\mathcal{Q}$ are said to have the same polar sets if

$$P(A) = 0 \text{ for all } P \in \mathcal{P} \iff Q(A) = 0 \text{ for all } Q \in \mathcal{Q},$$

for every measurable set $A \subseteq \Omega$. Equivalently, a property holds $\mathcal{P}$ -quasi-surely if and only if it holds $\mathcal{Q}$ -quasi-surely.

The ambiguity set $\mathcal{P}$ satisfies robust no-arbitrage, denoted $NA(\mathcal{P})$, if no self-financing trading strategy yields a nonnegative terminal payoff that is strictly positive on a set of positive probability under at least one $P \in \mathcal{P}$. Let $\mathcal{H}$ be the set of all predictable $\mathbb{R}^d$ -valued processes, representing trading strategies. Given a strategy $H \in \mathcal{H}$, the corresponding wealth process (starting from initial capital x) is given by the discrete-time integral

$$x + H \cdot S = x + \sum_{u=1}^T H_u \Delta S_u,$$

where $\Delta S_u = S_u - S_{u-1}$ is the price change, and $H_u \Delta S_u$ represents the inner product of the trading strategy H_u and the price increment ΔS_u over the d -dimensional price vector.

In incomplete markets under model uncertainty, dynamic trading in the underlying assets alone is insufficient to hedge a claim at a competitive price. Following Bouchard and Nutz (2015), we therefore allow for *semistatic* hedging strategies: a dynamic position $H \in \mathcal{H}$ in the stocks combined with a static position $h \in \mathbb{R}^e$ in a finite collection of options $g = (g_1, \dots, g_e)$ available for trading at time $t = 0$. Each option is normalized to have zero initial price, $g_i(0) = 0$, so that the portfolio value of these options, for a vector $h \in \mathbb{R}^e$, is

$$h \cdot g = \sum_{i=1}^e h_i \cdot g_i.$$

A pair $(H, h) \in \mathcal{H} \times \mathbb{R}^e$ is called a semistatic hedging strategy, where H represents dynamic stock trading and h represents the static option portfolio. As in any mathematical finance model, a no-arbitrage condition is necessary to make the model viable.

Definition 3.1 (Robust No-Arbitrage). *The condition Robust No-Arbitrage $NA(\mathcal{P})$ holds if for all $(H, h) \in \mathcal{H} \times \mathbb{R}^e$:*

$$H \cdot S_T + h \cdot g \geq 0 \quad \mathcal{P}\text{-q.s.} \implies H \cdot S_T + h \cdot g = 0 \quad \mathcal{P}\text{-q.s.}$$

When $\mathcal{P}$ is a singleton this reduces to the classical No-Arbitrage condition.

As in the classical case, an equivalence is established between no-arbitrage and the existence of martingale measures Q which are absolutely continuous with respect to the measures in the ambiguity set $\mathcal{P}$, i.e $Q(A) = 0$ for any $\mathcal{P}$ -polar set A . This relationship is denoted as $Q \ll \mathcal{P}$.

However, Bouchard and Nutz (2015) use a stronger notion of absolute continuity in their model:

$$Q \lll \mathcal{P} \quad \text{if there exists } P \in \mathcal{P} \text{ such that } Q \ll P,$$

i.e., there exists at least one measure P in the ambiguity set such that Q is absolutely continuous with respect to P . They then define the set $\mathcal{Q}$ which they refer to as the family of martingale measures

$$\mathcal{Q} = \left\{ Q \lll \mathcal{P} : Q \text{ is a martingale measure and } \mathbb{E}_Q[g^i] = 0 \text{ for } i = 1, \dots, e \right\}.$$

3.2 Robust superhedging: the robust primal problem

As in the classical setting of Section 2.2, the central pricing question is the minimal initial capital required to hedge a contingent claim. The key difference under model uncertainty is that the superhedging inequality must now hold $\mathcal{P}$ -quasi-surely, that is, simultaneously under every plausible model $P \in \mathcal{P}$ rather than under a single reference measure.

Theorem 3.1 (Robust Superhedging Theorem). *Let robust no-arbitrage hold, and let $\xi : \Omega \rightarrow \mathbb{R}$ be upper semianalytic. Then the minimal superhedging price:*

$$\pi(\xi) = \inf\{x \in \mathbb{R} : \exists(H, h) \in \mathcal{H} \times \mathbb{R}^e \text{ such that } x + H \cdot S_T + h \cdot g \geq \xi \quad \mathcal{P} - q.s.\},$$

satisfies

$$\pi(\xi) = \sup_{Q \in \mathcal{Q}} E_Q[\xi] \in (-\infty, \infty],$$

and there exist semistatic strategies $(H, h) \in \mathcal{H} \times \mathbb{R}^e$ such that $\pi(\xi) + H \cdot S_T + h \cdot g \geq \xi \quad \mathcal{P} - q.s.$

Note that the supremum over $\mathcal{Q}$ is not obtained in general: a maximum might not exist (uniqueness of equivalent martingale measure and thus market completeness is not guaranteed in this model).

When $\mathcal{P} = \{P\}$ is a singleton and no options are traded ($e = 0$), this reduces to the classical superhedging price. The quasi-sure inequality enforces the hedge against *every* model in the ambiguity set: this is the source of robustness, and also the source of conservativeness when $\mathcal{P}$ is enlarged by imposing rectangularity.

3.3 The robust dual problem

The dual representation of the robust primal problem expresses the robust superhedging price as a supremum over a set $\mathcal{Q}$ of martingale measures. In the classical setting, the dual domain consists of measures equivalent to the reference measure P . Without a single reference measure, this notion of equivalence must be reformulated, and the appropriate replacement is the relation $\lll$ defined by Bouchard and Nutz (2015) as

$$Q \lll \mathcal{P} \quad \text{if } \exists P \in \mathcal{P} \text{ such that } Q \ll P.$$

This means a pricing measure is admissible if it is absolutely continuous with respect to *at least one* plausible model in $\mathcal{P}$, not necessarily all of them. Using this definition, the set of martingale measures consistent with $\mathcal{P}$ is

$$\mathcal{Q} = \{Q \lll \mathcal{P} : S \text{ is a } Q\text{-martingale and } \mathbb{E}_Q[g_i] = 0 \text{ for } i = 1, \dots, e\}.$$

The condition $\mathbb{E}_Q[g_i] = 0$ ensures consistency with the zero initial price of the traded options.

As in the classical setting, Bouchard and Nutz (2015) have shown that the absence of arbitrage implies the existence of an equivalent martingale measure, though the definition of the dual domain changes slightly in the robust setting. In this case the First Fundamental Theorem of Asset Pricing states that $NA(\mathcal{P})$ is equivalent to $\mathcal{Q}$ being sufficiently rich, in the sense that any $\mathcal{Q}$ -polar set is $\mathcal{P}$ -polar (this is the notion of equivalence between $\mathcal{Q}$ and $\mathcal{P}$ in this setting.) The robust analog to the First Fundamental Theorem of Asset Pricing due to Bouchard and Nutz (2015) is as follows:

Theorem 3.2 (Robust First Fundamental Theorem of Asset Pricing). *The following are equivalent*

- (i) $NA(\mathcal{P})$ holds;
- (ii) for all $P \in \mathcal{P}$, there exists $Q \in \mathcal{Q}$ such that $P \ll Q$;
- (iii) $\mathcal{P}$ and $\mathcal{Q}$ have the same polar sets.

Condition (iii) implies that $\mathcal{Q}$ is nonempty under $NA(\mathcal{P})$. Indeed, if $\mathcal{Q}$ were empty, every measurable set would be $\mathcal{Q}$ -polar (the empty supremum), hence $\mathcal{P}$ -polar by (iii), contradicting the fact that measures in $\mathcal{P}$ have total mass one. The dual problem of the robust superhedging problem is therefore well-posed.

Theorem 3.3 (Robust superhedging duality (Bouchard and Nutz, 2015, Theorem 4.9)). *Under the no-arbitrage and measurability assumptions of Bouchard and Nutz (2015), the robust superhedging price of a contingent claim ξ is given by*

$$\pi_{\mathcal{P}}(\xi) = \inf \{x \in \mathbb{R} : \exists (H, h) \in \mathcal{H} \times \mathbb{R}^e \text{ such that } x + (H \cdot S)_T + h \cdot g \geq \xi \quad \mathcal{P}\text{-q.s.}\},$$

If ξ is upper semianalytic and satisfies the required integrability conditions, then

$$\pi_{\mathcal{P}}(\xi) = \sup_{Q \in \mathcal{Q}} \mathbb{E}_Q[\xi].$$

Thus the robust superhedging price admits a dual characterization as the worst-case expectation over martingale measures consistent with the ambiguity set $\mathcal{P}$ and the static calibration constraints.

3.4 Dynamic programming under model uncertainty

Following Bouchard and Nutz (2015), for each $t \in \{0, \dots, T-1\}$ and each history $\omega^t \in \Omega_t$, the conditional ambiguity set at (t, ω^t) is defined as

$$\mathcal{P}_t(\omega^t) = \{P(S_{t+1} \in \cdot \mid S_0 = s_0, \dots, S_t = s_t) : P \in \mathcal{P}\} \subseteq \mathfrak{P}(\mathbb{R}^d).$$

That is, $\mathcal{P}_t(\omega^t)$ collects all one-step conditional laws of S_{t+1} given the observed history ω^t that are attainable under some $P \in \mathcal{P}$. We assume throughout that $\omega^t \mapsto \mathcal{P}_t(\omega^t)$ is universally measurable, in the sense of (Bouchard and Nutz, 2015, Assumption 2.3), so that selection theorems apply. Since Ω is Polish, every $P \in \mathcal{P}$ admits a disintegration along the filtration:

$$P = P_0 \otimes P_1(\cdot \mid \omega^1) \otimes \dots \otimes P_{T-1}(\cdot \mid \omega^{T-1}),$$

where each kernel

$$P_t(\cdot \mid \omega^t) \in \mathcal{P}_t(\omega^t)$$

specifies the conditional law of S_{t+1} given the history up to time t . The global measure P is uniquely determined by its initial law and the family of conditional kernels $(P_t)_{t=0}^{T-1}$. Conversely,

the ambiguity set $\mathcal{P}$ is recovered globally from the conditional family $(\mathcal{P}_t(\omega^t))_{t, \omega^t}$ together with the consistency requirement that the kernels jointly arise from a single path measure in $\mathcal{P}$. We also introduce the corresponding conditional sets of martingale measures. For each $t \in \{0, \dots, T-1\}$ and each history $\omega^t \in \Omega_t$, define

$$\mathcal{Q}_t(\omega^t) = \left\{ Q \in \mathfrak{P}(\mathbb{R}^d) : Q \ll P \text{ for some } P \in \mathcal{P}_t(\omega^t), \quad \mathbb{E}_Q[S_{t+1} - s_t] = 0 \right\}.$$

This is the set of one-step martingale measures consistent with the conditional ambiguity set $\mathcal{P}_t(\omega^t)$. Under robust no-arbitrage $NA(\mathcal{P})$, Bouchard and Nutz (2015, Theorem 4.5) establish that $\mathcal{Q}_t(\omega^t) \neq \emptyset$ for $\mathcal{P}$ -quasi every history ω^t .

The classical superhedging problem under a single measure P can be solved by backward induction, since the conditional distributions of the price process depend only on the current state. Under non-dominated uncertainty, this structure no longer exists: the conditional ambiguity sets $\mathcal{P}_t(\omega)$ may depend on the entire past history $\omega = (\omega_1, \dots, \omega_t)$, and admissible conditional distributions cannot in general be selected independently across time and histories. The following dynamic programming principle, due to Bouchard and Nutz (2015), shows that a recursion is nevertheless available, but it operates over conditional ambiguity sets that inherit the path-dependence of $\mathcal{P}$.

Proposition 1 (Dynamic Programming Principle (Bouchard and Nutz, 2015, Section 4.3)). *Assume $NA(\mathcal{P})$ holds and let $\xi : \Omega \rightarrow \mathbb{R}$ be an upper semianalytic function bounded from above. For each $t = 0, \dots, T-1$ and $\omega \in \Omega_t$, let $\mathcal{Q}_t(\omega)$ denote the set of one-step martingale measures consistent with the conditional ambiguity set $\mathcal{P}_t(\omega)$.*

Define recursively the value functions $(V_t)_{t=0}^T$ by

$$V_T = \xi,$$

and for $t = T-1, \dots, 0$,

$$V_t(\omega) = \sup_{Q \in \mathcal{Q}_t(\omega)} \mathbb{E}_Q[V_{t+1}(\omega, \cdot)]$$

Then:

1. V_t is upper semianalytic for all t ,
2. For each t , there exists a universally measurable function $H_t : \Omega_t \rightarrow \mathbb{R}^d$ such that

$$V_t(\omega) + H_t(\omega) \cdot \Delta S_{t+1}(\omega, \cdot) \geq V_{t+1}(\omega, \cdot) \quad \mathcal{P}_t(\omega)\text{-}q.s.$$

3. The strategy $H = (H_t)_{t=0}^{T-1}$ satisfies

$$V_0 + (H \cdot S)_T \geq \xi \quad \mathcal{P}\text{-}q.s.$$

4. $V_0 = \pi(\xi)$

Proof. The proof follows the arguments of (Bouchard and Nutz, 2015, Section 4.3), which we detail for completeness.

For any upper semianalytic function f , define

$$\mathcal{E}_t(f)(\omega) = \sup_{Q \in \mathcal{Q}_t(\omega)} \mathbb{E}_Q[f(\omega, \cdot)] \tag{1}$$

By (Bouchard and Nutz, 2015, Lemma 4.10), for each t there exists a universally measurable

function $y_t : \Omega_t \rightarrow \mathbb{R}^d$ such that

$$\mathcal{E}_t(f)(\omega) + y_t(\omega) \cdot \Delta S_{t+1}(\omega, \cdot) \geq f(\omega, \cdot) \quad \mathcal{P}_t(\omega)\text{-q.s.}$$

for all $\omega \in \Omega_t$.

Define recursively

$$V_T = \xi, \quad V_t = \mathcal{E}_t(V_{t+1}), \quad t = T-1, \dots, 0.$$

Applying (1) with $f = V_{t+1}$ yields, for each t ,

$$V_t(\omega) + y_t(\omega) \cdot \Delta S_{t+1}(\omega, \cdot) \geq V_{t+1}(\omega, \cdot) \quad \mathcal{P}_t(\omega)\text{-q.s.} \quad (2)$$

Fix $P \in \mathcal{P}$. By the structure of the model (see (Bouchard and Nutz, 2015, Section 1.2)), P admits a decomposition

$$P = P_0 \otimes P_1(\cdot|\omega_1) \otimes \dots \otimes P_{T-1}(\cdot|\omega_{T-1}),$$

where each kernel $P_t(\cdot|\omega) \in \mathcal{P}_t(\omega)$.

For a fixed t , the inequality (2) holds $P_t(\omega)$ -almost surely for each ω . Since y_t and V_t are universally measurable by (Bouchard and Nutz, 2015, Lemma 4.10), the set

$$\{V_t + y_t \cdot \Delta S_{t+1} < V_{t+1}\}$$

is jointly measurable in (ω, ω') , so by Fubini's theorem applied to the product measure P , it follows that

$$V_t + y_t \cdot \Delta S_{t+1} \geq V_{t+1} \quad P\text{-a.s.}$$

Since $P \in \mathcal{P}$ was arbitrary, we conclude this holds for all $P \in \mathcal{P}$ and the property holds $\mathcal{P}$ -quasi surely

$$V_t + y_t \cdot \Delta S_{t+1} \geq V_{t+1} \quad \mathcal{P}\text{-q.s.} \quad (3)$$

We now show through backward induction that for all $t = 0, \dots, T$,

$$V_t + \sum_{u=t}^{T-1} y_u \cdot \Delta S_{u+1} \geq V_T \quad \mathcal{P}\text{-q.s.} \quad (4)$$

Base case: For $t = T-1$, this follows directly from (3)

$$V_{T-1} + y_{T-1} \cdot \Delta S_T \geq V_T.$$

Inductive step: Assume (4) holds for $t+1$. Then, using (3),

$$V_t + y_t \cdot \Delta S_{t+1} \geq V_{t+1}.$$

Combining with the induction hypothesis yields

$$V_t + y_t \cdot \Delta S_{t+1} + \sum_{u=t+1}^{T-1} y_u \cdot \Delta S_{u+1} \geq V_T,$$

which proves (4) at time t .

Define the predictable strategy $H \in \mathcal{H}$ by

$$H_{t+1} = y_t, \quad t = 0, \dots, T-1,$$

so that

$$(H \cdot S)_T = \sum_{t=0}^T H_t \cdot \Delta S_t.$$

Taking $t = 0$ in (4) and recalling that $V_T = \xi$, we obtain

$$V_0 + (H \cdot S)_T \geq \xi \quad \mathcal{P}\text{-q.s.}$$

Finally, by (Bouchard and Nutz, 2015, Lemma 4.13 and Theorem 4.9),

$$V_0 = \sup_{Q \in \mathcal{Q}} \mathbb{E}_Q[\xi] = \pi(\xi),$$

which completes the proof. $\square$

In the dominated case, the recursion above reduces to the classical dynamic programming principle under a fixed probability measure. In the non-dominated setting, however, the conditional ambiguity sets $\mathcal{P}_t(\omega)$ depend on the entire history and cannot, in general, be decomposed into independent components.

This dependence reflects the non-rectangular structure of the ambiguity set and prevents a simplification of the recursion into a Bellman equation involving a pointwise supremum over local transition laws. In particular, the recursion above is path-dependent and involves quasi-sure inequalities, in contrast to classical Bellman equations. This distinction is at the heart of the computational challenges associated with non-rectangular ambiguity sets.

The framework described so far serves as the basis for the following extensions. However, it is abstract and does not specify how the ambiguity set $\mathcal{P}$ is constructed. In the following section, we introduce data-driven ambiguity sets based on statistical distances, and study how their structural properties, in particular rectangularity or the lack thereof, affect superhedging prices.

3.5 Standing assumptions

Throughout the remainder of this thesis, we work in discrete time on a finite horizon $T \in \mathbb{N}$. The uncertainty is modeled on the canonical path space $\Omega = (\mathbb{R}^d)^{T+1}$ equipped with its Borel σ -algebra $\mathcal{F} = \mathcal{B}(\Omega)$. An element

$$\omega = (s_0, \dots, s_T) \in \Omega$$

represents a realization of the price trajectory of the d -dimensional asset process $S = (S_t)_{t=0}^T$, where the canonical coordinate process is defined by $S_t(\omega) = s_t$. For each $t = 0, \dots, T$, we define the natural filtration

$$\mathcal{F}_t = \sigma(S_0, \dots, S_t),$$

and denote by $\Omega_t = (\mathbb{R}^d)^{t+1}$ the corresponding history space up to time t . Since Ω is a Polish space, all history spaces Ω_t are Polish as well.

Trading strategies are assumed to be predictable processes

$$H = (H_t)_{t=0}^{T-1},$$

where each

$$H_t : \Omega_t \rightarrow \mathbb{R}^d$$

is $\mathcal{F}_t$ -measurable. The associated gains-from-trading process is

$$(H \cdot S)_T = \sum_{t=0}^{T-1} H_t \cdot (S_{t+1} - S_t).$$

All statements involving quasi-sure inequalities are understood in the sense that the property holds P -almost surely for every $P \in \mathcal{P}$.

3.6 Ambiguity sets

An ambiguity set $\mathcal{P} \subseteq \mathcal{P}(\Omega)$ is a family of probability distributions on a common measurable space, which represents the set of probability distributions we consider in our robust optimization problem, analogous to the set of multiple priors $\mathcal{P}$ of Bouchard and Nutz (2015). In particular, an ambiguity set is a nonempty subset $\mathcal{P} \subseteq \mathcal{P}(\Omega)$.

Each $P \in \mathcal{P}$ specifies a joint law of the entire price trajectory $(S_0, \dots, S_T)$. In particular, for any measurable set $A \subseteq \Omega$, the quantity $P(A)$ represents the probability of observing a trajectory in A .

By disintegration, each $P \in \mathcal{P}$ induces a family of one-step conditional distributions

$$P(S_{t+1} \in \cdot \mid S_0, \dots, S_t), \quad t = 0, \dots, T-1,$$

which describe the evolution of the process given its past history. However, these conditional distributions are not independent objects: they must be jointly consistent with a single underlying path measure P .

In particular, the ambiguity set $\mathcal{P}$ is specified at the level of full path laws rather than collections of one-step transition kernels. As a consequence, admissible conditional distributions at different times cannot in general be selected independently, since they must arise from the same global probability measure. This global consistency constraint is the source of non-rectangularity in the ambiguity sets considered in this thesis.

In data-driven settings, the ambiguity set is constructed around an empirical reference measure. Let $(\omega^{(i)})_{i=1}^n \subset \Omega$ be n observed sample paths and define the empirical measure

$$\hat{P}_n = \frac{1}{n} \sum_{i=1}^n \delta_{\omega^{(i)}}.$$

Ambiguity sets are then defined as neighborhoods of $\hat{P}_n$ with respect to a suitable statistical distance. The sample paths $\omega^{(i)}$ underlying $\hat{P}_n$ can arise from several sources. The most common, and the one we adopt in the numerical experiments, is Monte Carlo simulation under a calibrated model: a parametric stochastic process is fitted to historical return data, and n independent trajectories of horizon T are sampled from the calibrated distribution. Alternatives include bootstrap resampling of historical returns and direct partitioning of a long historical series into consecutive paths, although the latter only yields approximately exchangeable rather than independent samples due to regime persistence and autocorrelation in financial time series.

Let $\mathcal{F}$ be a class of measurable functions $f : \Omega \rightarrow \mathbb{R}$ such that all expectations below are

well-defined. The *integral probability metric* (IPM) associated with $\mathcal{F}$ is defined by

$$d_{\mathcal{F}}(Q, \hat{P}_n) = \sup_{f \in \mathcal{F}} \left| \mathbb{E}_Q[f] - \mathbb{E}_{\hat{P}_n}[f] \right|, \quad Q \in \mathcal{P}(\Omega).$$

Assuming that $\mathcal{F}$ is symmetric and uniformly bounded, $d_{\mathcal{F}}$ defines a pseudometric on $\mathcal{P}(\Omega)$ (Müller, 1997; Sriperumbudur et al., 2012). The corresponding ambiguity set is given by

$$\mathcal{P}(\rho) = \left\{ Q \in \mathcal{P}(\Omega) : d_{\mathcal{F}}(Q, \hat{P}_n) \leq \rho \right\}.$$

This formulation admits a statistical interpretation: $\mathcal{P}$ contains all distributions that cannot be distinguished from $\hat{P}_n$ based on expectations over the function class $\mathcal{F}$.

Definition 3.2 (Wasserstein Distance). *Let (Ω, d_{Ω}) be a Polish space. For $p \in [1, +\infty)$, the Wasserstein p -distance between two probability measures P and Q on Ω with finite p -moments is*

$$W_p(P, Q) = \left(\inf_{\Gamma \in \Pi(P, Q)} \int_{\Omega \times \Omega} d_{\Omega}(\omega, \tilde{\omega})^p \Gamma(d\omega, d\tilde{\omega}) \right)^{1/p}$$

where $\Pi(P, Q)$ denotes the set of joint probability measures on $\Omega \times \Omega$ with marginals P and Q .

In particular, we work with the Wasserstein-1 distance in this thesis. This choice is canonical in distributionally robust optimization (Esfahani and Kuhn, 2018; Kuhn et al., 2025), and particularly natural in the context of contingent claim pricing. Vanilla European options (calls, puts, and their linear combinations) are Lipschitz functions of the terminal price, and many path-dependent claims of practical interest are Lipschitz in the path under suitable choices of d_{Ω} .

The corresponding Wasserstein-1 ambiguity set is

$$\mathcal{P}^{W_1}(\rho) = \left\{ Q \in \mathcal{P}(\Omega) : W_1(Q, \hat{P}_n) \leq \rho \right\}.$$

If d_{Ω} has finite first moment under $\hat{P}_n$, then $\mathcal{P}^{W_1}$ is nonempty, convex, and weakly compact. Moreover, the Kantorovich–Rubinstein duality which holds only for $p = 1$, which yields the representation

$$W_1(Q, \hat{P}_n) = \sup_{\|f\|_{\text{Lip}} \leq 1} \left(\mathbb{E}_Q[f] - \mathbb{E}_{\hat{P}_n}[f] \right),$$

so it becomes apparent that the Wasserstein-1 distance is an integral probability metric corresponding to the class of 1-Lipschitz functions. The Kantorovich–Rubinstein duality further yields the bound

$$\sup_{Q \in \mathcal{P}^{W_1}(\rho)} \mathbb{E}_Q[\xi] - \mathbb{E}_{\hat{P}_n}[\xi] \leq L_{\xi} \cdot \rho,$$

for any payoff ξ with Lipschitz constant L_{ξ} . This bound relates the conservativeness of robust pricing directly to the regularity of the payoff and the Wasserstein radius.

If d_{Ω} has finite first moment under $\hat{P}_n$, then $\mathcal{P}^W(\rho)$ is nonempty, convex, and weakly compact. Moreover, established concentration results (Esfahani and Kuhn, 2018) guarantee that the unknown data-generating distribution P^* lies in $\mathcal{P}^W(\rho_n(\alpha))$ with probability at least $1 - \alpha$, for an appropriate choice of radius $\rho_n(\alpha)$ that vanishes as $n \rightarrow \infty$. This finite-sample guarantee is one of the primary reasons for the prevalence of Wasserstein ambiguity sets in data-driven decision-making.

In the remainder of the thesis we work with two complementary classes of ambiguity sets, which together illustrate the rectangularity gap from different angles.

The first is the Wasserstein-1 ball $\mathcal{P}^{W_1}(\rho)$ defined above. This is a nonparametric ambiguity model that captures distributional uncertainty over the full path law, makes no structural assumption about the dynamics, and is the natural setting for data-driven robust pricing.

The second is a parametric, moment-based ambiguity model in which the price process is a random walk with drift μ and unknown diffusion $\sigma \in [\underline{\sigma}, \bar{\sigma}]$. This model admits closed-form computations and serves as a tractable illustrative example in which both rectangular and non-rectangular versions of the ambiguity set can be compared more easily.

3.7 Rectangular vs non-rectangular ambiguity

We now formalize the distinction between rectangular and non-rectangular ambiguity sets and formally define the rectangularity gap. The distinction is fundamental for the robust superhedging problem since whether the ambiguity set is rectangular or not determines whether the worst-case pricing problem admits a tractable dynamic programming formulation, and whether the set of adverse scenarios coincides with what one would consider economically plausible.

Recall that each measure $P \in \mathcal{P}$ in the ambiguity set specifies a joint law of the entire price trajectory $(S_0, \dots, S_T)$ on the canonical path space $\Omega = (\mathbb{R}^d)^{T+1}$ equipped with the Borel σ -algebra $\mathcal{F} = \mathcal{B}(\Omega)$. Since Ω is Polish, each $P \in \mathcal{P}$ induces a family of one-step conditional distributions (transition kernels)

$$P(S_{t+1} \in \cdot \mid S_0 = s_0, \dots, S_t = s_t), \quad t = 0, \dots, T-1,$$

which denotes the conditional probabilities over the next price given the price history $\omega_t = (s_0, \dots, s_t)$ observed up until that point.

The structural question we now address is whether the conditional kernels at different times can be combined freely into a single global model, or whether they are linked by global consistency constraints imposed by the underlying path measure.

Definition 3.3 (Rectangular Ambiguity Set). *An ambiguity set $\mathcal{P} \subseteq \mathcal{P}(\Omega)$ is rectangular if admissible one-step conditional distributions can be chosen independently at each time step $t = 0, \dots, T-1$ and for all histories $\omega_t = S_0, \dots, S_t$. Let*

$$\mathcal{K}_t(\omega_t) = \{P(S_{t+1} \in \cdot \mid \omega_t) : P \in \mathcal{P}\}$$

denote the set of one-step conditional distributions attainable under $\mathcal{P}$. We say that $\mathcal{P}$ is rectangular if for every family of measurable selectors

$$K_t : \Omega_t \rightarrow \mathcal{P}(\mathbb{R}^d), \quad K_t(\omega_t) \in \mathcal{K}_t(\omega_t) \text{ for all } \omega_t \in \Omega_t, t = 0, \dots, T-1$$

the path measure generated recursively from the initial state S_0 together with the kernels $(K_t)_{t=0}^{T-1}$ belongs to $\mathcal{P}$.

In other words, rectangularity is the property that local feasibility implies global feasibility: if a one-step conditional distribution is admissible at one time and another one-step transition law is admissible at a different time, then these local choices may be combined freely into a single global model over different time steps with the choice of kernel remaining independent over each. This implies time consistency holds, i.e., a policy that is optimal from the perspective of time 0 remains optimal when re-evaluated at any later time t given the history $\omega_t = (s_0, \dots, s_t)$.

Definition 3.4 (Non-rectangular ambiguity set). *An ambiguity set $\mathcal{P} \subseteq \mathcal{P}(\Omega)$ is non-rectangular if it is not rectangular as defined by Definition 3.3*

The restriction to rectangular ambiguity sets to facilitate computational tractability was introduced by Iyengar (2005) and Nilim and El Ghaoui (2005). When the ambiguity set is rectangular, the problem admits a Bellman recursion as in Proposition 1. Note that in our setting, rectangularity does not by itself imply a Markov structure; in general, the admissible kernels may still depend on the full history $\omega_t = (s_0, \dots, s_t)$. When the ambiguity set is rectangular and suitable measurability assumptions hold, the dual value process admits the recursive representation

$$V_T(\omega_T) = \xi(\omega_T),$$

and for $t = T - 1, \dots, 0$,

$$V_t(\omega_t) = \sup_{Q \in \mathcal{Q}_t^{\text{rec}}(\omega_t)} \mathbb{E}_Q[V_{t+1}(\omega_t, S_{t+1})].$$

Equivalently, for the primal problem

$$V_t(\omega_t) = \inf_{H_t \in \mathbb{R}^d} \sup_{P \in \mathcal{P}_t^{\text{rec}}(\omega_t)} \mathbb{E}_P[V_{t+1}(\omega_t, S_{t+1}) - H_t \cdot (S_{t+1} - S_t)]. \quad (5)$$

The rectangular superhedging price is then

$$\pi^{\text{rec}}(\xi) = V_0(S_0).$$

By contrast, under non-rectangular ambiguity, admissible one-step kernels are linked by global consistency restrictions: they must arise from a single underlying path measure. For this reason, the worst-case choice cannot in general be made independently at each stage, and the path-dependent recursion above does not in general admit a Bellman recursion based on independent local optimizations over one-step conditional laws. To make this precise, we introduce a decomposability notion for kernel selections. This is the property that allows one, under additional measurability and integrability assumptions, to interchange pointwise maximization and integration; see (Theorem 14.60 Rockafellar and Wets, 1998). Decomposability alone is not sufficient, but it is the key structural ingredient.

Rectangularity is the structural property that permits independent recombination of admissible one-step conditional kernels across time and histories. This recombination property induces decomposability of the admissible kernel selection space, which in turn allows local pointwise optimization to be implemented through measurable selector arguments. Under suitable measurability and integrability assumptions, this yields the dynamic programming recursion underlying rectangular robust control problems.

Definition 3.5 (Decomposable spaces (Rockafellar and Wets, 1998, Definition 14.59)). *A space $\mathcal{X}$ of measurable functions $x : T \rightarrow \mathbb{R}^n$ is decomposable in association with a measure μ on $\mathcal{A}$ if, for every function $x_0 \in \mathcal{X}$, every set $A \in \mathcal{A}$ with $\mu(A) < \infty$, and every bounded measurable function $x_1 : A \rightarrow \mathbb{R}^n$, the space $\mathcal{X}$ also contains the function $x : T \rightarrow \mathbb{R}^n$ defined by*

$$x(t) = \begin{cases} x_0(t), & \text{for } t \in T \setminus A, \\ x_1(t), & \text{for } t \in A. \end{cases}$$

In our setting, the measurable space $(T, \mathcal{A})$ is replaced by $(\Omega_t, \mathcal{F}_t)$, and the role of $\mathcal{X}$ is played by a family of measurable kernel selections taking values in the Polish space $\mathcal{P}(\mathbb{R}^d)$ endowed with the topology of weak convergence. The integrability constraint $\mu(A) < \infty$ is automatically satisfied for every Borel set $A \subseteq \Omega_t$ when μ is any probability measure on Ω_t . The boundedness

of the measurable function is also automatically satisfied due to conditional probabilities law being bounded by 1.

Definition 3.6 (Decomposability of admissible kernel selections). *Let*

$$\mathfrak{K}_t = \left\{ K : \Omega_t \rightarrow \mathcal{P}(\mathbb{R}^d) \text{ measurable} : K(\omega_t) \in \mathcal{K}_t(\omega_t) \ \forall \omega_t \in \Omega_t \right\}.$$

We say that $\mathfrak{K}_t$ is decomposable if for every $K^0 \in \mathfrak{K}_t$, every $A \in \mathcal{F}_t$, and every measurable kernel $K^1 : \Omega_t \rightarrow \mathcal{P}(\mathbb{R}^d)$ satisfying

$$K^1(\omega_t) \in \mathcal{K}_t(\omega_t) \quad \text{for all } \omega_t \in A,$$

the spliced kernel

$$K(\omega_t) = \begin{cases} K^0(\omega_t), & \omega_t \in \Omega_t \setminus A \\ K^1(\omega_t), & \omega_t \in A, \end{cases}$$

belongs to $\mathfrak{K}_t$.

Proposition 2 (Rectangularity implies decomposability of kernel selections). *Let $\mathcal{P}^{\text{rec}} \subseteq \mathcal{P}(\Omega)$ be rectangular in the sense of Definition 3.3. Then, for every $t = 0, \dots, T-1$, the set $\mathfrak{K}_t$ of admissible measurable kernel selections is decomposable.*

Proof. Fix $t \in \{0, \dots, T-1\}$. Let $K^0 \in \mathfrak{K}_t$, let $A \in \mathcal{F}_t$, and let $K^1 : \Omega_t \rightarrow \mathcal{P}(\mathbb{R}^d)$ be measurable with

$$K^1(\omega_t) \in \mathcal{K}_t(\omega_t), \quad \text{for all } \omega_t \in A.$$

Define the spliced kernel

$$K(\omega_t) = \begin{cases} K^0(\omega_t), & \omega_t \in \Omega_t \setminus A \\ K^1(\omega_t), & \omega_t \in A. \end{cases}$$

First, we verify that the pasted kernel K is well-defined as an admissible measurable selector. To see measurability, let $B \in \mathcal{B}(\mathbb{R}^d)$. Since K^0 and K^1 are measurable kernels, the maps

$$\omega_t \mapsto K^0(\omega_t, B), \quad \omega_t \mapsto K^1(\omega_t, B),$$

are $\mathcal{F}_t$ -measurable. Since $A \in \mathcal{F}_t$, the indicator functions $\mathbf{1}_A$ and $\mathbf{1}_{A^c}$ are also $\mathcal{F}_t$ -measurable. Hence

$$K(\omega_t, B) = \mathbf{1}_{A^c}(\omega_t) K^0(\omega_t, B) + \mathbf{1}_A(\omega_t) K^1(\omega_t, B)$$

is $\mathcal{F}_t$ -measurable as a sum and product of measurable functions. Next, we verify pointwise admissibility. If $\omega_t \notin A$, then

$$K(\omega_t) = K^0(\omega_t).$$

Since $K^0 \in \mathcal{K}_t$, we have

$$K^0(\omega_t) \in \mathcal{K}_t(\omega_t).$$

Hence $K(\omega_t) \in \mathcal{K}_t(\omega_t)$ for all $\omega_t \notin A$.

If $\omega_t \in A$, then

$$K(\omega_t) = K^1(\omega_t).$$

By assumption,

$$K^1(\omega_t) \in \mathcal{K}_t(\omega_t),$$

for all $\omega_t \in A$. Therefore $K(\omega_t) \in \mathcal{K}_t(\omega_t)$ also on A .

Combining the two cases, we obtain

$$K(\omega_t) \in K_t(\omega_t), \quad \text{for all } \omega_t \in \Omega_t.$$

Thus the pasted kernel K is both measurable and pointwise admissible at every history $\omega_t \in \Omega_t$, and by construction

$$K(\omega_t) \in \mathcal{K}_t(\omega_t), \quad \text{for all } \omega_t \in \Omega_t.$$

To show that $K \in \mathfrak{K}_t$, it remains to verify that K is admissible as a time- t kernel under some measure in $\mathcal{P}^{\text{rec}}$. For each $u \neq t$, choose an arbitrary measurable selector

$$\widehat{K}_u(\omega_u) \in \mathcal{K}_u(\omega_u)$$

For instance, one may take the time- u kernel induced by some fixed reference measure in $\mathcal{P}^{\text{rec}}$. Since $\mathcal{P}^{\text{rec}}$ is rectangular, the path measure generated by the family of kernels

$$(\widehat{K}_0, \dots, \widehat{K}_{t-1}, K, \widehat{K}_{t+1}, \dots, \widehat{K}_{T-1})$$

belongs to $\mathcal{P}^{\text{rec}}$. By construction, its time- t conditional kernel is K . Hence $K \in \mathfrak{K}_t$, and $\mathfrak{K}_t$ is decomposable. $\square$

Decomposability allows local pointwise maximization over admissible kernels to be implemented through measurable selector arguments. Under suitable measurability and integrability assumptions, it allows the interchange of pointwise suprema and conditional expectations in the backward recursion; see Rockafellar and Wets (1998, Theorem 14.60). Consequently, under rectangularity, optimization over global measures can be represented by recursive pointwise optimization over local kernels, so that

$$\sup_{P \in \mathcal{P}^{\text{rec}}} \mathbb{E}_P[\xi] = \sup_{K_0 \in \mathcal{K}_0} \mathbb{E}_{K_0} \left[\sup_{K_1(\omega_1) \in \mathcal{K}_1(\omega_1)} \mathbb{E}_{K_1(\omega_1)} \left[\dots \sup_{K_{T-1}(\omega_{T-1}) \in \mathcal{K}_{T-1}(\omega_{T-1})} \mathbb{E}_{K_{T-1}(\omega_{T-1})} [\xi(\omega_T)] \right] \right].$$

Under non-rectangular ambiguity, this identity fails in general. Although each conditional law may be individually attainable at a given history, the pointwise maximizers chosen at different histories need not be jointly realizable under any single measure in $\mathcal{P}^{\text{nr}}$. Hence history-by-history optimization ranges over a larger pasted class than the original admissible set, and one only has

$$\sup_{P \in \mathcal{P}^{\text{nr}}} \mathbb{E}_P[\xi] \leq \sup_{K_0 \in \mathcal{K}_0} \mathbb{E}_{K_0} \left[\sup_{K_1(\omega_1) \in \mathcal{K}_1(\omega_1)} \mathbb{E}_{K_1(\omega_1)} \left[\dots \sup_{K_{T-1}(\omega_{T-1}) \in \mathcal{K}_{T-1}(\omega_{T-1})} \mathbb{E}_{K_{T-1}(\omega_{T-1})} [\xi(\omega_T)] \right] \right],$$

with strict inequality in general. Furthermore, this failure of pasting may be interpreted as a form of time inconsistency (Epstein and Schneider, 2003). Under non-rectangular ambiguity, the continuation problem re-evaluated after a given history is solved over conditional laws that may be locally admissible at that history but incompatible with the conditional laws selected after other histories. Hence the history-by-history worst-case rule need not coincide with any ex-ante globally admissible model in $\mathcal{P}^{\text{nr}}$. In this sense, Bellman separability fails because local optimization is not dynamically consistent with the original ambiguity set.

In order to compare robust superhedging prices under rectangular and non-rectangular ambiguity, we define the rectangular hull of a non-rectangular ambiguity set as the smallest rectangular ambiguity set that contains the original non-rectangular set.

Definition 3.7 (Rectangular hull of a non-rectangular ambiguity set). *Given an arbitrary non-rectangular ambiguity set $\mathcal{P}^{\text{nr}} \subseteq \mathcal{P}(\Omega)$, one can construct a rectangular ambiguity set $\mathcal{P}^{\text{rec}}$, which we call the rectangular hull of $\mathcal{P}^{\text{nr}}$, that approximates $\mathcal{P}^{\text{nr}}$ and shares the same attainable*

conditional probability distributions (kernels) . Define

$$\mathcal{K}_t^{nr}(\omega_t) = \{P(S_{t+1} \in \cdot | \omega_t) : P \in \mathcal{P}^{nr}\}$$

the set of attainable kernels for the non-rectangular ambiguity set $\mathcal{P}^{nr}$ for each time $t = 0, \dots, T-1$ and each history $(s_0, \dots, s_t)$. The corresponding rectangular hull is then defined as the set of all path measures $\tilde{P}$, whose one-step kernels belong to this set of attainable kernels:

$$\mathcal{P}^{rec} = \left\{ \tilde{P} \in \mathcal{P}(\Omega) : \tilde{P}(S_{t+1} \in \cdot | S_0 = s_0, \dots, S_t = s_t) \in \mathcal{K}_t^{nr}(s_0, \dots, s_t) \text{ for all } t, (s_0, \dots, s_t) \right\}.$$

Since the kernels are chosen independently from $\mathcal{K}_t^{nr}$ at each time step, $\mathcal{P}^{rec}$ is rectangular by construction

The construction of the rectangular hull can be summarized informally as follows: one keeps all one-step conditional laws that are attainable under the original set, while removing the global consistency restrictions that couple these kernels across time. The result is a rectangular outer approximation of $\mathcal{P}^{nr}$.

Proposition 3 (Inclusion of non-rectangular set in its rectangular hull). *Let $\mathcal{P}^{nr} \subseteq \mathcal{P}(\Omega)$ be any ambiguity set with rectangular hull $\mathcal{P}^{rec}$ as defined in Definition 3.7. Then*

$$\mathcal{P}^{nr} \subseteq \mathcal{P}^{rec}.$$

Proof. Fix $P \in \mathcal{P}^{nr}$. For each $t = 0, \dots, T-1$ and each history $(s_0, \dots, s_t)$, the conditional distribution

$$P(S_{t+1} \in \cdot | S_0 = s_0, \dots, S_t = s_t)$$

is, by Definition 3.7, an element of $\mathcal{K}_t^{nr}(s_0, \dots, s_t)$.

Since P is uniquely determined by its initial law and these conditional kernels, and since each kernel lies in $\mathcal{K}_t^{nr}(s_0, \dots, s_t)$, the measure P satisfies the defining condition of $\mathcal{P}^{rec}$. Hence any measure P in $\mathcal{P}^{nr}$ is also in the rectangular hull $\mathcal{P}^{rec}$, and the claim follows. $\square$

Definition 3.8 (Rectangularity Gap). *Let $\mathcal{P}^{nr} \subseteq \mathcal{P}(\Omega)$ be any ambiguity set with rectangular hull $\mathcal{P}^{rec}$ as defined in Definition 3.7. The robust superhedging price under the ambiguity set $\mathcal{P}^{nr}$ is then given by*

$$\pi^{nr}(\xi) = \inf\{x \in \mathbb{R} : \exists H \in \mathcal{H} \text{ such that } x + H \cdot S_T \geq \xi \quad \mathcal{P}^{nr} - q.s.\}.$$

and the robust superhedging price under the ambiguity set $\mathcal{P}^{rec}$ is given by:

$$\pi^{rec}(\xi) = \inf\{x \in \mathbb{R} : \exists H \in \mathcal{H} \text{ such that } x + H \cdot S_T \geq \xi \quad \mathcal{P}^{rec} - q.s.\}.$$

Then, the rectangularity gap is defined as

$$\mathcal{G}(\xi) = \pi^{rec}(\xi) - \pi^{nr}(\xi).$$

In economic terms the non-rectangular ambiguity set $\mathcal{P}^{nr}$ represents the agent's actual uncertainty about the model: this may concern a structural parameter that is fixed throughout the horizon, or a global feature of the path distribution that is not known precisely. The rectangular hull $\mathcal{P}^{rec}$, by contrast, allows nature to re-select the local model at each time step (after each observed history, which may correspond to scenarios that are economically implausible and lead to a much more extreme worst-case distribution than under the non-rectangular set. The rectangularity gap $\mathcal{G}(\xi)$ measures the cost of this enlargement of the ambiguity set; it is the additional initial

capital one must commit if one prices under the rectangularized model rather than the original model. We argue that the rectangularity gap $\mathcal{G}(\xi)$ in asset pricing is nontrivial and later present two numerical simulations meant to illustrate the potential severity of the rectangularity gap.

Proposition 4 (Rectangularity gap is nonnegative). *Let $\mathcal{P}^{nr} \subseteq \mathcal{P}(\Omega)$ be any ambiguity set with rectangular hull $\mathcal{P}^{rec}$ as defined in Definition 3.7. Let $\pi^{nr}(\xi)$ be the robust superhedging price under $\mathcal{P}^{nr}$ and $\pi^{rec}(\xi)$ the robust superhedging price under $\mathcal{P}^{rec}$. Then*

$$\mathcal{G}(\xi) = \pi^{rec}(\xi) - \pi^{nr}(\xi) \geq 0.$$

Proof. The primal problem under an ambiguity set $\mathcal{P}$ is given by:

$$\pi(\xi) = \inf \{x \in \mathbb{R} : \exists H \in \mathcal{H} \text{ such that } x + (H \cdot S)_T \geq \xi \text{ } \mathcal{P}\text{-q.s.}\}.$$

The feasible sets for the primal problem under $\mathcal{P}^{nr}$ and $\mathcal{P}^{rec}$ are

$$A^{nr} = \{(H, x) : x + (H \cdot S)_T \geq \xi \text{ } \mathcal{P}^{nr}\text{-q.s.}\},$$

$$A^{rec} = \{(H, x) : x + (H \cdot S)_T \geq \xi \text{ } \mathcal{P}^{rec}\text{-q.s.}\}.$$

Since $\mathcal{P}^{nr} \subseteq \mathcal{P}^{rec}$, any constraint holding $\mathcal{P}^{rec}$ -q.s. also holds $\mathcal{P}^{nr}$ -q.s. Hence the $\mathcal{P}^{rec}$ -q.s. constraint is more restrictive than the $\mathcal{P}^{nr}$ -q.s. constraint, so the feasible set A^{rec} is contained in A^{nr} :

$$A^{rec} \subseteq A^{nr}.$$

Therefore,

$$\pi^{nr}(\xi) = \inf_{(H,x) \in A^{nr}} x \quad \text{and} \quad \pi^{rec}(\xi) = \inf_{(H,x) \in A^{rec}} x.$$

Since the smaller feasible set leads to a larger infimum in the rectangular case,

$$\pi^{rec}(\xi) \geq \pi^{nr}(\xi).$$

The dual problem for an ambiguity set $\mathcal{P}$ is defined as

$$\pi(\xi) = \sup_{Q \in \mathcal{Q}} \mathbb{E}_Q[\xi],$$

where

$$\mathcal{Q} = \{Q : \exists P \in \mathcal{P}, Q \ll P, S \text{ is a } Q\text{-martingale}\}.$$

The dual domains for the pricing problem under $\mathcal{P}^{nr}$ and $\mathcal{P}^{rec}$ respectively are

$$\mathcal{Q}^{nr} = \{Q : \exists P \in \mathcal{P}^{nr}, Q \ll P, Q \text{ martingale}\},$$

$$\mathcal{Q}^{rec} = \{Q : \exists P \in \mathcal{P}^{rec}, Q \ll P, Q \text{ martingale}\}.$$

By definition, for all $Q \in \mathcal{Q}^{nr}$, there exists $P^{nr} \in \mathcal{P}^{nr}$ such that

$$Q \ll P^{nr}.$$

Since by Proposition 3, $\mathcal{P}^{nr} \subseteq \mathcal{P}^{rec}$, it follows that

$$P^{nr} \in \mathcal{P}^{rec},$$

which implies

$$Q \in \mathcal{Q}^{\text{rec}}, \quad \text{for all } Q \in \mathcal{Q}^{\text{nr}}.$$

Hence,

$$\mathcal{Q}^{\text{nr}} \subseteq \mathcal{Q}^{\text{rec}}.$$

Then we have

$$\pi^{\text{nr}}(\xi) = \sup_{Q \in \mathcal{Q}^{\text{nr}}} \mathbb{E}_Q[\xi] \quad \text{and} \quad \pi^{\text{rec}}(\xi) = \sup_{Q \in \mathcal{Q}^{\text{rec}}} \mathbb{E}_Q[\xi].$$

By monotonicity of the supremum with respect to set inclusion,

$$\sup_{Q \in \mathcal{Q}^{\text{rec}}} \mathbb{E}_Q[\xi] \geq \sup_{Q \in \mathcal{Q}^{\text{nr}}} \mathbb{E}_Q[\xi].$$

Therefore,

$$\pi^{\text{rec}}(\xi) \geq \pi^{\text{nr}}(\xi).$$

□

3.8 Illustrative examples

Example 1 (Parametric ambiguity model via unknown volatility). We consider a one-dimensional price process $(S_t)_{t=0}^T$ evolving according to

$$S_{t+1} = S_t + \mu + \sigma \varepsilon_{t+1}, \quad t = 0, \dots, T-1$$

where

- $S_0 \in \mathbb{R}$ and $\mu \in \mathbb{R}$ are given;
- $(\varepsilon_t)_{t=1}^T$ are i.i.d. random variables satisfying

$$P(\varepsilon_t = 1) = P(\varepsilon_t = -1) = \frac{1}{2};$$

- the volatility parameter σ is unknown but belongs to a compact interval

$$\Theta = [\underline{\sigma}, \bar{\sigma}] \subset (0, \infty).$$

For each $\sigma \in \Theta$, let $P^\sigma \in \mathcal{P}(\Omega)$ denote the probability measure induced by the above dynamics. The ambiguity set is defined as

$$\mathcal{P}^{\text{nr}} = \{P^\sigma : \sigma \in \Theta\}.$$

The ambiguity set $\mathcal{P}^{\text{nr}}$ is non-rectangular because the same parameter σ governs the transition law at all time steps. In particular, for any $P^\sigma \in \mathcal{P}^{\text{nr}}$, the conditional distribution of S_{t+1} given $(S_0, \dots, S_t)$ depends on σ , and this dependence is identical across all t . Under some P^σ , the conditional law at time t of the price at time $t+1$ is given by

$$P^\sigma(S_{t+1} | S_0, \dots, S_t) = \frac{1}{2} \delta_{S_t + \mu + \sigma} + \frac{1}{2} \delta_{S_t + \mu - \sigma}.$$

Thus, the one-step conditional distributions cannot be chosen independently across time: they must all correspond to the same value of σ . This is the source of non-rectangularity.

To see precisely why a Bellman recursion based on local optimization fails, suppose for simplicity that $\Theta = \{\underline{\sigma}, \bar{\sigma}\}$. At a given history ω_t^1 , the worst-case continuation value for the claim may be

attained by the high-volatility kernel corresponding to $\bar{\sigma}$, while at another history ω_t^2 it may be attained by the low-volatility kernel corresponding to $\underline{\sigma}$. A stagewise Bellman optimization would then attempt to choose $\bar{\sigma}$ after ω_t^1 and $\underline{\sigma}$ after ω_t^2 . However, no measure in $\mathcal{P}^{nr} = \{P^\sigma : \sigma \in \Theta\}$ can implement both choices simultaneously, because every admissible measure is generated by a single volatility parameter σ fixed over the entire horizon. Hence the pointwise maximizing continuation kernels are not jointly attainable under any single global model. The obstruction is therefore the failure of Bellman separability since local worst-case kernels cannot be pasted together into a globally admissible measure.

A natural rectangular approximation is obtained by replacing the single parameter σ with a sequence of parameters $(\sigma_t)_{t=0}^{T-1}$ satisfying

$$\sigma_t(S_0, \dots, S_t) \in \Theta \quad \text{for all } t \in \{0, \dots, T-1\}.$$

The corresponding dynamics become

$$S_{t+1} = S_t + \mu + \sigma_t(S_0, \dots, S_t) \varepsilon_{t+1}.$$

This yields a rectangular ambiguity set $\mathcal{P}^{rec}$ that contains $\mathcal{P}^{nr}$ as a subset. Economically, this allows nature to re-select the volatility after observing the realized path, which enlarges the set of adverse scenarios and leads to more conservative superhedging prices. Consequently,

$$\mathcal{G}(\xi) = \pi^{rec} - \pi^{nr} \geq 0.$$

The rectangular version of this ambiguity set is the discrete time analog of the uncertain volatility model of Avellaneda et al. (1995).

We now want to price a derivative with some terminal payoff $\Phi(S_T)$. As mentioned before, the rectangular problem admits a Bellman recursion and in our numerical simulations we can use dynamic programming to find the rectangular price. Let $V_t(s)$ denote the minimal capital required at time t and price s to superhedge a payoff $\Phi(S_T)$. The problem then admits the recursion:

$$\begin{aligned} V_T(s) &= \Phi(s), \\ V_t(s) &= \inf_H \sup_{\sigma \in \Theta} \mathbb{E}_{P^\sigma}[V_{t+1}(s + \mu + \sigma \varepsilon) - H(\mu + \sigma \varepsilon)], \end{aligned}$$

and the rectangular robust superhedging price for a payoff Φ is then given by

$$\pi^{rec}(\Phi) = V_0(S_0).$$

Since we cannot obtain such a recursion for the non-rectangular model in our simulations, we use an LP approximation using a discretized version of the non-rectangular ambiguity set. However, this approach works only for small horizons since the constraints grow exponentially in the horizon. The ambiguity set contains a continuum of possible volatility values in the interval $[\underline{\sigma}, \bar{\sigma}]$ which would give an infinite-dimensional linear program. We instead discretize the ambiguity set by considering M possible values of σ :

$$\Theta_M = \{\sigma^1, \dots, \sigma^M\} \quad \text{such that} \quad \sigma^m \in [\underline{\sigma}, \bar{\sigma}] \quad \forall m = 1, \dots, M,$$

which results in the discrete ambiguity set

$$\mathcal{P}_M = \{P^{\sigma^1}, \dots, P^{\sigma^M}\}.$$

The larger the value of M chosen, the closer the approximation to the original continuous ambiguity set. For any $\sigma^m \in \Theta_M$, and a sign sequence $\varepsilon = (\varepsilon_1, \dots, \varepsilon_T) \in \{-1, +1\}^T$ define the price process:

$$S_0^{m,\varepsilon} = S_0, \quad S_{t+1}^{m,\varepsilon} = S_t^{m,\varepsilon} + \mu + \sigma^m \varepsilon_{t+1} \quad t = 0, \dots, T-1.$$

The corresponding terminal payoff is then $\Phi(S_T^{m,\varepsilon})$, and given an initial capital x and predictable strategy H , the terminal wealth on the path (m, ε) is

$$W_T(m, \varepsilon) = x + \sum_{t=0}^{T-1} H_t(\varepsilon_1, \dots, \varepsilon_t) \cdot \Delta S_{t+1}^{m,\varepsilon}.$$

The discretized robust superhedging problem is then given by the linear program

$$\begin{aligned} \pi_{LP}^{nr} = \min_{x, H} \quad & x \\ \text{s.t.} \quad & W_T(m, \varepsilon) \geq \Phi(S_T^{m,\varepsilon}) \quad \forall m = 1, \dots, M, \quad \forall \varepsilon \in \{-1, +1\}^T \end{aligned}$$

This is a linear program because the objective is linear in x , and the superhedging constraints, one for each volatility level σ^m and each sign path $\varepsilon \in \{-1, +1\}^T$, are affine in x and in the hedge variables $h_t(\varepsilon_1, \dots, \varepsilon_t)$. Since at each time step the price can move up or down, the predictable strategy is parametrized by one hedge decision per node of the binary tree. The number of such nodes is

$$1 + \sum_{t=1}^{T-1} 2^t = 2^T - 1,$$

so the LP has $2^T - 1$ hedge variables plus the initial capital variable x , giving 2^T decision variables in total. Because the constraint must hold for all M volatility levels σ^m and for all 2^T possible sign sequences $\varepsilon \in \{-1, +1\}^T$, the number of constraints is $M \cdot 2^T$.

The quantity π_{LP}^{nr} is the exact robust superhedging price for the discretized ambiguity set. We expect that as M increases this approximation approaches the continuous ambiguity set price π^{nr}

$$\lim_{M \rightarrow \infty} \pi_{LP}^{nr} = \pi^{nr}.$$

Since the size of the problem grows exponentially in T due to the 2^T path constraints, this makes the LP suitable as a benchmark for short horizons. To address large horizons, we introduce the MDP formulation and the actor-critic algorithm which scales better.

Example 2 (Nonparametric ambiguity model via Wasserstein ball). We now consider a data-driven ambiguity model defined directly on the space of price paths. Since different price paths cannot be observed in practice, one can think of the different price paths as generated by a Monte Carlo simulation arising from some modeling assumption. We use the Wasserstein model in this thesis as a theoretical example illustrating non-rectangularity beyond the parametric volatility case, but do not report numerical results for it. This is because both the rectangular DP recursion and non-rectangular LP are difficult to implement and computationally expensive for the Wasserstein model at the problem sizes considered in this thesis.

The Wasserstein ball is a natural choice of ambiguity set in this data-driven setting because it admits a finite-sample guarantee where the radius $\varepsilon_n(\beta)$ can be chosen so that the true data-generating measure P^* lies in $\mathcal{P}^W(\varepsilon_n(\beta))$ with probability at least $1 - \beta$, with $\varepsilon_n(\beta) \rightarrow 0$ as $n \rightarrow \infty$ (Esfahani and Kuhn, 2018).

Suppose we observe n sample paths

$$\omega^{(i)} = (s_0^{(i)}, \dots, s_T^{(i)}) \in \Omega \quad i = 1, \dots, n$$

The empirical measure which we use as a reference for building the Wasserstein ball is given by

$$\hat{P}_n = \frac{1}{n} \sum_{i=1}^n \delta_{\omega^{(i)}}$$

i.e., we take the measure that assigns equal probabilities to all paths in the Borel σ -algebra of Ω as the reference point.

Let $d_\Omega : \Omega \times \Omega \rightarrow [0, \infty)$ be a ground metric on the path space, for instance

$$d_\Omega(\omega, \tilde{\omega}) = \sum_{t=0}^T \alpha_t |S_t - \tilde{S}_t|,$$

for some weights $\alpha_t > 0$. The ambiguity set is defined as a Wasserstein ball

$$\mathcal{P}^W(\rho) = \left\{ Q \in \mathcal{P}(\Omega) : W(Q, \hat{P}_n) \leq \rho \right\},$$

where W denotes the Wasserstein distance induced by d_Ω .

The ambiguity set $\mathcal{P}^W(\rho)$ is non-rectangular because the Wasserstein constraint is imposed on the joint law of the entire trajectory $(S_0, \dots, S_T)$. In particular, admissible perturbations of the distribution at different time steps are coupled through a single global transport constraint.

As a consequence, the conditional distributions

$$P(S_{t+1} \in \cdot \mid S_0, \dots, S_t),$$

cannot be chosen independently across time and histories, since they must arise from a single measure $P \in \mathcal{P}^W(\rho)$.

To see precisely why a Bellman recursion based on local optimization fails, suppose that at one history ω_t^1 the worst-case continuation value is attained by a local perturbation that shifts mass between two particular observed paths, while at another history ω_t^2 it is attained by a different local perturbation. A stagewise Bellman optimization would attempt to apply both perturbations simultaneously. However, the total transport cost is a single global budget bounded by ρ so it is generally not possible to implement both local worst-case perturbations under a single coupling that satisfies $W(P, \hat{P}_n) \leq \rho$. The pointwise maximizing continuation kernels are therefore not jointly attainable under any single measure in $\mathcal{P}^W(\rho)$. As in the parametric example, the obstruction is the failure of Bellman separability: local worst-case kernels cannot be pasted together into a globally admissible measure.

A rectangular hull can be constructed by considering the set of all one-step conditional distributions that arise under some $P \in \mathcal{P}^W(\rho)$, and allowing these to be recombined independently across time.

This yields a larger ambiguity set $\mathcal{P}^{W, \text{rec}}(\rho)$ satisfying

$$\mathcal{P}^W(\rho) \subseteq \mathcal{P}^{W, \text{rec}}(\rho).$$

As in the parametric case, this approximation removes global consistency constraints and therefore leads to more conservative robust superhedging prices.

We now want to price a derivative with terminal payoff $\Phi(S_T)$. Under the rectangular approximation, the problem admits a dynamic programming formulation. Let $V_t(\omega_t)$ denote the minimal capital required at time t after observing history $\omega_t = (s_0, \dots, s_t)$ to superhedge the payoff. Then

$$V_T(\omega_T) = \Phi(s_T),$$

and for $t = T - 1, \dots, 0$,

$$V_t(\omega_t) = \inf_{H \in \mathbb{R}} \sup_{K \in \mathcal{K}_t^W(\omega_t)} \mathbb{E}_K [V_{t+1}(\omega_{t+1}) - H(S_{t+1} - S_t)].$$

The rectangular robust superhedging price is then given by

$$\pi^{W, \text{rec}}(\Phi) = V_0(S_0).$$

Such a Bellman recursion is generally not available for the original non-rectangular Wasserstein ambiguity set, because the one-step kernels must arise jointly from a single path measure satisfying the global constraint

$$W_{d_\Omega}(P, \hat{P}_n) \leq \rho.$$

As in the sigma interval case, there is a continuum of measures inside a Wasserstein ball of radius ρ , in numerical simulations we use a finite-dimensional approximation of the non-rectangular Wasserstein model.

Specifically, we restrict attention to probability measures supported on the observed sample paths

$$P^\theta = \sum_{i=1}^n \theta_i \delta_{\omega^{(i)}}, \quad \theta \in \Delta_n = \left\{ \theta \in \mathbb{R}_+^n : \sum_{i=1}^n \theta_i = 1 \right\}.$$

The empirical measure corresponds to

$$\hat{\theta} = \left(\frac{1}{n}, \dots, \frac{1}{n} \right).$$

The finite-dimensional Wasserstein ambiguity set is then

$$\Theta = \left\{ \theta \in \Delta_n : W_{d_\Omega}(P^\theta, \hat{P}_n) \leq \rho \right\}.$$

Hence the finite-support version of the ambiguity set is

$$\mathcal{P}_\rho^n = \left\{ P^\theta : \theta \in \Theta \right\}.$$

Since both P^θ and $\hat{P}_n$ are discrete measures, the Wasserstein distance admits the finite-dimensional optimal transport representation

$$W_{d_\Omega}(P^\theta, \hat{P}_n) = \min_{\Gamma \in \mathbb{R}_+^{n \times n}} \left\{ \sum_{i,j=1}^n \Gamma_{ij} d_\Omega(\omega^{(i)}, \omega^{(j)}) : \sum_{j=1}^n \Gamma_{ij} = \theta_i, \sum_{i=1}^n \Gamma_{ij} = \frac{1}{n} \right\}.$$

Equivalently,

$$\Theta = \left\{ \theta \in \Delta_n : \exists \Gamma \in \mathbb{R}_+^{n \times n} \text{ such that } \sum_{j=1}^n \Gamma_{ij} = \theta_i, \sum_{i=1}^n \Gamma_{ij} = \frac{1}{n}, \sum_{i,j=1}^n \Gamma_{ij} d_\Omega(\omega^{(i)}, \omega^{(j)}) \leq \rho \right\}.$$

As in the volatility interval example, we derive a finite-dimensional LP benchmark by restricting attention to ambiguity measures supported on the observed paths and imposing the superhedging constraint pathwise. Since

$$\mathcal{P}_\rho^n = \{P^\theta : \theta \in \Theta\}$$

is supported on the observed paths

$$\omega^{(1)}, \dots, \omega^{(n)},$$

a strategy superhedges under every measure in $\mathcal{P}_\rho^n$ if it superhedges pathwise on every observed trajectory $\omega^{(i)}$ that carries positive mass under at least one $\theta \in \Theta$.

For each path $\omega^{(i)}$, define

$$\Delta S_{t+1}^{(i)} = S_{t+1}^{(i)} - S_t^{(i)}, \quad t = 0, \dots, T-1.$$

The terminal wealth generated by initial capital x and predictable strategy H is

$$W_T(\omega^{(i)}) = x + \sum_{t=0}^{T-1} H_t(\omega^{(i)}) \Delta S_{t+1}^{(i)}.$$

Hence the finite-support Wasserstein superhedging benchmark is

$$\begin{aligned} \pi_{LP}^{nr}(\Phi) = \min_{x, H} \quad & x \\ \text{s.t.} \quad & x + \sum_{t=0}^{T-1} H_t(\omega^{(i)}) \Delta S_{t+1}^{(i)} \geq \Phi(S_T^{(i)}), \quad i = 1, \dots, n, \\ & H_t(\omega^{(i)}) = H_t(\omega^{(j)}) \quad \text{whenever} \quad (S_0^{(i)}, \dots, S_t^{(i)}) = (S_0^{(j)}, \dots, S_t^{(j)}), \\ & i, j = 1, \dots, n, \quad t = 0, \dots, T-1. \end{aligned}$$

As in the sigma interval model, the continuum of admissible models is replaced by a finite set of representative sample paths, and the superhedging inequalities are imposed pathwise.

The optimal value $\pi_{LP}^{nr}(\Phi)$ does not depend on the Wasserstein radius ρ : it equals the pathwise superhedging price on the empirical support for *every* $\rho \geq 0$. This is a known feature of Wasserstein DRO under a finite nominal support: worst-case measures under a Wasserstein constraint with discrete nominal remain supported on (or near) the nominal points (Esfahani and Kuhn, 2018), so a quasi-sure criterion on the empirical support cannot distinguish radii once every nominal point is reachable.

Consequently, the LP is best interpreted not as the exact non-rectangular Wasserstein superhedging price, but as a *pathwise lower bound* on it

$$\pi_{LP}^{\text{path}}(\Phi) \leq \pi^{nr}(\Phi),$$

with equality whenever the worst-case measure in $\mathcal{P}^W(\rho)$ is supported on the observed paths. Therefore, in order to compare the superhedging prices under Wasserstein ambiguity sets of different sizes in the numerical simulations, we use the reinforcement learning algorithm introduced in the next section to compute the price under non-rectangular ambiguity. The number of hedge variables equals the number of distinct history nodes observed in the sample paths. This number is at most

$$\sum_{t=0}^{T-1} n = nT,$$

and may be smaller when different paths share the same history. The Wasserstein constraint introduces n^2 transport variables Γ_{ij} . Thus, unlike the volatility-grid LP whose number of path constraints grows as $M \cdot 2^T$, the finite-support Wasserstein benchmark scales polynomially in n and T , although the number of sample paths required for an accurate approximation may still be large. As the number of sample paths increases and the empirical measure better approximates the underlying path law, this benchmark approximates the continuous path-space Wasserstein robust problem. As in the parametric volatility example, the benchmark is useful for moderate problem sizes, while the actor-critic formulation is introduced to handle larger-scale

non-rectangular ambiguity models.

4 Markov Decision Processes

Since computing the price under non-rectangular ambiguity with the discretized ambiguity set is still computationally expensive for large horizons, we reformulate the superhedging problem as a robust Markov decision process in order to use the reinforcement learning algorithm developed by Li et al. (2026) to compute an approximation of the superhedging price in our numerical illustrations.

One can think of an MDP as an agent that is faced with the problem of choosing a policy that influences the behavior of a probabilistic system through time. At each time the system occupies a state and the agent may choose an action from a set of allowable actions in order to influence the evolution of the system.

Formally, a Markov decision process is a 4-tuple $(\mathcal{S}, \mathcal{A}, Q, r)$, where

- $\mathcal{S}$ is the **state space**, a set of possible states;
- $\mathcal{A}$ is the **action space**, representing the set of actions available in each state;
- Q is the **transition kernel**,

$$Q : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$$

where $\Delta(\mathcal{S})$ denotes the probability simplex over $\mathcal{S}$. For $s, s' \in \mathcal{S}$ and $a \in \mathcal{A}$,

$$Q(s' \mid s, a)$$

gives the probability of transitioning to state s' given current state s and action a .

- r is the **reward function**,

$$r : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \rightarrow \mathbb{R}$$

where $r(s, a, s')$ is the reward received when taking action a in state s and transitioning to state s' .

A **policy** is a rule for choosing actions. A time-invariant, Markovian, deterministic policy is a mapping $\pi : \mathcal{S} \rightarrow \mathcal{A}$, while more generally a stochastic policy is given by

$$\pi(\cdot \mid s) \in \Delta(\mathcal{A}),$$

and $\pi(a \mid s)$ denotes the probability of choosing action a in state s .

The process is called "Markov" as a reflection of the fact that the transition probabilities and rewards depend on the past only through the current state and action (Puterman, 2005). Formally,

$$Q(s_{t+1} \mid s_0, a_0, \dots, s_t, a_t) = Q(s_{t+1} \mid s_t, a_t).$$

In this thesis, we restrict attention to discrete-time, finite-horizon MDPs. In this setting, the system evolves according to

$$s_{t+1} \sim Q(\cdot \mid s_t, a_t)$$

and a randomized policy is given by a sequence $\pi = \{\pi_t\}_{t=0}^{T-1}$, of time-varying policies

$$\pi_t : \mathcal{S} \rightarrow \Delta(\mathcal{A})$$

Given a policy $\pi = \{\pi_t\}_{t=0}^{T-1}$, the objective of the agent is to maximize the expected cumulative reward over the finite horizon. Starting from an initial state $s_t = s$, the value function associated with policy π at time t is defined as

$$V_t^\pi(s) = \mathbb{E}^\pi \left[\sum_{k=t}^{T-1} r(s_k, a_k, s_{k+1}) \middle| s_t = s \right]$$

where the expectation is taken with respect to the probability law induced by the policy π and the transition kernel Q . The optimal value function is defined as

$$V_t^*(s) = \sup_{\pi \in \Pi} V_t^\pi(s)$$

for all $s \in \mathcal{S}$ and $t = 0, \dots, T$. The set Π denotes all randomized Markovian policies.

An optimal policy $\pi^* = \{\pi_t^*\}_{t=0}^{T-1}$ is a policy that attains this supremum, i.e.,

$$V_t^*(s) = V_t^{\pi^*}(s), \quad \text{for all } s \in \mathcal{S}, t = 0, \dots, T.$$

The robust analog to an MDP arises when one considers the transition kernels to be unknown and considers a set of possible conditional distributions at each step given by a set of prior joint laws called the ambiguity set (Wiesemann et al., 2013; Iyengar, 2005). The robust MDP framework was introduced by Iyengar (2005) and Nilim and El Ghaoui (2005), both of whom imposed rectangularity from the outset so that the robust problem admits a Bellman recursion and can be solved by dynamic programming. This assumption became the standard in the literature that followed, and Wiesemann et al. (2013) later showed that the robust MDP problem is strongly NP-hard when the ambiguity set is non-rectangular, which explains why rectangularity has remained the default. Recently Li et al. (2026) have presented a provably convergent algorithm for non-rectangular MDPs.

We consider a discrete-time, finite-horizon MDP defined on a finite state space $\mathcal{S}$ and a finite action space $\mathcal{A}$. Time evolves over $t = 0, 1, \dots, T$. The transition kernel Q is unknown and must be estimated. Following Wiesemann et al. (2013), we model this uncertainty via an ambiguity set $\mathcal{P}$. Each $Q \in \mathcal{P}$ specifies transition probabilities $Q(\cdot | s, a)$. The true transition kernel is unknown but is assumed to lie in $\mathcal{P}$, although in the case of the Wasserstein ambiguity set we can give a guarantee of the probability with which this holds depending on the radius of the ambiguity set. Given a policy π , its worst-case value is defined as

$$V_t^\pi(s) = \inf_{Q \in \mathcal{P}} \mathbb{E}^{Q, \pi} \left[\sum_{k=t}^{T-1} r(s_k, a_k, s_{k+1}) \middle| s_t = s \right].$$

The robust control problem becomes

$$V_t^*(s) = \sup_{\pi} \inf_{Q \in \mathcal{P}} \mathbb{E}^{Q, \pi} \left[\sum_{k=t}^{T-1} r(s_k, a_k, s_{k+1}) \right].$$

This formulation corresponds to a two-player game between a decision maker choosing π and an adversary choosing the worst-case model $Q \in \mathcal{P}$.

Definition 4.1 (Rectangular ambiguity). *The ambiguity set $\mathcal{P}$ is called*

- ***(s, a) -rectangular** if*

$$\mathcal{P} = \prod_{(s, a) \in \mathcal{S} \times \mathcal{A}} \mathcal{P}_{s, a}, \quad \mathcal{P}_{s, a} \subseteq \Delta(\mathcal{S});$$

- *s-rectangular if*

$$\mathcal{P} = \prod_{s \in \mathcal{S}} \mathcal{P}_s, \quad \mathcal{P}_s \subseteq \prod_{a \in \mathcal{A}} \Delta(\mathcal{S}).$$

Rectangularity ensures that uncertainty decomposes across states (and possibly actions). This property is crucial for tractability and dynamic programming. Non-rectangular ambiguity sets lead to loss of dynamic programming structure, potential need for history-dependent policies and therefore computational intractability. In contrast, under rectangular ambiguity, the value function satisfies a robust Bellman recursion

$$V_t(s) = \sup_{a \in \mathcal{A}} \inf_{Q(\cdot|s,a) \in \mathcal{P}_{s,a}} \mathbb{E}_{s' \sim Q(\cdot|s,a)} [r(s, a, s') + V_{t+1}(s')].$$

This recursion highlights the stagewise nature of worst-case optimization.

Remark 2 (State augmentation and equivalent formulations). *The MDP framework admits several equivalent formulations that differ in how the reward function is specified. The reward may depend on the current state and action, $r(s, a)$; on the current state, action, and next state, $r(s, a, s')$; or even only on the current state, $r(s)$. These formulations are equivalent in the sense that any one can be reduced to another by suitably augmenting the state space.*

For instance, a reward of the form $r(s, a, s')$ can be reformulated as a reward depending only on the current state by augmenting the state to include the previous state and action: defining $\tilde{s}_{t+1} = (s_t, a_t, s_{t+1})$, the reward becomes a function of $\tilde{s}_{t+1}$ alone. Conversely, $r(s, a)$ is recovered from $r(s, a, s')$ by taking the conditional expectation

$$r(s, a) = \mathbb{E}_{s' \sim Q(\cdot|s,a)} [r(s, a, s')].$$

The choice of formulation is therefore one of convenience rather than generality.

4.1 Superhedging as a Markov decision process

In this section, we reformulate the robust superhedging problem as a Markov decision process. This connection allows us to interpret dynamic trading under model uncertainty as a sequential decision-making problem and provides a foundation for using reinforcement learning methods for obtaining an approximation to the optimization problem in our numerical simulations. Recall the robust superhedging problem

$$\pi(\xi) = \inf \{x \in \mathbb{R} : \exists H \text{ such that } x + (H \cdot S)_T \geq \xi \quad \mathcal{P} - q.s.\},$$

where

$$(H \cdot S)_T = \sum_{t=0}^{T-1} H_t \cdot \Delta S_{t+1}.$$

It is natural to think of the price of the underlying $S = (S_t)_{t=0}^T$ as the state of the MDP, the trading strategies H as the actions, and the portfolio value or wealth

$$W_t = x + \sum_{u=0}^t H_u \cdot \Delta S_u(\omega_u)$$

as the reward function. The transition law would then be given by $P(\cdot|S_t, H_t)$ for some $P \in \mathcal{P}$, and the policy by $\pi(\cdot|S_t)$. In our case specifically we augment the state variable at each time to include not only the price history but also the time and wealth up to that point in time. However, Wiesemann et al. (2013) have shown that when the ambiguity set $\mathcal{P}$ is non-rectangular

the robust MDP problem is strongly NP-hard, even when $\mathcal{P}$ is a convex polyhedron. Therefore, we need to relax the problem by introducing a penalty term λ such that if we choose λ large enough, the penalized problem is equivalent to the original superhedging problem.

For any pair (x, H) , define the shortfall random variable

$$L_{x,H}(\omega) = (\xi(\omega) - x - (H \cdot S)_T(\omega))^+.$$

The shortfall represents the amount by which the terminal wealth falls short of the claim payoff depending on ω . We can rewrite the superhedging condition

$$x + (H \cdot S)_T \geq \xi \quad \mathcal{P} - q.s.$$

in terms of shortfall

$$L_{x,H} = 0 \quad \mathcal{P} - q.s.$$

Then, the superhedging problem can be expressed as

$$\pi(\xi) = \inf_{x,H} \{x : L_{x,H} = 0 \quad \mathcal{P} - q.s.\}.$$

Since the positive part of a number vanishes iff the number is ≤ 0 , the superhedging condition $L_{x,H} = 0 \quad \mathcal{P} - q.s.$ can be expressed as the worst-case expectation over the measures in $\mathcal{P}$

$$L_{x,H} = 0 \quad \mathcal{P} - q.s. \quad \Leftrightarrow \quad \sup_{P \in \mathcal{P}} \mathbb{E}_P[L_{x,H}] = 0.$$

Then the superhedging constraint is equivalent to the worst-case expected shortfall over the measures in the set $\mathcal{P}$ and the robust superhedging problem admits the representation

$$\pi(\xi) = \inf_{x,H} \left\{ x : \sup_{P \in \mathcal{P}} \mathbb{E}_P[(\xi - x - (H \cdot S)_T)^+] = 0 \right\}.$$

We introduce the penalized version

$$\pi_\lambda(\xi) = \inf_{x,H} \left\{ x + \lambda \sup_{P \in \mathcal{P}} \mathbb{E}_P[(\xi - x - (H \cdot S)_T)^+] \right\},$$

where $\lambda > 0$ is a penalty parameter weighting the worst-case expected shortfall against the initial capital x .

The original robust superhedging price $\pi(\xi)$ corresponds to a seller who is unwilling to bear any residual risk: the terminal wealth must dominate the claim payoff under every model in the ambiguity set, and any shortfall, however unlikely, is prohibited. In practice, however, sellers typically tolerate some residual risk in exchange for charging a lower price. The penalty parameter $\lambda > 0$ then acts as the price the seller assigns to one unit of expected shortfall under the worst-case model with small values of λ corresponding to a risk-tolerant seller who is willing to accept significant downside in exchange for a lower initial capital, while large values of λ correspond to a near-superhedging regime in which residual shortfall is heavily penalized.

As $\lambda \rightarrow \infty$, violations of the superhedging constraint become increasingly costly and this formulation approximates the robust superhedging price. Since for every λ one has

$$\pi_\lambda(\xi) \leq \pi(\xi)$$

and the map $\lambda \mapsto \pi_\lambda(\xi)$ is non-decreasing, as increasing λ makes the penalty term weakly larger

for every fixed pair (x, H) . Under additional regularity assumptions ensuring an exact penalty property, one may further obtain

$$\lim_{\lambda \rightarrow \infty} \pi_\lambda(\xi) = \pi(\xi).$$

In the absence of such assumptions, $\pi_\lambda(\xi)$ should be viewed as a surrogate relaxation of the robust superhedging price rather than an equivalent reformulation. For the purposes of this thesis we treat λ as a sufficiently large but fixed parameter, and use the penalized problem as the target objective for the MDP.

We now formulate the penalized problem as a Markov decision process. We define the augmented state variable

$$Z_t = (t, X_t, W_t)$$

where

- t is time;
- $X_t = (S_0, \dots, S_t)$ is the price history up to time t ;
- W_t is the current wealth.

The full state space of the MDP is

$$\mathcal{Z} = \{0, \dots, T\} \times \mathcal{X} \times \mathbb{R},$$

where $\mathcal{X}$ is the state space of the price history X_t

$$\mathcal{X} = \bigcup_{t=0}^T (\mathbb{R}^d)^{t+1}.$$

At time t , the agent's actions are the choices of trading strategies

$$A_t = H_t \in \mathbb{R}^d$$

and the action space is then given by $\mathcal{A} = \mathbb{R}^d$. The dynamics of the state and wealth variables then evolve according to

$$X_{t+1} = (X_t, S_{t+1}), \quad W_{t+1} = W_t + H_t \cdot \Delta S_{t+1}.$$

For each $P \in \mathcal{P}$, the conditional law of S_{t+1} given X_t defines a transition kernel. Assume the payoff has the form $\xi = g(S_T)$.

We now specify the cost function so that the resulting cumulative cost coincides with the penalty term in the penalized problem. Assume the contingent claim payoff depends only on the terminal price, $\xi = g(S_T)$ for some measurable $g: \mathbb{R}^d \rightarrow \mathbb{R}$. Define the per-stage cost

$$c((t, X_t, W_t), A_t) = 0 \quad \text{for } t < T,$$

and at the terminal time

$$c((T, X_T, W_T), A_T) = \gamma^{-T} \lambda (g(s_T) - W_T)_+,$$

where s_T denotes the terminal component of the price history x and $\gamma \in (0, 1]$ is a discount factor. The factor γ^{-T} exactly cancels the discounting applied at time T in the cumulative cost,

so that for any policy π and any model $P \in \mathcal{P}$,

$$\mathbb{E}^{\pi, P} \left[\sum_{t=0}^{\infty} \gamma^t c(Z_t, A_t) \right] = \lambda \mathbb{E}^{\pi, P} [(g(S_T) - W_T)_+] = \lambda \mathbb{E}^{\pi, P} [(\xi - W_T)_+].$$

Defining the value function

$$V_{\pi}^P(z) = \mathbb{E}^{\pi, P} \left[\sum_{t=0}^{\infty} \gamma^t c(Z_t, A_t) \mid Z_0 = z \right],$$

the surrogate problem can be rewritten as the robust MDP

$$\min_{\pi} \sup_{P \in \mathcal{P}} V_{\pi}^P(Z_0), \quad Z_0 = (0, S_0, x_0), \quad (6)$$

where the initial wealth x_0 enters as a decision variable through the penalized objective: the agent chooses both the initial capital x_0 and the dynamic trading strategy. In practice we minimize jointly over x_0 and π , recovering the surrogate price $\pi_{\lambda}(\xi)$ as the optimal value of the robust MDP.

4.2 Preservation of (non-) rectangularity

We now examine whether the MDP formulation retains the properties of the original problem, in particular whether the original non-rectangular robust superhedging problem reformulated as an MDP remains non-rectangular.

We show that rectangularity is preserved under the lift to the history-augmented state space: a non-rectangular ambiguity set on path space induces a non-rectangular MDP, while the rectangular hull $\mathcal{P}^{\text{rec}}$ induces an s -rectangular MDP.

Recall that the lifted state variable is

$$Z_t = (t, X_t, W_t), \quad X_t = (S_0, \dots, S_t),$$

and that the action is the portfolio decision $A_t = H_t \in \mathcal{A} = \mathbb{R}^d$. For a fixed path measure $P \in \mathcal{P}$, the transition from a state $z = (t, X_t, W_t)$ under action $a \in \mathcal{A}$ proceeds in two steps. First, the next price S_{t+1} is drawn from the one-step conditional law induced by P given the history X_t . Second, the wealth is updated deterministically through the self-financing condition

$$W_{t+1} = W_t + a \cdot (S_{t+1} - S_t).$$

Each path measure $P \in \mathcal{P}$ therefore induces a controlled transition kernel

$$Q^P(\cdot \mid z, a)$$

on the lifted state space, and the path-space ambiguity set $\mathcal{P}$ induces the corresponding MDP ambiguity set

$$\mathcal{Q} = \{Q^P : P \in \mathcal{P}\}.$$

Suppose first that $\mathcal{P}$ is non-rectangular in the path-space sense of Definition 3.4. Then the induced family of MDP transition kernels $\mathcal{Q}$ is in general also non-rectangular. Although one may still characterize the locally feasible one-step conditional laws for S_{t+1} given the current history x_t , these local conditionals cannot in general be recombined independently across states: they must arise jointly from a single global path measure $P \in \mathcal{P}$. The induced robust MDP

therefore fails the standard rectangularity assumptions of the robust MDP literature.

Suppose now instead that the original ambiguity set $\mathcal{P}^{\text{nr}}$ is replaced by its rectangular hull $\mathcal{P}^{\text{rec}}$. For each lifted state $z = (t, X_t, W_t)$, let

$$\mathcal{U}(z) = \{Q(\cdot | z, a) : Q \text{ is induced by some admissible one-step conditional law of } S_{t+1} | X_t\}$$

denote the set of admissible next-state transition kernels at z . Under $\mathcal{P}^{\text{rec}}$, the conditional law of S_{t+1} given X_t may be selected independently for each history X_t , and the corresponding next-state kernel may therefore likewise be selected independently for each lifted state z . The action a influences the next wealth level through the self-financing update $W_{t+1} = W_t + a \cdot (S_{t+1} - S_t)$, but does not affect the law of S_{t+1} itself. The uncertainty set is therefore naturally indexed by the current state (equivalently, the current history) but not by the action. The induced robust MDP is consequently s -rectangular, and not (s, a) -rectangular on the state space.

4.3 Actor–critic algorithm

For non-rectangular ambiguity sets the reformulation of the penalized robust superhedging problem as the robust MDP

$$\min_{\pi} \sup_{P \in \mathcal{P}} V_{\pi}^P(Z_0)$$

inherits the non-rectangular structure of the underlying ambiguity set $\mathcal{P}$, so the standard robust Bellman recursion is not available. We solve it numerically with the actor–critic method of Li et al. (2026), which is a convergent policy-gradient method for robust MDPs with non-rectangular uncertainty sets.

For the MDP implementation we discretize the price and wealth variables as follows:

$$\mathcal{S} = \{S_1, \dots, S_{N_S}\}, \quad \mathcal{W} = \{W_1, \dots, W_{N_W}\},$$

to obtain the discretized state space

$$\mathcal{Z} = \{(t, S_i, W_j) : t = 0, \dots, T, S_i \in \mathcal{S}, W_j \in \mathcal{W}\}.$$

Additionally, in order to make the problem tractable, we discretize the action space and restrict the possible actions to an interval as follows: $\mathcal{A} = \{1, \dots, A_{N_A}\} \subset [-H_{\max}, H_{\max}]$. We now have a finite state space $\mathcal{Z}$ and a finite action grid $\mathcal{A}$ with $|\mathcal{Z}| = T \times N_S \times N_W$ and $|\mathcal{A}| = N_A$. We parametrize the ambiguity set as

$$\mathcal{P} = \{P^{\theta} : \theta \in \Theta\},$$

where the meaning of the parameter θ depends on the example

- In the sigma interval example $\theta = \sigma \in \Theta = [\underline{\sigma}, \bar{\sigma}]$ is the unknown constant volatility.
- In the Wasserstein ball example $\theta \in \Theta \subset \Delta_N$ is a probability weight vector on the observed sample paths, restricted to the Wasserstein ball of radius ρ around the empirical measure $\hat{P}_n$.

A stationary policy is a map

$$\pi(a | z), \quad z \in \mathcal{Z}, a \in \mathcal{A}, \quad \sum_{a \in \mathcal{A}} \pi(a | z) = 1,$$

and the set of admissible policies is $\Pi = \prod_{z \in \mathcal{Z}} \Delta(\mathcal{A})$, where $\Delta(\mathcal{A})$ denotes the probability

simplex. For a fixed policy π and a fixed model P^θ , the value function is

$$V_\pi^{P^\theta}(z) = \mathbb{E}^{\pi, P^\theta} \left[\sum_{t=0}^{\infty} \gamma^t c(Z_t, A_t) \mid Z_0 = z \right].$$

We denote the corresponding Q -function by

$$Q_\pi^{P^\theta}(z, a) = c(z, a) + \gamma \sum_{z'} P^\theta(z' \mid z, a) V_\pi^{P^\theta}(z').$$

The robust objective is

$$J(\pi) = \sup_{\theta \in \Theta} V_\pi^{P^\theta}(Z_0),$$

and the goal is to compute $\min_{\pi \in \Pi} J(\pi)$.

The algorithm alternates between two steps at each iteration $k = 0, 1, \dots$

1. a **critic step**, which approximately solves the robust policy evaluation problem for the current policy π^k ,

$$\theta^k \in \arg \max_{\theta \in \Theta} V_{\pi^k}^{P^\theta}(Z_0);$$

2. an **actor step**, which updates the policy by a projected policy-gradient step using the worst-case model P^{θ^k} identified by the critic.

The names "actor" and "critic" come from the reinforcement learning literature (Konda and Tsitsiklis, 1999). The actor is the agent choosing the policy $\pi(a|z)$ (in our context this is the trading strategy), and the critic evaluates the policy under the worst-case model by estimating either the value function $V(z)$ or the Q -function $Q(z, a)$. In the robust setting considered here, the critic must additionally search over the ambiguity set to identify that worst-case model. The actor and critic alternate because the policy and the worst-case model depend on each other. A change in the policy generally changes which model in $\mathcal{P}$ is worst-case, and conversely the optimal policy depends on which model is being defended against. Alternating updates resolve this coupling.

Critic step: robust policy evaluation

For a fixed policy π^k , the critic solves

$$\max_{\theta \in \Theta} F(\theta),$$

where

$$F(\theta) = V_{\pi^k}^{P^\theta}(Z_0).$$

The objective $F(\theta)$ is the expected discounted cost incurred by following the fixed policy π^k under the candidate transition model P^θ , viewed as a function of the model parameter θ . In the superhedging setting, the policy π^k represents the hedging strategy, while $F(\theta)$ corresponds to the resulting superhedging price under model P^θ .

Li et al. (2026) show that, under suitable smoothness assumptions, F is differentiable in θ with gradient

$$\nabla_\theta F(\theta) = \sum_{z, a, z'} d_{\pi^k}^{P^\theta}(z) \pi^k(a \mid z) Q_{\pi^k}^{P^\theta}(z, a) \nabla_\theta P^\theta(z' \mid z, a), \quad (7)$$

where $d_{\pi^k}^{P^\theta}$ is the discounted state-visitation distribution under policy π^k and model P^θ , defined as the unique solution of

$$d = (1 - \gamma)\mu_0 + \gamma d P_{\pi^k, \theta},$$

with μ_0 denoting the distribution of the initial state Z_0 and

$$P_{\pi^k, \theta}(z' | z) = \sum_a \pi^k(a | z) P^\theta(z' | z, a)$$

the state transition matrix induced by the policy and model. Each term in (7) has a natural interpretation:

- $d_{\pi^k}^{P^\theta}(z)$ is the *discounted state-visitation distribution*, which measures how frequently state z is visited under policy π^k and model P^θ , weighted by the discount factor γ .
- $\pi^k(a | z)$ is the policy given by the actor which gives the probability of taking action a in state z under the current policy.
- $Q_{\pi^k}^{P^\theta}(z, a)$ is the Q-function, which gives the expected discounted cost incurred when starting from state z , taking action a , and thereafter following policy π^k under model P^θ .
- $\nabla_\theta P^\theta(z' | z, a)$ measures the sensitivity of the transition probability to perturbations in the model parameter θ .

The gradient (7) therefore weights each transition (z, a, z') according to how frequently it is visited under π^k , how costly it is, and how strongly its probability changes with θ . Intuitively, the adversary updates θ in directions that increase the likelihood of costly and frequently visited transitions. Since F is generally nonconvex in θ , Li et al. (2026) propose two stochastic optimization methods for the critic step: projected Langevin dynamics and the Frank–Wolfe method.

Algorithm 1 Projected Langevin dynamics for the robust policy evaluation problem (Li et al., 2026, Algorithm 3.1)

Require: Initial iterate $\theta^{(0)} \in \Theta$, Gibbs parameter $\beta > 1$, step size $\eta > 0$, iteration number $M \in \mathbb{N}$

- 1: **for** $m = 0, 1, \dots, M - 1$ **do**
- 2: Sample $w_{m+1} \sim \mathcal{N}(0, I_q)$
- 3: $\theta^{(m+1)} \leftarrow \text{Proj}_\Theta \left(\theta^{(m)} + \eta \nabla_\theta V_\pi^{P^\theta}(z_0) \Big|_{\theta=\theta^{(m)}} + \sqrt{2\eta/\beta} w_{m+1} \right)$
- 4: **end for**
- 5: **return** $\theta^{(M)}$

In our examples, the parameter θ is one-dimensional in the volatility-interval model ($\theta = \sigma \in [\underline{\sigma}, \bar{\sigma}]$, so $q = 1$) and n -dimensional in the Wasserstein model ($\theta \in \Theta \subset \Delta_n$, so $q = n$). In the first case $F(\sigma)$ is convex on $[\underline{\sigma}, \bar{\sigma}]$ and the worst-case parameter can be found by direct endpoint comparison. However in the second case the exponential dependence of the projected Langevin dynamics' iteration count on $q = n$, combined with the cost of projecting onto Θ at every iteration, makes projected Langevin impractical. We therefore adopt Frank–Wolfe (Algorithm2) as the critic in our implementation.

Algorithm 2 Frank–Wolfe algorithm for the robust policy evaluation problem (Li et al., 2026, Algorithm 3.2)

Require: Initial iterate $\theta^{(0)} \in \Theta$, positive stepsizes $\{\alpha_m\}_{m=0}^\infty$, tolerance $\epsilon > 0$

- 1: $m \leftarrow 0$
- 2: **repeat**
- 3: Find an ϵ -optimal solution θ_ϵ of the linear subproblem

$$\max_{\theta \in \Theta} \langle \nabla_{\theta} V_{\pi}^{P^{\theta'}}(Z_0), \theta - \theta' \rangle \quad \text{with } \theta' = \theta^{(m)}$$

- 4: $\theta^{(m+1)} \leftarrow (1 - \alpha_m) \theta^{(m)} + \alpha_m \theta_\epsilon$
 - 5: $m \leftarrow m + 1$
 - 6: **until** $\langle \nabla_{\theta} V_{\pi}^{P^{\theta^{(m)}}}(Z_0), \theta_\epsilon - \theta^{(m)} \rangle \leq \epsilon$
 - 7: **return** $\hat{\theta} = \theta^{(m)}$
-

For the experiments in this thesis we use Frank–Wolfe, both because the parameter sets Θ in our examples are convex and compact and because the linear subproblem in each example admits a particularly simple structure.

Actor step: projected policy-gradient update

Given the worst-case model P^{θ^k} identified by the critic, the actor updates the policy using a gradient step. Define the objective function of the actor

$$J_k(\pi) = V_{\pi}^{P^{\theta^k}}(Z_0).$$

By the policy-gradient theorem (Sutton et al., 2000; Sutton and Barto, 2018), the gradient with respect to $\pi(a | z)$ is

$$\nabla_{\pi(a|z)} J_k(\pi) = d_{\pi}^{P^{\theta^k}}(z) Q_{\pi}^{P^{\theta^k}}(z, a), \quad (8)$$

where the discounted visitation distribution $d_{\pi}^{P^{\theta^k}}$ is again obtained from the linear system described above. The actor update is

$$\pi^{k+1} = \text{Proj}_{\Pi} \left(\pi^k - \eta \nabla_{\pi} J_k(\pi^k) \right),$$

where $\eta > 0$ denotes the step size.

Because the policy $\pi(a|z)$ is a probability, we must put constraints on the gradient step to ensure that the probabilities are always nonnegative and sum to 1. In order to ensure the policies chosen by the actor are valid, after taking the gradient step we project back onto the probability simplex. Projection essentially finds the nearest point back inside the feasible set based on Euclidean distance. The feasible set $\Pi = \prod_{z \in \mathcal{Z}} \Delta(\mathcal{A})$ for $\pi(a|z)$ is the set of all stochastic policies where $\Delta(\mathcal{A})$ denotes the probability simplex

$$\Delta(\mathcal{A}) = \left\{ p \in \mathbb{R}^{|\mathcal{A}|} : p_a \geq 0, \quad \sum_{a \in \mathcal{A}} p_a = 1 \right\}.$$

Since Π factorizes across states, the projection decouples into one simplex projection per state. The typical gradient step is the unconstrained gradient step

$$\tilde{\pi}_z = \pi^k(\cdot | z) - \eta \nabla_{\pi(\cdot|z)} J_k(\pi^k),$$

the projected policy is obtained by solving

$$\pi^{k+1}(\cdot | z) = \arg \min_{p \in \Delta(\mathcal{A})} \|p - \tilde{\pi}_z\|_2^2.$$

This Euclidean simplex projection is solved efficiently using the sorting-based algorithm of Duchi et al. (2008), which runs in $\mathcal{O}(|\mathcal{A}| \log |\mathcal{A}|)$ time per state. The sorting-based projection is computed as follows. Let

$$v = \tilde{\pi}_z.$$

Sort the components of the unconstrained policy update vector in descending order

$$v_{(1)} \geq v_{(2)} \geq \dots \geq v_{(|\mathcal{A}|)}.$$

Then find the number m of sorted components whose values remain positive after enforcing the probability-sum constraint

$$m = \max \left\{ j : v_{(j)} - \frac{1}{j} \left(\sum_{i=1}^j v_{(i)} - 1 \right) > 0 \right\}.$$

Next, compute the normalization shift

$$\tau = \frac{1}{m} \left(\sum_{i=1}^m v_{(i)} - 1 \right).$$

Finally, perform the componentwise projection

$$\pi^{k+1}(a|z) = \max\{v_a - \tau, 0\}.$$

Thus, τ acts as a threshold where components with $v_a \leq \tau$ are clipped to zero, while components with $v_a > \tau$ are shifted by τ and remain positive.

Combining the actor and critic updates yields an iterative robust actor–critic method that alternates between improving the policy and updating the adversarial transition model. The complete procedure is summarized in Algorithm 3 (Li et al., 2026).

Algorithm 3 Actor–critic algorithm for the robust policy improvement problem (Li et al., 2026, Algorithm 4.1)

Require: Iteration number $K \in \mathbb{N}$, step size $\eta > 0$, tolerance $\epsilon > 0$

1: Initialize $\pi^{(0)}(a | Z) = 1/|\mathcal{A}|$ for all $Z \in \mathcal{Z}$, $a \in \mathcal{A}$, and set $k \leftarrow 0$

2: **while** $k \leq K - 1$ **do**

3: **Critic:** find $\theta^{(k)} \in \Theta$ such that $V_{\pi^{(k)}}^{P^{\theta^{(k)}}}(Z_0) \geq V_{\pi^{(k)}}^*(Z_0) - \epsilon$ $\triangleright$ e.g. via Algorithm 2

4: **Actor:** $\pi^{(k+1)} \leftarrow \text{Proj}_{\Pi} \left(\pi^{(k)} - \eta \nabla_{\pi} V_{\pi^{(k)}}^{P^{\theta^{(k)}}}(Z_0) \right)$

5: $k \leftarrow k + 1$

6: **end while**

7: **return** $\pi^{(K)}$

Li et al. (2026) establish convergence guarantees for both the actor and critic steps of the algorithm. Under suitable smoothness assumptions, the critic step converges to a global maximizer of the robust policy evaluation problem when projected Langevin dynamics is used. When the Frank–Wolfe method is employed instead, convergence to a stationary point is obtained at rate $O(k^{-1/2})$ for nonconvex objectives.

The actor step performs projected gradient descent on a smoothed version of the robust objective. Combined with sufficiently accurate critic updates and appropriate step-size choices, the overall actor–critic algorithm produces an ε -stationary solution in $\mathcal{O}(\varepsilon^{-4})$ iterations, meaning that after roughly ε^{-4} iterations the first-order optimality error is at most ε .

Overall, the quality of the resulting policy depends on the accuracy with which the critic step solves the robust policy evaluation problem. To the best of our knowledge, these results provide the first provably convergent for robust MDPs with non-rectangular ambiguity sets, and support its use as a scalable approximation method for the robust superhedging problem.

4.4 Illustrative examples

Example 3 (Parametric ambiguity model via unknown volatility). We continue with the ambiguity model from Example 1, and formulate the superhedging problem under volatility uncertainty as an MDP. Recall the one-dimensional price dynamics

$$S_{t+1} = S_t + \mu + \sigma \varepsilon_{t+1}, \quad t = 0, \dots, T-1$$

where $(\varepsilon_t)_{t=1}^T$ are i.i.d random variables satisfying

$$P(\varepsilon_t = 1) = P(\varepsilon_t = -1) = \frac{1}{2},$$

and $\sigma \in \Theta = [\underline{\sigma}, \bar{\sigma}]$ is the unknown constant volatility. The corresponding non-rectangular ambiguity set is

$$\mathcal{P}^{\text{nr}} = \{P^\sigma : \sigma \in \Theta\}.$$

Following the construction outlined earlier in this section, we define the augmented state Z_t

$$Z_t = (t, X_t, W_t), \quad X_t = (S_0, \dots, S_t),$$

with state space

$$\mathcal{Z} = \{0, \dots, T\} \times \mathcal{X} \times \mathbb{R},$$

where

$$\mathcal{X} = \bigcup_{t=0}^T \mathbb{R}^{t+1}.$$

The action $A_t = H_t \in \mathbb{R}$ is the position in the risky asset over $[t, t+1)$ so $\mathcal{A} = \mathbb{R}$. For each $\sigma \in \Theta$, the price increment $S_{t+1} - S_t$ has conditional law

$$P^\sigma(S_{t+1} \in \cdot | X_t) = \frac{1}{2} \delta_{S_t + \mu + \sigma} + \frac{1}{2} \delta_{S_t + \mu - \sigma},$$

which, together with the deterministic wealth update,

$$W_{t+1} = W_t + H_t \cdot (S_{t+1} - S_t),$$

induce a controlled transition kernel $Q^\sigma(\cdot | z, a)$ on $\mathcal{Z}$. The corresponding MDP ambiguity set is

$$\mathcal{Q}^{\text{nr}} = \{Q^\sigma : \sigma \in \Theta\}.$$

The ambiguity set $\mathcal{P}^{\text{nr}}$ is non-rectangular on the path space because a single parameter σ governs the conditional law at every history and time step. We now show that the non-rectangularity is inherited by the induced MDP ambiguity set $\mathcal{Q}^{\text{nr}}$.

For a lifted state $z = (t, X_t, W_t)$ with $t < T$, let

$$\mathcal{U}(z) = \{Q^\sigma(\cdot \mid z, a) : \sigma \in \Theta, a \in \mathcal{A}\}$$

denote the set of admissible next-state transition kernels at z . Conditional on z and the action a , the next state

$$Z_{t+1} = (t+1, (X_t, S_{t+1}), W_t + a(S_{t+1} - S_t))$$

is a deterministic function of S_{t+1} , whose conditional law is

$$\frac{1}{2}\delta_{S_t+\mu+\sigma} + \frac{1}{2}\delta_{S_t+\mu-\sigma}.$$

The set $\mathcal{U}(z)$ is therefore indexed by $\sigma \in \Theta$ alone: the local ambiguity is one-dimensional and given by the choice of volatility level. Crucially, the same parameter σ indexes $\mathcal{U}(z)$ at *every* state z , since σ is constant in time under any $P^\sigma \in \mathcal{P}^{\text{nr}}$.

Suppose now, for the sake of contradiction, that $\mathcal{Q}^{\text{nr}}$ were s -rectangular, i.e.,

$$\mathcal{Q}^{\text{nr}} = \prod_{z \in \mathcal{Z}} \mathcal{Q}_z^{\text{nr}}, \quad \mathcal{Q}_z^{\text{nr}} \subseteq \prod_{a \in \mathcal{A}} \Delta(\mathcal{Z}).$$

Then the choice of admissible kernel at one state would be independent of the choice of admissible kernel at any other state. In particular, for two distinct states z_1 and z_2 representing different histories, rectangularity would allow the simultaneous choice

$$\sigma(z_1) = \bar{\sigma}, \quad \sigma(z_2) = \underline{\sigma}, \quad \underline{\sigma} < \bar{\sigma}.$$

However, by construction of $\mathcal{Q}^{\text{nr}}$ every admissible kernel set is generated by a single parameter $\sigma \in \Theta$ which remains constant across states

$$\mathcal{Q}^{\text{nr}} = \{Q^\sigma : \sigma \in \Theta\}$$

Hence the kernel used at z_1 uniquely determines the kernel used at every other state, so choosing volatility $\bar{\sigma}$ forces the same choice at z_2 . Therefore, the mixed selection above is not admissible and $\mathcal{Q}^{\text{nr}}$ is non-rectangular. The rectangular hull $\mathcal{P}^{\text{rec}}$ of $\mathcal{P}^{\text{nr}}$ corresponds, in this example, to allowing the volatility to be reselected as a measurable function of the history,

$$S_{t+1} = S_t + \mu + \sigma_t(X_t) \varepsilon_{t+1}, \quad \sigma_t(X_t) \in \Theta,$$

which is exactly the discrete-time analog of the uncertain volatility model used by Avellaneda et al. (1995). The induced MDP ambiguity set $\mathcal{Q}^{\text{rec}}$ then admits the product structure

$$\mathcal{Q}_z^{\text{rec}} = \{Q^\sigma(\cdot \mid z, \cdot) : \sigma \in \Theta\}, \quad \mathcal{Q}^{\text{rec}} = \prod_{z \in \mathcal{Z}} \mathcal{Q}_z^{\text{rec}},$$

since under $\mathcal{P}^{\text{rec}}$ the volatility may be chosen independently at each state. This lifted ambiguity set is s -rectangular, and not (s, a) -rectangular, since the volatility does not depend on the action. The robust MDP under $\mathcal{Q}^{\text{rec}}$ therefore admits the standard robust Bellman recursion

$$V_t(X_t, W_t) = \inf_{H \in \mathbb{R}} \sup_{\sigma \in \Theta} \mathbb{E}^\sigma[V_{t+1}(X_{t+1}, W_{t+1}) \mid X_t, W_t],$$

which is precisely the dynamic programming recursion used in the LP benchmark of Example 1. By contrast, the non-rectangular MDP under $\mathcal{Q}^{\text{nr}}$ does not admit such a recursion, motivating the use of the actor-critic algorithm.

Example 4 (Nonparametric ambiguity model via Wasserstein ball). We now specialize the general robust MDP formulation to the nonparametric Wasserstein ambiguity model of Example 2. As in the sigma interval case, we show that the non-rectangularity of the underlying ambiguity set $\mathcal{P}^W(\rho)$ is preserved when lifting the problem to the history-augmented state space. Recall that the ambiguity set is the Wasserstein ball

$$\mathcal{P}^W(\rho) = \left\{ Q \in \mathcal{P}(\Omega) : W_{d_\Omega}(Q, \hat{P}_n) \leq \rho \right\}$$

centered at the empirical measure $\hat{P}_n = \frac{1}{n} \sum_{i=1}^n \delta_{\omega^{(i)}}$ derived from n observed sample paths, $\omega^{(i)} = (s_0^{(i)}, \dots, s_T^{(i)})$. For numerical tractability we work with the finite-support version

$$\mathcal{P}_\rho^n = \{P^\theta : \theta \in \Theta\}, \quad P^\theta = \sum_{i=1}^n \theta_i \delta_{\omega^{(i)}},$$

where $\Theta \subset \Delta_n$ is the set of probability weight vectors θ such that $W_{d_\Omega}(P^\theta, \hat{P}_n) \leq \rho$. We define the augmented state

$$Z_t = (t, X_t, W_t), \quad X_t = (S_0, \dots, S_t),$$

with state space

$$\mathcal{Z} = \{0, \dots, T\} \times \mathcal{X} \times \mathbb{R}, \quad \mathcal{X} = \bigcup_{t=0}^T \mathbb{R}^{t+1}.$$

The action $A_t = H_t \in \mathbb{R}$ is the position in the risky asset over $[t, t+1)$, so $\mathcal{A} = \mathbb{R}$. For each $\theta \in \Theta$, the measure P^θ induces one-step conditional distributions

$$P^\theta(S_{t+1} \in \cdot \mid X_t = x_t) = \frac{\sum_{i: (s_0^{(i)}, \dots, s_t^{(i)}) = x_t} \theta_i \delta_{s_{t+1}^{(i)}}}{\sum_{i: (s_0^{(i)}, \dots, s_t^{(i)}) = x_t} \theta_i},$$

whenever the denominator is strictly positive. The wealth process evolves according to the self-financing dynamics

$$W_{t+1} = W_t + H_t(S_{t+1} - S_t).$$

Together, the conditional law of S_{t+1} and the deterministic wealth update induce a controlled transition kernel

$$Q^\theta(\cdot \mid z, a)$$

on the augmented state space $\mathcal{Z}$. The corresponding ambiguity set of transition kernels is therefore

$$\mathcal{Q}^{\text{nr}} = \{Q^\theta : \theta \in \Theta\}.$$

The ambiguity set $\mathcal{P}^W(\rho)$ is non-rectangular on path space because the Wasserstein constraint

$$W_{d_\Omega}(Q, \hat{P}_n) \leq \rho$$

imposes a single global transport budget coupling the entire trajectory distribution.

We now show that this coupling persists after lifting the problem to the MDP formulation.

Fix a lifted state

$$z = (t, X_t, W_t), \quad t < T,$$

and define the corresponding set of admissible transition kernels

$$\mathcal{U}(z) = \left\{ Q^\theta(\cdot \mid z, a) : \theta \in \Theta, a \in \mathcal{A} \right\}.$$

Conditional on (z, a) , the next state satisfies

$$Z_{t+1} = (t + 1, (X_t, S_{t+1}), W_t + a(S_{t+1} - S_t)),$$

so the law of Z_{t+1} is entirely determined by the conditional distribution of S_{t+1} under the same global weight vector θ . Crucially, the parameter θ cannot be selected independently at different histories. Every local transition kernel must arise from a single global measure

$$P^\theta = \sum_{i=1}^n \theta_i \delta_{\omega(i)}$$

satisfying the shared transport constraint

$$W_{d_\Omega}(P^\theta, \hat{P}_n) \leq \rho.$$

Suppose, for contradiction, that the lifted ambiguity set were s -rectangular:

$$\mathcal{Q}^{\text{nr}} = \prod_{z \in \mathcal{Z}} \mathcal{Q}_z^{\text{nr}}, \quad \mathcal{Q}_z^{\text{nr}} \subseteq \prod_{a \in \mathcal{A}} \Delta(\mathcal{Z}).$$

Then admissible local kernels could be recombined independently across states. In particular, consider two distinct histories

$$z_1 = (t, X_t^{(1)}, W_t^{(1)}), \quad z_2 = (t, X_t^{(2)}, W_t^{(2)}),$$

and choose two weight vectors

$$\theta^{(1)}, \theta^{(2)} \in \Theta$$

such that

- $\theta^{(1)}$ realizes an almost worst-case perturbation at z_1 , using nearly all of the transport budget ρ ;
- $\theta^{(2)}$ realizes an almost worst-case perturbation at z_2 , also using nearly all of the budget.

Rectangularity would allow these local perturbations to be combined into a single global transition law. However, the resulting path measure would require a transport cost exceeding ρ , since the two perturbations consume the Wasserstein budget independently rather than jointly. Hence no single measure in $\mathcal{P}^{\text{W}}(\rho)$ can realize both local perturbations simultaneously. Consequently, the recombined kernel does not belong to $\mathcal{Q}^{\text{nr}}$, contradicting the assumed product structure. Therefore, $\mathcal{Q}^{\text{nr}}$ is non-rectangular.

The rectangular hull of $\mathcal{P}^{\text{W}}(\rho)$ is obtained by retaining all one-step conditional laws attainable under some

$$P \in \mathcal{P}^{\text{W}}(\rho),$$

while removing the global coupling induced by the transport constraint.

For each history X_t , define the set of attainable conditional distributions

$$\mathcal{K}_t^{\text{W}}(X_t) = \left\{ P(S_{t+1} \in \cdot \mid X_t) : P \in \mathcal{P}^{\text{W}}(\rho) \right\}.$$

The rectangularized ambiguity set

$$\mathcal{P}^{\text{W}, \text{rec}}(\rho)$$

consists of all path measures whose one-step kernels belong to $\mathcal{K}_t^{\text{W}}(X_t)$ at every history.

The induced MDP ambiguity set becomes

$$\mathcal{Q}^{\text{rec}} = \prod_{z \in \mathcal{Z}} \mathcal{Q}_z^{\text{rec}},$$

where

$$\mathcal{Q}_z^{\text{rec}} = \left\{ Q^K(\cdot \mid z, \cdot) : K \in \mathcal{K}_t^{\text{W}}(X_t) \right\}.$$

Since the local kernels may now be selected independently at each history, the lifted ambiguity set is s -rectangular. The robust control problem therefore admits the Bellman recursion

$$V_t(X_t, W_t) = \inf_{H \in \mathbb{R}} \sup_{K \in \mathcal{K}_t^{\text{W}}(X_t)} \mathbb{E}_K [V_{t+1}(X_{t+1}, W_{t+1}) \mid X_t, W_t].$$

Rectangularization removes the global transport-budget constraint and allows the adversary to choose the worst-case local perturbation independently at every history. This enlarges the ambiguity set and generally leads to more conservative superhedging prices. By contrast, under the original non-rectangular ambiguity set $\mathcal{Q}^{\text{nr}}$, the adversary must allocate a single finite transport budget across all histories simultaneously. This global coupling prevents Bellman separability and motivates the actor–critic approach developed in the previous subsection.

5 Numerical Experiments

The Python code for the simulations in this section can be found here: [GitHub](#) Sections 3 and 4

introduced two illustrative ambiguity models alongside the theoretical framework. Example 1 considered a parametric volatility-interval model in which the price process is a random walk with unknown but constant volatility $\sigma \in [\underline{\sigma}, \bar{\sigma}]$, and Example 2 considered a nonparametric Wasserstein ball around an empirical measure on the path space.

In this section we report numerical results for the parametric volatility model from Example 1 only. Preliminary experiments with finite-support Wasserstein ambiguity confirmed that the rectangular approximation is considerably more difficult to implement reliably, because the global transport constraint affects both the initial path weights and the conditional transition probabilities after every history. Since a stable and verifiable implementation would require a separate treatment of these conditional likelihood effects, the numerical section focuses on the parametric volatility model, for which the non-rectangular LP and rectangular DP can be computed and validated directly.

Although the sigma interval model is stylized, it captures a fundamental form of model uncertainty: the volatility level is unknown, but fixed over the trading horizon. This is economically natural when volatility is viewed as a latent structural parameter estimated from data. The non-rectangular ambiguity set reflects this interpretation by requiring one common volatility parameter across all histories. Its rectangular hull, by contrast, allows the adversary to choose a different volatility level at each node of the tree. The resulting price gap therefore measures the additional conservativeness introduced solely by rectangularization, rather than by changes in the underlying ambiguity specification.

We simulate robust superhedging prices for two different payoff classes in order to examine how payoff regularity affects the rectangularity gap. The first is a digital call payoff, which is discontinuous at the strike, while the second is a European call payoff, which is continuous and 1-Lipschitz in the terminal underlying price S_T . The payoff regularity proves to be an important factor in determining the asymptotic behavior of the rectangularity gap both as the horizon T

grows and as the ambiguity radius shrinks to zero.

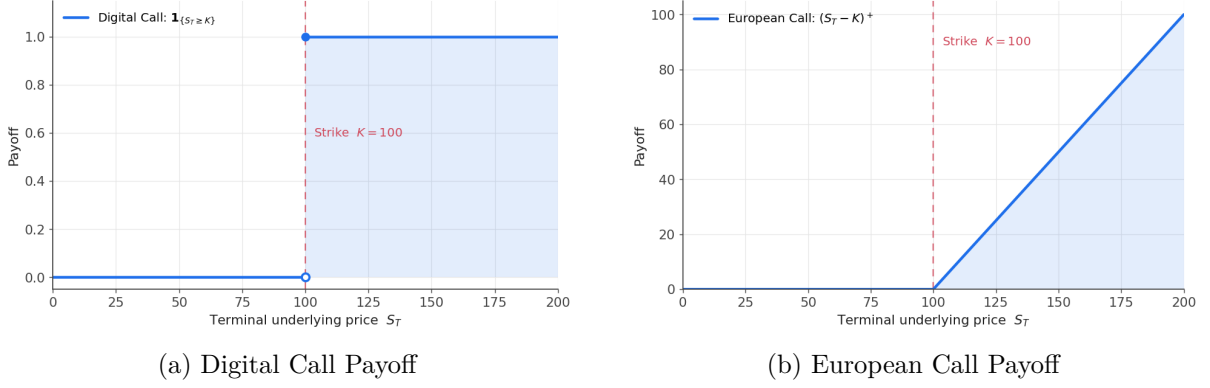


Figure 1: Option payoff functions.

Figure 2 shows how the classical (non-robust) price of each option behaves as the horizon increases, to give a baseline for the horizon comparisons in the robust setting. Longer horizons lead to higher prices in both cases, since more timesteps allow paths to reach higher terminal values, but the two payoffs differ in their long-run behavior: the European call price keeps growing with T , while the digital call price saturates at 1.

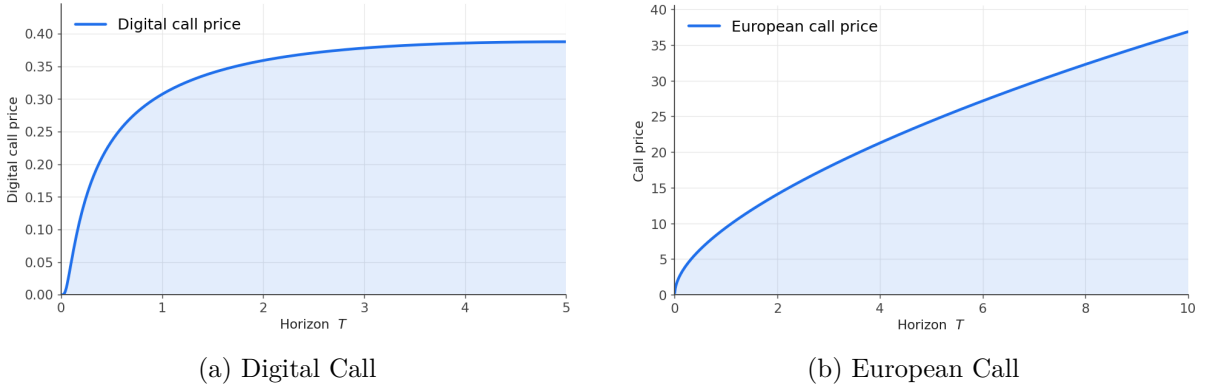


Figure 2: Option price vs. horizon.

5.1 Parametric ambiguity model with unknown volatility

We now compare the robust superhedging price under the non-rectangular sigma-interval ambiguity set $\mathcal{P}^{nr} = \{P_\sigma : \sigma \in \Theta\}$ with $\Theta = [\underline{\sigma}, \bar{\sigma}]$ against its rectangular hull $\mathcal{P}^{rec}$, in which the volatility may be reselected as a measurable function of the history at each node of the binary tree. While robust optimization is already known to be conservative due to its worst-case nature, these results show that rectangularity introduces an additional layer of conservativeness by enlarging the adversary's feasible model class beyond the original globally consistent ambiguity set. We computed the rectangular price using the exact Bellman recursion

$$V_T(s) = \Phi(s),$$

$$V_t(s) = \inf_H \sup_{\sigma \in \Theta} \mathbb{E}_{P^\sigma}[V_{t+1}(s + \mu + \sigma\varepsilon) - H(\mu + \sigma\varepsilon)],$$

and the rectangular robust superhedging price for a payoff Φ is then given by

$$\pi^{rec}(\Phi) = V_0(S_0),$$

where the terminal condition is given by

- $\max(S_T - K, 0)$ for a European call option;
- $\mathbf{1}_{\{S_T \geq K\}}$ for a digital call option.

The non-rectangular price is computed in two complementary ways. First, the discretized linear program of Example 1, sampling $M = 45$ volatility levels from Θ and imposing the superhedging constraint pathwise on the 2^T sign sequences:

$$\begin{aligned} \pi_{LP}^{nr} &= \min_{x, H} x \\ \text{s.t. } x + \sum_{t=0}^{T-1} H_t(\varepsilon_1, \dots, \varepsilon_t) \Delta S_{t+1}^{m, \varepsilon} &\geq \Phi(S_T^{m, \varepsilon}), \quad \forall m = 1, \dots, M, \quad \forall \varepsilon \in \{-1, +1\}^T. \end{aligned}$$

Second, we compute the price π_{AC}^{nr} using the actor–critic algorithm of Section 4, which scales beyond the $M \cdot 2^T$ constraint regime of the LP and is our primary tool for larger T . Since the actor–critic algorithm is a stochastic policy-gradient method, the value $\hat{\pi}^{(k)} = V_{\pi^{(k)}}^{P_{\theta^{(k)}}}(Z_0)$ reported at iteration k is subject to two sources of fluctuation: the stochasticity of the gradient estimates used in the actor update, and the inexactness of the critic’s solution to the inner robust policy-evaluation problem at each iteration. As a consequence, the raw per-iteration prices oscillate around the underlying trend even after the actor has effectively converged to a near-stationary policy. To report a more stable estimate of the converged price, we compute a trailing moving average

$$\bar{\pi}^{(k)} = \frac{1}{w} \sum_{j=0}^{w-1} \hat{\pi}^{(k-j)}, \quad k = w - 1, \dots, K,$$

where the window size w is set to 10 iterations and K is the total number of actor–critic iterations.

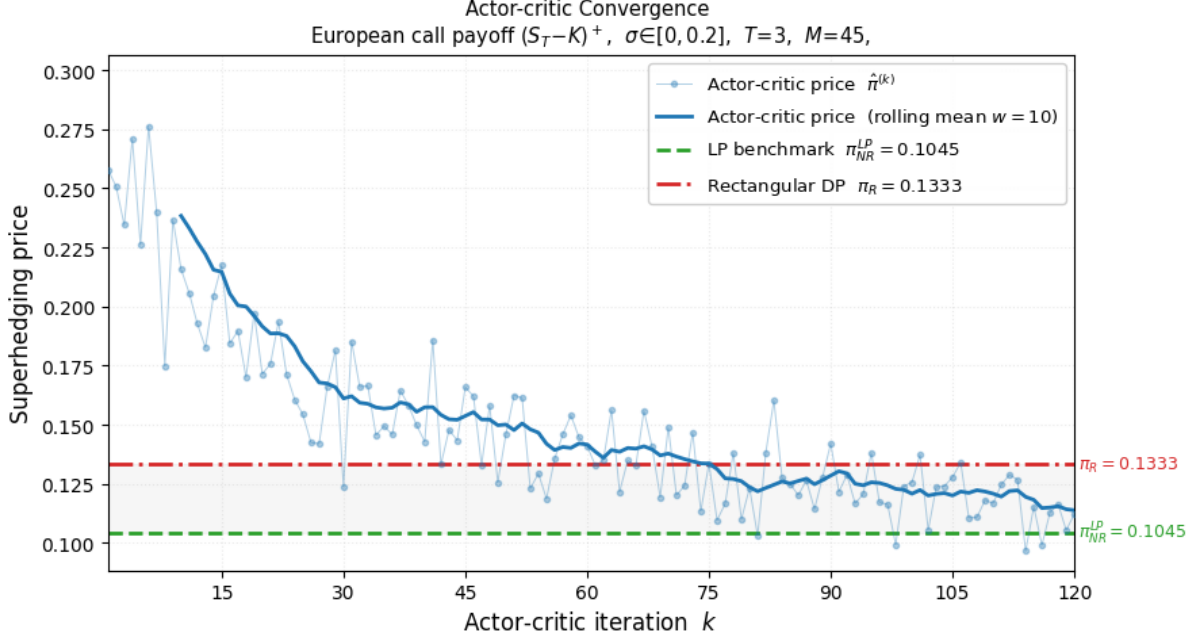


Figure 3: actor-critic convergence European call

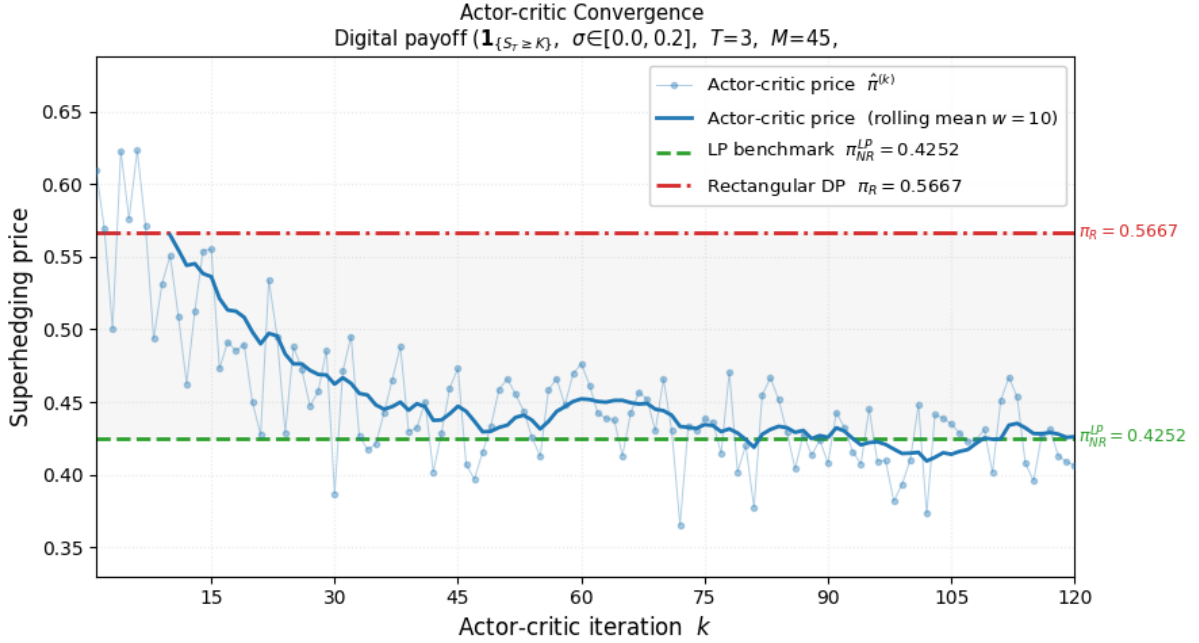


Figure 4: actor-critic convergence digital call

Figures 3 and 4 report the actor-critic iteration history at $T = 3$ compared against the rectangular price obtained from the DP and the non-rectangular price obtained from the discretized ambiguity set LP. The moving average $\bar{\pi}^{(k)}$ descends from its initial value and stabilizes in a neighborhood of the LP benchmark π_{NR}^{LP} over the second half of the run, remaining strictly below the rectangular DP price π_R throughout.

5.2 Numerical setup

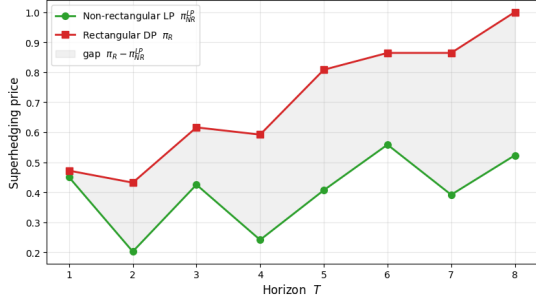
Table 1 collects the parameters used in the experiments below. All experiments use `numpy` random seed 0. The number of variables M in the discretized ambiguity set is a factor in the size of the discretized non-rectangular LP, which has $M \cdot 2^T$ pathwise constraints. Since M scales the LP linearly and T scales it exponentially, a larger M becomes costly when combined with larger horizons. We chose $M = 45$ as a compromise since it is fine enough that the LP price is stable to further refinement of the volatility grid, and coarse enough that the LP remains solvable across the horizon range. The action and wealth grids for the MDP are chosen to be wide enough that the hedge does not bind at the boundary in any of our runs and the wealth process is fine enough to resolve the optimal hedge to within the precision needed to track the LP benchmark. The actor–critic iteration count $K = 120$ is supported by Figures 3 and 4, which show the moving average stabilizes well before the run terminates. We used a grid search to select a large enough penalty parameter λ so that further increases did not change the converged actor–critic price by more than a negligible amount.

Group	Parameter	Value
Market	Initial price S_0	1.0
	Strike K	1.1
	Drift μ	0.02
	Horizon T (default)	3
Ambiguity set	Volatility interval $[\underline{\sigma}, \bar{\sigma}]$	$[0, 0.2]$
	LP discretization size M	45
MDP discretization	Action grid $\mathcal{A}$	13 points on $[-2, 2]$
	Wealth grid $\mathcal{W}$	31 points on $[-1, 2]$
Actor–critic	Iterations K	120
	Actor step size η	0.05

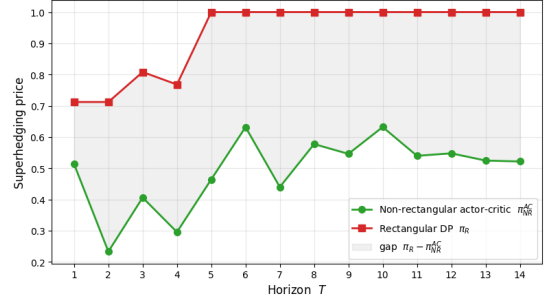
Table 1: Parameters used in the numerical experiments

Note that none of the three prices π^{rec} , π_{LP}^{nr} and π_{AC}^{nr} are the exact robust superhedging price under the continuous sigma-interval model. The π_{LP}^{nr} is exact for the discretized ambiguity set $\Theta_M = \{\sigma^1, \dots, \sigma^M\}$ with $M = 45$, but is only an approximation to the price under the continuous interval $[\underline{\sigma}, \bar{\sigma}]$, and we expect convergence as M grows. The rectangular price π^{rec} uses backward induction on an exact price lattice, but the inner minimization over actions is still computed on a finite grid. The actor–critic non-rectangular price π_{AC}^{nr} optimizes over a discretized state, wealth, and action grid using Monte Carlo estimates of the critic’s value function and output, and is reported as a moving average to dampen the resulting stochastic oscillations.

5.3 Digital call option



(a) Robust superhedging prices vs. horizon, digital call, $\sigma \in [0, 0.2]$, $M = 45$



(b) actor-critic price vs. horizon, digital call, $\sigma \in [0, 0.2]$

Figure 5: Digital call

Figure 5 shows that the rectangular price rises rapidly with the horizon and saturates at the maximum payoff value of 1 from around $T = 5$, while the non-rectangular prices saturate at a significantly lower level. Under the rectangular hull, the volatility may be reselected at every node of the binary tree as a function of the observed history, so the worst-case price process can be steered to keep S_T just below the strike K along each branch. As T grows, this flexibility forces the worst-case expectation of the indicator payoff arbitrarily close to 1. Under the non-rectangular set, the volatility is a single constant $\sigma \in [\underline{\sigma}, \bar{\sigma}]$ fixed throughout the horizon, so the non-rectangular price saturates at the worst-case probability that $\mathbf{1}_{\{S_T \geq K\}}$ over a single σ , which remains strictly below 1 for the parameters considered since no single volatility level can push every path's terminal value across the strike simultaneously.

The saturation of both prices is expected, and is already visible in the classical non-robust pricing of the digital call shown in Figure 2a. The same mechanism operates in the robust setting under both ambiguity specifications, with the rectangular price saturating at 1 and the non-rectangular price at a strictly lower level, so the rectangularity gap itself saturates as T grows. This is particularly visible in Figure 5b, where the actor-critic extension up to $T = 14$ shows the gap stabilizing once the rectangular price hits its ceiling.

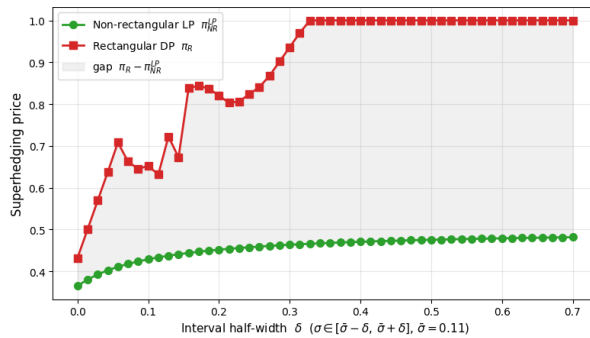
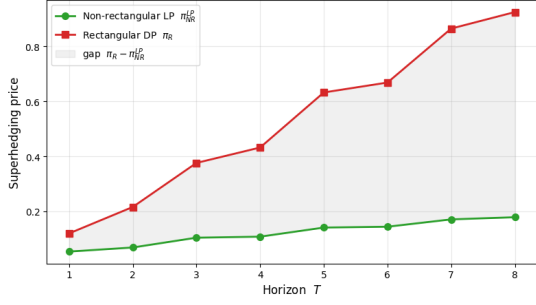


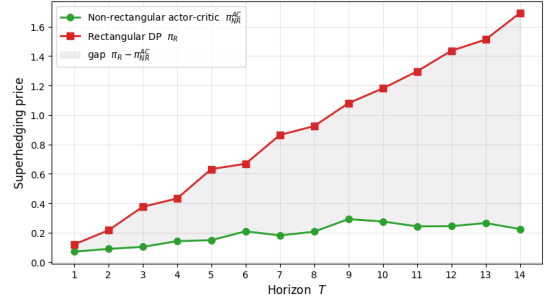
Figure 6: Robust superhedging prices vs. interval half-width, digital call, $T = 3$, $M = 45$

Figure 6 shows the dependence on the ambiguity radius for a fixed horizon. The gap grows with the size of the interval until the rectangular price saturates at 1, and the non-rectangular price increases and saturates at a much lower level. This is the radius analog of the horizon behavior in Figure 5: the bounded payoff caps the gap, and rectangularity simply reaches the cap sooner.

5.4 European call option



(a) Robust superhedging prices vs. horizon, European call, $\sigma \in [0.0, 0.2]$, $M = 45$



(b) actor-critic price vs. horizon, digital call, $\sigma \in [0, 0.2]$

Figure 7: European call

The European call payoff $\max(S_T - K, 0)$ is 1-Lipschitz in the terminal price and unbounded, and the resulting rectangularity gap behaves qualitatively differently from the digital case. The growth of both prices with horizon is expected, and is already visible in the classical non-robust pricing of the European call shown in Figure 2b. Since the payoff is unbounded, the price under any single measure grows without bound in T . Figure 7 shows that both the rectangular and non-rectangular price grow approximately linearly with T , although at different rates, which leads to a widening gap as horizon increases. Unlike in the case of the digital call, the European call payoff is unbounded, neither price has a ceiling to saturate at, and the stagewise advantage that rectangularity grants the adversary appears to compound across time steps rather than dissipating. The actor-critic extension in Figure 7b extends the comparison up to $T = 14$ and shows the same qualitative pattern persisting well beyond the range where the LP is feasible.

For the digital call, the payoff is bounded above by 1, so even a maximally flexible local adversary can only push the worst-case expected payoff up to that ceiling. Rectangularization reaches this ceiling quickly, around $T = 5$ in Figure 5, after which the gap saturates because there is no further damage the local adversary can do. The economic implication is that rectangularization is most damaging for Lipschitz claims with unbounded growth in the terminal price, of which the European call is a typical example. For bounded payoffs such as digital calls, rectangularization remains wasteful but the damage is capped by the payoff itself.

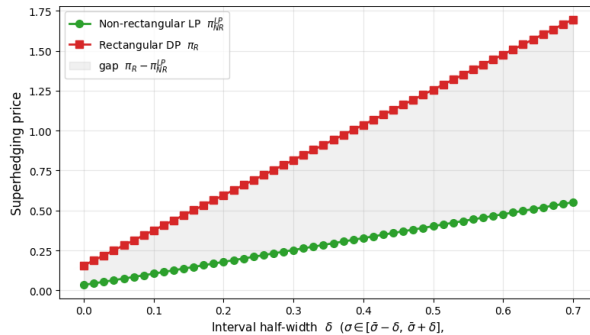


Figure 8: Robust superhedging prices vs. interval half-width, European call, $T = 3$, $M = 45$

Figure 8 examines the dependence on the ambiguity radius for a fixed horizon. Both prices increase monotonically with the size of the interval, but the rectangular price increases markedly faster, and the resulting gap grows steeply in the radius over the full range considered. Unlike the

digital case in Figure 6, where the gap eventually flattens as the rectangular price saturates at 1, here the gap continues to grow across the full range of radii considered because the European call payoff is unbounded. As the agent widens its ambiguity set, the rectangular price diverges from the non-rectangular price at a roughly constant rate per unit of radius, while total robust capital is itself growing in ρ . Rectangularization therefore becomes a larger fraction of total robust capital as the agent becomes more cautious, not a smaller one. The more conservative agent pays twice for caution, once for the ambiguity it is actually hedging and again for the additional premium due to the rectangularization.

The evidence from these numerical experiments points to the size of the rectangularity gap being large enough to matter for pricing under this model. Figure 7 shows that the rectangular price exceeds the non-rectangular price by a factor that grows with the horizon, reaching roughly three times the non-rectangular price at $T = 14$ in Figure 7b, and Figure 8 shows the gap widening monotonically in the ambiguity radius at fixed horizon. Although both π_{LP}^{nr} and π_{AC}^{nr} are approximations rather than the exact non-rectangular price under the continuous sigma-interval, the magnitudes involved are larger than what could be purely explained by discretization or Monte Carlo error. As a consequence of pricing under the rectangular hull model, a seller would be charging a premium that is unlikely to be sustainable, as a competitor pricing under the non-rectangular set could undercut the rectangular price by a wide margin while still hedging the claim against every model in the agent's actual ambiguity set. The gap therefore quantifies the cost of using rectangularity purely as a computational simplification rather than as a description of the agent's beliefs, and the experiments suggest that this cost is far from negligible for unbounded Lipschitz payoffs over moderate to long horizons.

Overall, the numerical experiments suggest that assuming rectangularity of the ambiguity set can lead to a substantial overestimation of robust superhedging prices. In the case of European call options, the effect appears to become increasingly severe with increasing horizon and increasing radius of the ambiguity set. The behavior of the rectangularity gap appears to be dependent on the payoff. For the bounded digital payoff, the gap increases initially but eventually saturates because the payoff itself is bounded above. In order to validate these findings, we propose some theoretical results on the rectangularity gap in the parametric ambiguity set setting in the following section.

6 Theoretical Properties of the Rectangularity Gap

The numerical experiments of Section 5 suggest two asymptotic features of the rectangularity gap in the parametric volatility model: the gap vanishes as the interval half-width ρ shrinks to zero at a fixed horizon, and it may grow as the horizon T increases at fixed $\rho > 0$. In this section, we establish both statements analytically within the volatility interval model under certain conditions.

6.1 Consistency

We now show that in the specific case of the sigma-interval volatility, for a fixed horizon as ρ shrinks to zero, so does the rectangularity gap. We then follow up with a more general result which requires stronger assumptions. Let

$$S_{t+1} = S_t + \mu + \sigma_t \varepsilon_{t+1}, \quad t = 0, \dots, T-1,$$

where $\mu \in \mathbb{R}$ is fixed and $(\varepsilon_t)_{t=0}^{T-1}$ are i.i.d. random variables such that

$$P(\varepsilon_t = 1) = P(\varepsilon_t = -1) = \frac{1}{2}, \quad t = 0, \dots, T-1.$$

The volatility ambiguity set we consider is the interval with half-width ρ around a reference volatility σ_0 : $\Theta = [\sigma_0 - \rho, \sigma_0 + \rho]$. For every $\sigma \in \Theta$, let P^σ denote the probability measure corresponding to the model in which the volatility is fixed and equal to σ at every time step. Define the non-rectangular ambiguity set

$$\mathcal{P}^{\text{nr}} = \{P^\sigma : \sigma \in \Theta\},$$

and the rectangular ambiguity set

$$\mathcal{P}^{\text{rec}} = \left\{P^\sigma : \sigma = (\sigma_t)_{t=0}^{T-1} \text{ predictable with } \sigma_t \in \Theta\right\}.$$

Define the path metric

$$d_\Omega(\omega, \tilde{\omega}) = \sum_{t=0}^T \alpha_t |S_t(\omega) - S_t(\tilde{\omega})|, \quad \alpha_t \geq 0, \quad (9)$$

and define

$$C_T = \sum_{t=1}^T \alpha_t t.$$

The metric d_Ω is a weighted L^1 -distance on path space. It measures the distance between two price paths by comparing the stock prices at each time and summing the weighted absolute differences. This is a natural choice for path-dependent claims, since it captures how discrepancies between two trajectories accumulate over the entire time horizon. The constant C_T will be used to control the total accumulation of these pathwise errors over time.

The European call payoff $\Phi(\omega) = (S_T - K)^+$ is 1-Lipschitz with respect to d_Ω whenever $\alpha_T \geq 1$, since $|\Phi(\omega) - \Phi(\omega')| \leq |S_T(\omega) - S_T(\omega')|$. The simplest choice $\alpha_T = 1$, $\alpha_t = 0$ for $t < T$ then gives $C_T = T$, so the bound below specializes to a constant times $T \cdot \rho$. Path-dependent Lipschitz claims are accommodated by choosing the weights α_t to match the payoff's pathwise sensitivity. Bounded but discontinuous payoffs such as the digital call are not covered by Proposition 5.

We assume the following throughout:

(A1) The horizon $T < \infty$ is fixed;

(A2) The no-arbitrage condition

$$|\mu| < \sigma_0 - \rho$$

holds;

(A3) Robust superhedging duality (Theorem 3.3) holds for both $\mathcal{P}^{\text{nr}}$ and $\mathcal{P}_\rho^{\text{rec}}$;

(A4) The payoff ξ is measurable and integrable under all martingale measures in the relevant dual domains.

Proposition 5 (Consistency of the rectangularity gap as $\rho \downarrow 0$). *If the claim payoff ξ is Lipschitz with constant L_ξ with respect to d_Ω as defined in (9), then for every finite horizon T ,*

$$0 \leq G_{T,\rho}(\xi) \leq L_\xi \cdot \left(1 + \frac{|\mu|}{\sigma_0 - \rho}\right) \cdot C_T \cdot \rho.$$

In particular,

$$\lim_{\rho \downarrow 0} G_{T,\rho}(\xi) = 0.$$

Before stating the result we sketch the strategy of the proof, since the argument that follows is long. In the dual formulation of the robust superhedging problem, the rectangularity gap is a difference between two suprema of expectations over different martingale-measure domains, and there is no direct way to bound such a difference. A standard approach is to introduce a coupling: a single joint distribution Π on the product path space $\Omega \times \Omega$ whose two marginals reproduce the two measures one wants to compare. Once such a coupling is constructed, we compare a reference path $(S_t^0)_{t=0}^T$ constructed from a measure Q^0 in the non-rectangular dual domain (the one generated by the volatility σ^0), and a path $(S_t)_{t=0}^T$ constructed from an arbitrary measure in the rectangular dual domain. The difference $E_Q[\Phi] - E_{Q^0}[\Phi]$ becomes an expectation under Π of $\Phi(S) - \Phi(S^0)$, which can be controlled using the Lipschitz property of Φ and a pathwise bound on the distance between the two coupled processes.

Proof. The lower bound follows from Proposition 4.

Fix $\sigma \in \Theta$ and let Q be a martingale measure absolutely continuous with respect to P^σ . Under P^σ , the increment

$$\Delta S_{t+1} = \mu + \sigma \varepsilon_{t+1}$$

takes the two values

$$\Delta S_{t+1} = \begin{cases} \mu + \sigma, & \varepsilon_{t+1} = +1, \\ \mu - \sigma, & \varepsilon_{t+1} = -1. \end{cases}$$

Write

$$q^+(\sigma) = Q(\varepsilon_{t+1} = +1 \mid \mathcal{F}_t), \quad q^-(\sigma) = Q(\varepsilon_{t+1} = -1 \mid \mathcal{F}_t).$$

Since Q is a martingale measure,

$$\mathbb{E}_Q[\Delta S_{t+1} \mid \mathcal{F}_t] = 0,$$

and therefore

$$q^+(\sigma)(\mu + \sigma) + q^-(\sigma)(\mu - \sigma) = 0.$$

Together with

$$q^+(\sigma) + q^-(\sigma) = 1,$$

this gives the unique solution

$$q^+(\sigma) = \frac{\sigma - \mu}{2\sigma}, \quad q^-(\sigma) = \frac{\sigma + \mu}{2\sigma}.$$

The no-arbitrage assumption (A2) $|\mu| < \sigma_0 - \rho \leq \sigma$ implies $q^\pm(\sigma) \in (0, 1)$, so P^σ admits a unique equivalent martingale measure denoted Q^σ . The same argument applies conditionally at each node in the rectangular model. Thus every $Q \in \mathcal{Q}_\rho^{\text{rec}}$ is parametrized by a predictable volatility process

$$(\sigma_t)_{t=0}^{T-1}, \quad \sigma_t \in \Theta,$$

with conditional probabilities

$$q_t^+ = \frac{\sigma_t - \mu}{2\sigma_t}, \quad q_t^- = \frac{\sigma_t + \mu}{2\sigma_t}.$$

Let $Q_0 = Q^{\sigma_0}$ denote the martingale measure corresponding to the constant-volatility model

with volatility σ_0 . Its conditional probabilities are

$$q_0^+ = \frac{\sigma_0 - \mu}{2\sigma_0}, \quad q_0^- = \frac{\sigma_0 + \mu}{2\sigma_0}.$$

Since $\sigma_0 \in \Theta$ for every $\rho < \sigma_0$, we have that the reference measure is in the non-rectangular ambiguity set

$$P^{\sigma_0} \in \mathcal{P}^{\text{nr}},$$

and hence

$$Q_0 \in \mathcal{Q}_\rho^{\text{nr}}.$$

By the robust superhedging duality assumption A3,

$$\pi^{\text{nr}}(\xi) = \inf \{x \in \mathbb{R} : \exists H \in \mathcal{H} \text{ such that } x + (H \cdot S)_T \geq \xi \quad \mathcal{P}^{\text{nr}}\text{-q.s.}\}.$$

is equivalent to

$$\pi^{\text{nr}}(\xi) = \sup_{Q \in \mathcal{Q}_\rho^{\text{nr}}} \mathbb{E}_Q[\xi] \geq \mathbb{E}_{Q_0}[\xi],$$

where the last inequality holds because $Q_0 \in \mathcal{Q}_\rho^{\text{nr}}$.

We now construct a joint probability distribution Π on the product path space $\Omega \times \Omega$. The price process corresponding to an arbitrary measure $Q \in \mathcal{Q}_\rho^{\text{rec}}$ is

$$S = (S_t)_{t=0}^T.$$

The price process corresponding to the reference measure $Q_0 \in \mathcal{Q}_\rho^{\text{nr}}$ is

$$S^0 = (S_t^0)_{t=0}^T.$$

We choose Π so that its first marginal is Q and its second marginal is Q_0 . In other words,

$$\Pi(S \in A) = Q(A), \quad \Pi(S^0 \in B) = Q_0(B),$$

for all measurable sets $A, B \subseteq \Omega$. The distribution Π is constructed step by step. Define the joint filtration

$$\mathcal{F}_t^\Pi = \sigma(S_0, \dots, S_t, S_0^0, \dots, S_t^0),$$

which contains the information generated by both paths up to time t . Suppose the joint distribution has already been constructed up to time t . We now specify the conditional joint distribution of the next signs

$$(\varepsilon_{t+1}, \varepsilon_{t+1}^0) \in \{-1, +1\}^2$$

given $\mathcal{F}_t^\Pi$. The first sign must have the conditional distribution required by the arbitrary rectangular model

$$\Pi(\varepsilon_{t+1} = +1 \mid \mathcal{F}_t^\Pi) = q_t^+.$$

The second sign must have the conditional distribution required by the reference model

$$\Pi(\varepsilon_{t+1}^0 = +1 \mid \mathcal{F}_t^\Pi) = q_0^+.$$

Among all joint distributions with these two marginals, we choose one for which the two signs disagree with the smallest possible conditional probability. For Bernoulli random variables, this minimum disagreement probability is $|q_t^+ - q_0^+|$. Thus we choose the conditional joint distribution

so that

$$\Pi(\varepsilon_{t+1} \neq \varepsilon_{t+1}^0 \mid \mathcal{F}_t^\Pi) = |q_t^+ - q_0^+|. \quad (10)$$

Equivalently, the conditional joint distribution may be written explicitly as follows:

$$\begin{aligned} \Pi(\varepsilon_{t+1} = +1, \varepsilon_{t+1}^0 = +1 \mid \mathcal{F}_t^\Pi) &= \min(q_t^+, q_0^+), \\ \Pi(\varepsilon_{t+1} = +1, \varepsilon_{t+1}^0 = -1 \mid \mathcal{F}_t^\Pi) &= (q_t^+ - q_0^+)^+, \\ \Pi(\varepsilon_{t+1} = -1, \varepsilon_{t+1}^0 = +1 \mid \mathcal{F}_t^\Pi) &= (q_0^+ - q_t^+)^+, \end{aligned}$$

and

$$\Pi(\varepsilon_{t+1} = -1, \varepsilon_{t+1}^0 = -1 \mid \mathcal{F}_t^\Pi) = \min(q_t^-, q_0^-).$$

This conditional joint distribution has the correct first marginal $q_t^\pm$, the correct second marginal $q_0^\pm$, and satisfies

$$\Pi(\varepsilon_{t+1} \neq \varepsilon_{t+1}^0 \mid \mathcal{F}_t^\Pi) = |q_t^+ - q_0^+|.$$

The minimum-disagreement coupling is the natural choice for obtaining a tighter bound since under any other coupling the two paths would diverge faster. The next increments are then defined by

$$\Delta S_{t+1} = \mu + \sigma_t \varepsilon_{t+1},$$

and

$$\Delta S_{t+1}^0 = \mu + \sigma_0 \varepsilon_{t+1}^0.$$

Thus

$$S_{t+1} = S_t + \Delta S_{t+1}, \quad S_{t+1}^0 = S_t^0 + \Delta S_{t+1}^0.$$

Repeating this construction for $t = 0, \dots, T-1$ defines a joint probability distribution Π on $\Omega \times \Omega$. By construction, the first marginal law of S is Q , while the second marginal law of S^0 is Q_0 .

On the coupled space $\Omega \times \Omega$, the drift terms in the two increments cancel

$$\Delta S_{t+1} - \Delta S_{t+1}^0 = (\mu + \sigma_t \varepsilon_{t+1}) - (\mu + \sigma_0 \varepsilon_{t+1}^0) = \sigma_t \varepsilon_{t+1} - \sigma_0 \varepsilon_{t+1}^0.$$

Rewrite this as

$$\Delta S_{t+1} - \Delta S_{t+1}^0 = (\sigma_t - \sigma_0) \varepsilon_{t+1} + \sigma_0 (\varepsilon_{t+1} - \varepsilon_{t+1}^0).$$

Since $|\varepsilon_{t+1}| = 1$,

$$\left| \Delta S_{t+1} - \Delta S_{t+1}^0 \right| \leq |\sigma_t - \sigma_0| + \sigma_0 \left| \varepsilon_{t+1} - \varepsilon_{t+1}^0 \right|.$$

Since $\varepsilon_{t+1}, \varepsilon_{t+1}^0 \in \{-1, +1\}$, we have

$$\left| \varepsilon_{t+1} - \varepsilon_{t+1}^0 \right| = 2 \times \mathbf{1}_{\{\varepsilon_{t+1} \neq \varepsilon_{t+1}^0\}}.$$

Taking conditional expectations under Π and applying (10),

$$\mathbb{E}_\Pi \left[\left| \varepsilon_{t+1} - \varepsilon_{t+1}^0 \right| \mid \mathcal{F}_t^\Pi \right] = 2|q_t^+ - q_0^+|.$$

Now compute

$$\begin{aligned}
|q_t^+ - q_0^+| &= \left| \frac{\sigma_t - \mu}{2\sigma_t} - \frac{\sigma_0 - \mu}{2\sigma_0} \right| \\
&= \left| \frac{\mu(\sigma_t - \sigma_0)}{2\sigma_t\sigma_0} \right| \\
&\leq \frac{|\mu|}{2(\sigma_0 - \rho)\sigma_0} |\sigma_t - \sigma_0|,
\end{aligned}$$

where we used $\sigma_t \geq \sigma_0 - \rho > 0$. Substituting this estimate,

$$\mathbb{E}_\Pi \left[\left| \Delta S_{t+1} - \Delta S_{t+1}^0 \right| \middle| \mathcal{F}_t^\Pi \right] \leq \left(1 + \frac{|\mu|}{\sigma_0 - \rho} \right) |\sigma_t - \sigma_0|.$$

Since $|\sigma_t - \sigma_0| \leq \rho$, we obtain

$$\mathbb{E}_\Pi \left[\left| \Delta S_{t+1} - \Delta S_{t+1}^0 \right| \middle| \mathcal{F}_t^\Pi \right] \leq K_{\mu,\rho} \rho,$$

where

$$K_{\mu,\rho} = 1 + \frac{|\mu|}{\sigma_0 - \rho}.$$

Since the two processes have the same initial value $S_0 = S_0^0$,

$$S_t - S_t^0 = \sum_{u=0}^{t-1} \left(\Delta S_{u+1} - \Delta S_{u+1}^0 \right),$$

and therefore

$$|S_t - S_t^0| = \left| \sum_{u=0}^{t-1} (\Delta S_{u+1} - \Delta S_{u+1}^0) \right| \leq \sum_{u=0}^{t-1} |\Delta S_{u+1} - \Delta S_{u+1}^0|,$$

where the inequality follows from the triangle inequality. Taking Π -expectations and applying the law of iterated expectations¹ together with the conditional bound above,

$$\mathbb{E}_\Pi |S_t - S_t^0| \leq t K_{\mu,\rho} \rho.$$

We now use this pointwise-in- t bound to control the expected path metric $\mathbb{E}_\Pi[d_\Omega(S, S^0)]$. Recall that the path metric is defined as a weighted sum of pointwise discrepancies

$$d_\Omega(\omega, \tilde{\omega}) = \sum_{t=0}^T \alpha_t |S_t(\omega) - S_t(\tilde{\omega})|, \quad \alpha_t \geq 0.$$

Applied to the coupled processes (S, S^0) ,

$$d_\Omega(S, S^0) = \sum_{t=0}^T \alpha_t |S_t - S_t^0|.$$

Noting that the $t = 0$ term in the sum vanishes because both processes have the same initial value, and taking Π -expectations

$$\mathbb{E}_\Pi[d_\Omega(S, S^0)] = \sum_{t=0}^T \alpha_t \mathbb{E}_\Pi |S_t - S_t^0| \leq K_{\mu,\rho} \rho \sum_{t=1}^T \alpha_t t = K_{\mu,\rho} C_T \rho.$$

¹for any integrable X and sub- σ -algebra $\mathcal{G} \subseteq \mathcal{F}$, $\mathbb{E}[\mathbb{E}[X|\mathcal{G}]] = \mathbb{E}[X]$

Since ξ is L_ξ -Lipschitz with respect to d_Ω ,

$$|\xi(S) - \xi(S^0)| \leq L_\xi d_\Omega(S, S^0) \quad \Pi\text{-a.s.}$$

Taking Π -expectations and using the marginal property of Π , gives

$$\mathbb{E}_\Pi[\xi(S)] = \mathbb{E}_Q[\xi], \quad \mathbb{E}_\Pi[\xi(S^0)] = \mathbb{E}_{Q_0}[\xi],$$

since the expectation of a function of just one of the two coupled processes depends only on that process's marginal law and not on how the two are coupled, we obtain

$$|\mathbb{E}_Q[\xi] - \mathbb{E}_{Q_0}[\xi]| = |\mathbb{E}_\Pi[\xi(S) - \xi(S^0)]| \leq L_\xi \mathbb{E}_\Pi[d_\Omega(S, S^0)] \leq L_\xi K_{\mu,\rho} C_T \rho.$$

This step is where the coupling Π connects the joint distribution of (S, S^0) on the product space, and the difference of pricing expectations $\mathbb{E}_Q[\xi] - \mathbb{E}_{Q_0}[\xi]$ through its marginal property.

Since $Q \in \mathcal{Q}_\rho^{\text{rec}}$ was arbitrary, we obtained the same upper bound $\mathbb{E}_Q[\xi] \leq \mathbb{E}_{Q_0}[\xi] + L_\xi K_{\mu,\rho} C_T \rho$ for every Q in the rectangular pricing domain. Taking the supremum over $Q \in \mathcal{Q}_\rho^{\text{rec}}$ on the left and applying robust superhedging duality on the rectangular side gives

$$\pi_\rho^{\text{rec}}(\xi) = \sup_{Q \in \mathcal{Q}_\rho^{\text{rec}}} \mathbb{E}_Q[\xi] \leq \mathbb{E}_{Q_0}[\xi] + L_\xi K_{\mu,\rho} C_T \rho.$$

Recall that $\mathbb{E}_{Q_0}[\xi] \leq \pi_\rho^{\text{nr}}(\xi)$, since $Q_0 \in \mathcal{Q}_\rho^{\text{nr}}$ is one feasible measure in the supremum defining $\pi_\rho^{\text{nr}}(\xi)$. Substituting this inequality,

$$\pi_\rho^{\text{rec}}(\xi) \leq \pi_\rho^{\text{nr}}(\xi) + L_\xi K_{\mu,\rho} C_T \rho,$$

which rearranges to $G_{T,\rho}(\xi) = \pi_\rho^{\text{rec}}(\xi) - \pi_\rho^{\text{nr}}(\xi) \leq L_\xi K_{\mu,\rho} C_T \rho$. Combined with the lower bound $G_{T,\rho}(\xi) \geq 0$ from Proposition 4, we conclude

$$0 \leq G_{T,\rho}(\xi) \leq L_\xi K_{\mu,\rho} C_T \rho.$$

Finally, $K_{\mu,\rho} = 1 + |\mu|/(\sigma_0 - \rho)$ is bounded as $\rho \downarrow 0$ (it converges to $1 + |\mu|/\sigma_0$) and C_T is independent of ρ , so the right-hand side vanishes as $\rho \downarrow 0$, and therefore

$$\lim_{\rho \downarrow 0} G_{T,\rho}(\xi) = 0.$$

□

Proposition 5 is specific to the binary-tree volatility-interval model, where the martingale-measure structure is explicit and the coupling can be constructed step by step from Bernoulli sign processes. The same argument has a natural abstract counterpart in which the volatility coupling is replaced by an optimal-transport coupling on path space, and the explicit bound is replaced by a Wasserstein collapse condition on the dual domain. We state this version because it isolates the structural property that drives the consistency result, and provides a template for proving consistency results in other ambiguity models, in particular the Wasserstein ball of Example 2, where an analogous statement is conjectured but not proved in this thesis.

Under stronger assumptions, the consistency property extends to a more general setting. Fix a finite horizon $T \in \mathbb{N}$ and let

$$\Omega = (\mathbb{R}^d)^{T+1}, \quad S_t(\omega) = s_t, \quad \omega = (s_0, \dots, s_T),$$

be the canonical path space equipped with a Polish metric d_Ω and the natural filtration $(\mathcal{F}_t)_{t=0}^T$. For an ambiguity set $\mathcal{P} \subseteq \mathfrak{P}(\Omega)$, write

$$\mathcal{Q}(\mathcal{P}) = \left\{ Q : \exists P \in \mathcal{P} \text{ such that } Q \ll P, S \text{ is a } Q\text{-martingale} \right\},$$

Under the robust no-arbitrage and measurability assumptions of Bouchard and Nutz (2015), robust superhedging duality gives

$$\pi_{\mathcal{P}}(\xi) = \sup_{Q \in \mathcal{Q}(\mathcal{P})} \mathbb{E}_Q[\xi], \quad (11)$$

for every upper semianalytic claim ξ for which the expectation is finite.

For every ambiguity radius $\rho > 0$, let $\mathcal{P}_\rho^{\text{nr}}$ be the original non-rectangular ambiguity set and let $\mathcal{P}_\rho^{\text{rec}}$ be its rectangular hull or canonical rectangular outer approximation. The corresponding robust superhedging prices and rectangularity gap are

$$\pi_\rho^{\text{nr}}(\xi) = \pi_{\mathcal{P}_\rho^{\text{nr}}}(\xi), \quad \pi_\rho^{\text{rec}}(\xi) = \pi_{\mathcal{P}_\rho^{\text{rec}}}(\xi), \quad G_\rho(\xi) = \pi_\rho^{\text{rec}}(\xi) - \pi_\rho^{\text{nr}}(\xi).$$

Since $\mathcal{P}_\rho^{\text{nr}} \subseteq \mathcal{P}_\rho^{\text{rec}}$, monotonicity of the quasi-sure superhedging constraint, equivalently monotonicity of the dual domain, implies $G_\rho(\xi) \geq 0$. Fix $T < \infty$. Let $\mathcal{P}_0$ be a limiting ambiguity set and set

$$\mathcal{Q}_0 = \mathcal{Q}(\mathcal{P}_0), \quad \mathcal{Q}_\rho^{\text{nr}} = \mathcal{Q}(\mathcal{P}_\rho^{\text{nr}}), \quad \mathcal{Q}_\rho^{\text{rec}} = \mathcal{Q}(\mathcal{P}_\rho^{\text{rec}}).$$

Assume the following.

- (B1) Robust superhedging duality (11) holds for $\mathcal{P}_0$, $\mathcal{P}_\rho^{\text{nr}}$, and $\mathcal{P}_\rho^{\text{rec}}$ for all sufficiently small ρ .
- (B2) The limiting pricing domain is contained in the non-rectangular pricing domain and the non-rectangular pricing domain is contained in the rectangular pricing domain:

$$\mathcal{Q}_0 \subseteq \mathcal{Q}_\rho^{\text{nr}} \subseteq \mathcal{Q}_\rho^{\text{rec}}.$$

In particular, this holds if $\mathcal{P}_0 \subseteq \mathcal{P}_\rho^{\text{nr}} \subseteq \mathcal{P}_\rho^{\text{rec}}$.

- (B3) The rectangular pricing domains collapse to the limiting pricing domain in the one-sided Wasserstein sense

$$\Delta_\rho = \sup_{Q \in \mathcal{Q}_\rho^{\text{rec}}} \inf_{Q^0 \in \mathcal{Q}_0} W_1(Q, Q^0) \longrightarrow 0 \quad \text{as } \rho \downarrow 0. \quad (12)$$

Proposition 6 (Fixed-horizon consistency of the rectangularity gap). *If $\xi : \Omega \rightarrow \mathbb{R}$ is bounded and L_ξ -Lipschitz with respect to d_Ω , then*

$$0 \leq G_\rho(\xi) \leq L_\xi \Delta_\rho. \quad (13)$$

Consequently,

$$\lim_{\rho \downarrow 0} G_\rho(\xi) = 0.$$

Proof. The lower bound follows from $\mathcal{Q}_\rho^{\text{nr}} \subseteq \mathcal{Q}_\rho^{\text{rec}}$. Let

$$\pi_0(\xi) = \sup_{Q^0 \in \mathcal{Q}_0} \mathbb{E}_{Q^0}[\xi].$$

By assumption B2, $\pi_0(\xi) \leq \pi_\rho^{\text{nr}}(\xi)$. On the other hand, for any $Q \in \mathcal{Q}_\rho^{\text{rec}}$ and any $Q^0 \in \mathcal{Q}_0$, the

Kantorovich–Rubinstein inequality gives

$$\mathbb{E}_Q[\xi] \leq \mathbb{E}_{Q^0}[\xi] + L_\xi W_1(Q, Q^0).$$

Taking the infimum over $Q^0 \in \mathcal{Q}_0$ and then the supremum over $Q \in \mathcal{Q}_\rho^{\text{rec}}$ yields

$$\pi_\rho^{\text{rec}}(\xi) = \sup_{Q \in \mathcal{Q}_\rho^{\text{rec}}} \mathbb{E}_Q[\xi] \leq \pi_0(\xi) + L_\xi \Delta_\rho.$$

Therefore

$$G_\rho(\xi) = \pi_\rho^{\text{rec}}(\xi) - \pi_\rho^{\text{nr}}(\xi) \leq \pi_0(\xi) + L_\xi \Delta_\rho - \pi_0(\xi) = L_\xi \Delta_\rho,$$

which proves (13). assumption B3 gives the limit. $\square$

6.2 Divergence

The convergence assumption (12) is not merely a technical convenience. Rectangularization is an operation on conditional kernels, and conditional distributions fail to be continuous under weak or Wasserstein convergence in the absence of additional regularity; in particular, kernels after histories of vanishing probability can behave discontinuously. As a result, the intuitive claim that "the ambiguity set collapses to a singleton, so the gap must vanish" does not hold at the required level of generality unless one can also guarantee that the rectangular hull, or at least its martingale pricing domain, collapses to the same limit.

We now exhibit a setting in which the rectangularity gap grows linearly with the horizon. The result highlights a mechanism through which rectangularization can accumulate pricing distortions over time: under the rectangular model, volatility may be reselected locally at each time step, whereas the non-rectangular model enforces one globally consistent volatility parameter.

Fix $T \geq 2$, $\mu \in \mathbb{R}$, and volatility bounds $0 < a < b$. Let $S_0 = 0$, and consider the price process

$$S_{t+1} = S_t + \mu + \sigma_t \varepsilon_{t+1},$$

where $\varepsilon_t \in \{-1, 1\}$.

We consider the volatility-interval model with midpoint $\sigma_0 > 0$ and ambiguity radius $\rho > 0$, so that

$$a = \sigma_0 - \rho, \quad b = \sigma_0 + \rho, \quad 0 < a < b.$$

The non-rectangular ambiguity set is

$$\mathcal{P}_T^{\text{nr}} = \{P_T^\sigma : \sigma \in [a, b]\},$$

where, under P_T^σ ,

$$\Delta S_t = \mu + \sigma \varepsilon_t, \quad P_T^\sigma(\varepsilon_t = 1) = P_T^\sigma(\varepsilon_t = -1) = \frac{1}{2}.$$

Thus the volatility parameter is unknown but fixed over the whole horizon.

The corresponding rectangular model is

$$\mathcal{P}_T^{\text{rec}} = \{P_T^\sigma : \sigma_t \in [a, b] \text{ predictable}\},$$

where, under P_T^σ ,

$$\Delta S_t = \mu + \sigma_{t-1}\varepsilon_t, \quad P_T^\sigma(\varepsilon_t = 1 \mid \mathcal{F}_{t-1}) = P_T^\sigma(\varepsilon_t = -1 \mid \mathcal{F}_{t-1}) = \frac{1}{2}.$$

Proposition 7 (Linear divergence of the rectangularity gap as $T \rightarrow \infty$). *Define the drift-adjusted path-dependent payoff*

$$\Phi = \sum_{t=1}^{T-1} \left| (\Delta S_{t+1} - \mu)^2 - (\Delta S_t - \mu)^2 \right|.$$

Then

$$\mathcal{G}(\Phi) = (T-1)(b^2 - a^2) = 4\sigma_0\rho(T-1).$$

In particular, for this class of path-dependent claims, the rectangularity gap grows linearly in the horizon.

Proof. For every t , we have

$$\Delta S_t - \mu = (\mu + \sigma_{t-1}\varepsilon_t) - \mu = \sigma_{t-1}\varepsilon_t.$$

Therefore

$$(\Delta S_t - \mu)^2 = (\sigma_{t-1}\varepsilon_t)^2 = \sigma_{t-1}^2 \varepsilon_t^2.$$

Since $\varepsilon_t \in \{-1, 1\}$, we have $\varepsilon_t^2 = 1$. Hence

$$(\Delta S_t - \mu)^2 = \sigma_{t-1}^2.$$

Consequently, the payoff can be rewritten pathwise as

$$\Phi = \sum_{t=1}^{T-1} |\sigma_t^2 - \sigma_{t-1}^2|.$$

We first consider the non-rectangular model. In this case volatility is fixed globally, so

$$\sigma_t = \sigma \quad \text{for all } t.$$

Therefore

$$\sigma_t^2 - \sigma_{t-1}^2 = \sigma^2 - \sigma^2 = 0$$

for every $t = 1, \dots, T-1$. Hence

$$\Phi = \sum_{t=1}^{T-1} 0 = 0$$

under every $P_T^\sigma \in \mathcal{P}_T^{\text{nr}}$. Since $\Phi \geq 0$, this implies

$$\pi_T^{\text{nr}}(\Phi) = 0.$$

Now consider the rectangular model. Since $\sigma_t \in [a, b]$ and $0 < a < b$, we have

$$a^2 \leq \sigma_t^2 \leq b^2.$$

Thus, for each $t = 1, \dots, T-1$,

$$|\sigma_t^2 - \sigma_{t-1}^2| \leq b^2 - a^2.$$

Summing over $t = 1, \dots, T - 1$ gives

$$\Phi = \sum_{t=1}^{T-1} |\sigma_t^2 - \sigma_{t-1}^2| \leq \sum_{t=1}^{T-1} (b^2 - a^2).$$

Since the summand is constant and there are $T - 1$ terms,

$$\sum_{t=1}^{T-1} (b^2 - a^2) = (T - 1)(b^2 - a^2).$$

Therefore

$$\Phi \leq (T - 1)(b^2 - a^2)$$

pathwise under every rectangular model. Hence

$$\pi_T^{\text{rec}}(\Phi) \leq (T - 1)(b^2 - a^2).$$

To obtain the reverse inequality, choose the deterministic predictable volatility process

$$\sigma_t = \begin{cases} a, & t \text{ even,} \\ b, & t \text{ odd.} \end{cases}$$

This choice is admissible in the rectangular model. Then

$$\sigma_t^2 = \begin{cases} a^2, & t \text{ even,} \\ b^2, & t \text{ odd.} \end{cases}$$

Hence the sequence of squared volatilities alternates between a^2 and b^2 , so that

$$|\sigma_t^2 - \sigma_{t-1}^2| = b^2 - a^2$$

for every $t = 1, \dots, T - 1$. Therefore

$$\Phi = \sum_{t=1}^{T-1} (b^2 - a^2) = (T - 1)(b^2 - a^2)$$

pathwise under this admissible rectangular model. It follows that

$$\pi_T^{\text{rec}}(\Phi) \geq (T - 1)(b^2 - a^2).$$

Combining the upper and lower bounds yields

$$\pi_T^{\text{rec}}(\Phi) = (T - 1)(b^2 - a^2).$$

Since $\pi_T^{\text{nr}}(\Phi) = 0$, we conclude that

$$\mathcal{G}(\Phi) = (T - 1)(b^2 - a^2).$$

Finally, substituting

$$a = \sigma_0 - \rho, \quad b = \sigma_0 + \rho,$$

gives

$$b^2 - a^2 = (\sigma_0 + \rho)^2 - (\sigma_0 - \rho)^2 = 4\sigma_0\rho.$$

Hence

$$\mathcal{G}(\Phi) = 4\sigma_0\rho(T - 1).$$

□

This construction is built from the same volatility-interval structure as the numerical experiments: the non-rectangular model uses one unknown but fixed volatility parameter, while the rectangular model allows volatility to be selected locally as a predictable process. The proposition does not imply that every payoff in the volatility-interval model has a linearly divergent rectangularity gap. Rather, it shows that this model structure contains enough local switching freedom under rectangularization to generate a linear-in-horizon gap for a suitable path-dependent payoff. The example therefore provides an analytical mechanism consistent with the numerical findings. In the non-rectangular model, the intertemporal restriction

$$\sigma_1 = \sigma_2 = \dots = \sigma_T$$

rules out volatility switching. In the rectangular model, this restriction is removed, and the sequence

$$\sigma_1, \dots, \sigma_T$$

can vary over time. The payoff Φ is specifically designed to detect and accumulate this local variation. Hence, the example isolates one channel through which rectangularization can create increasing overpricing as the horizon grows.

7 Conclusion and Outlook

This thesis investigated whether the rectangularity assumption commonly imposed on ambiguity sets in robust asset pricing is a benign computational simplification or whether it introduces a substantive distortion of the resulting prices. We approached this question by formalizing the rectangular hull of a non-rectangular ambiguity set, defining the rectangularity gap $\mathcal{G}(\xi) = \pi^{\text{rec}}(\xi) - \pi^{\text{nr}}(\xi)$, and studying it both numerically and analytically in the parametric volatility-interval model. The evidence assembled in Sections 5 and 6 suggests that the gap can be substantial. It scales with the ambiguity radius, grows approximately linearly with the horizon for the unbounded Lipschitz payoffs we tested, and admits an explicit closed-form lower bound of order $4\sigma_0\rho(T - 1)$ for a suitably chosen path-dependent claim. Within the settings we study, the rectangularity assumption should therefore be understood as an economically substantive modeling choice rather than a purely technical relaxation. Several directions would sharpen these results and are natural starting points for further work.

The contrast between the digital-call simulations where the gap saturates once π^{rec} reaches the payoff ceiling, and the European-call simulations, where the gap appears to grow roughly linearly in T and monotonically in ρ , points to a finer dichotomy. We conjecture that for bounded upper semianalytic payoffs the gap satisfies $\mathcal{G}(\Phi) \leq \|\Phi\|_\infty$ uniformly in T , while for Lipschitz payoffs with linear growth at infinity the gap grows at most linearly in T at fixed $\rho > 0$. A precise statement separating the bounded and unbounded cases and identifying the regularity threshold at which the asymptotic regime changes is left for future work.

The numerical experiments in this thesis focused on the parametric volatility model because the rectangular DP and non-rectangular LP were difficult to implement reliably for the finite-

support Wasserstein ambiguity set, owing to the way the global transport constraint couples both the initial path weights and the conditional transition probabilities at every history. A clean implementation of the rectangular hull in the Wasserstein case would allow the numerical analysis to be replicated in a genuinely nonparametric setting. On the theoretical side, a Wasserstein analogue of the consistency as $\rho \downarrow 0$ would presumably replace the volatility coupling by an optimal-transport coupling between path measures with shared transport budget ρ , and the natural conjecture is that the gap is again of order ρ for payoffs that are Lipschitz with respect to the ground metric d_Ω .

Moreover, all results in this thesis are formulated in discrete time on a finite horizon, which made it possible to use the Bouchard and Nutz (2015) framework and to compare LP, DP and actor-critic prices on a common footing. The continuous-time analogue of the rectangularity gap comparing, for instance, the Avellaneda et al. (1995) uncertain volatility model against a non-rectangular version in which σ is an unknown but path-constant parameter, would be the natural next step. The coupling argument of the divergence with horizon result has a continuous-time counterpart through a synchronous coupling of two Itô processes with different volatilities, and the linear-in- T lower bound suggests that the continuous-time gap should scale with the quadratic-variation difference between the two models. Making these statements precise would require dealing with the non-dominated nature of the ambiguity set in continuous time, for which the framework of Biagini et al. (2017) provides a natural starting point.

A more fundamental limitation, which the rectangularity gap inherits rather than creates, is that superhedging prices are widely regarded as too conservative for practical use even in the classical setting with a single reference measure. In incomplete markets the superhedging price typically sits at the upper end of the no-arbitrage interval $[\inf_{Q \in \mathcal{M}} \mathbb{E}_Q[\xi], \sup_{Q \in \mathcal{M}} \mathbb{E}_Q[\xi]]$, and is often so far above any reasonable transaction price that it provides little guidance for actual hedging decisions. This is precisely the motivation for utility indifference pricing, good-deal bounds, and risk-measure-based pricing approaches. The non-rectangular price $\pi^{\text{nr}}(\xi)$ studied in this thesis is a strict improvement over its rectangular counterpart, but it remains a worst-case quantity and is subject to the same critique. The rectangularity gap quantifies one specific source of excess conservativeness, the one introduced by a tractability-driven structural assumption on the ambiguity set, and isolating this source seems valuable precisely because it can be removed without abandoning the worst-case framework. A complete treatment of conservativeness in robust pricing would combine the present analysis with a relaxation of the superhedging criterion itself, for instance by allowing a controlled probability of shortfall under the worst-case measure. The penalized MDP formulation of Section 4 is already a step in this direction, since the parameter λ implicitly trades off initial capital against expected shortfall under the worst-case model, and studying how the rectangularity gap behaves as λ varies would connect the two relaxations directly.

However, the rectangularity gap as a concept is not specific to superhedging. Rectangularity is a near-universal assumption across the robust MDP literature (Iyengar, 2005; Nilim and El Ghaoui, 2005; Wiesemann et al., 2013), and the robust economics and finance literature that builds on the multiple-priors framework of Epstein and Schneider (2003), where it is invoked to recover time consistency and dynamic programming. Its conservativeness is routinely acknowledged in passing, but the literature has paid comparatively little attention to when this side effect actually matters. The hardness results of Wiesemann et al. (2013) explain why rectangularity is imposed, not how much is lost by imposing it, and most applied work proceeds as if the loss were negligible by default.

A natural direction is to develop the rectangularity gap, or a computable estimator $\hat{\mathcal{G}}$ as a proxy for it, as a general diagnostic for robust sequential decision problems. The volatility-interval

results of the numerical experiments coupled with the analytical derivations already suggest what such a diagnostic might track. Payoff regularity, horizon length, and ambiguity radius each have first-order effects on the gap. Identifying analogous mechanisms in other settings such as robust inventory control, robust portfolio choice, and robust reinforcement learning, and characterizing when the gap is small versus large, would give practitioners a principled basis for an assumption they currently make by default.

Taken together, the results suggest that rectangularity should not be treated as a harmless technical assumption. It changes the economic interpretation of the uncertainty model by replacing global consistency constraints with a setting in which conditional laws can be recombined after each history. This enables dynamic programming, but at the cost of enlarging the set of priors. The rectangularity gap quantifies the pricing impact of this enlargement. Understanding when this cost is negligible and when it becomes economically significant is therefore important for deciding when rectangular approximations are justified in robust pricing and, more generally, in robust sequential decision problems.

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